AN INTEGRATED WEB AND MOBILE-BASED MANAGEMENT SYSTEM FOR ARNIS SPORTS PROGRAM: ARNIS CONNECT

A Capstone Project Presented to the Faculty of College of

Computer, Information and Communications Technology Department

Cebu Technological University - Main Campus

In Partial Fulfillment

Of the Requirements for the Degree

Bachelor of Science in Information Systems

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May 2026

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ABSTRACT

ACKNOWLEDGMENT

This study stands as a testament to the researchers' dedication, perseverance, and collaborative effort in successfully completing the Capstone Project entitled *"*Arnis Connect: An Integrated Web-Based Management System for Arnis Sports Program.*"* The research team extends their sincere gratitude to all individuals and institutions who contributed to the realization of this study.

First and foremost, to the Almighty God, for granting the researchers the wisdom, strength, and guidance needed to overcome the challenges encountered throughout the research process. His presence has been a constant source of inspiration and perseverance.

To Cebu Technological University, for providing the academic environment, resources, and support necessary for the completion of this project.

To our research adviser, Mr. Gerry Ritz R. Densing, for his invaluable guidance, continuous support, and dedication in reviewing and improving this manuscript. His constructive feedback greatly enhanced the quality and credibility of this study.

To the Lapu-Lapu Arnis de Abanico, for their warm accommodation, cooperation, and willingness to participate throughout the duration of this study. Their support and assistance greatly contributed to the successful completion of this research.

To the panelists, for their insightful comments, suggestions, and recommendations that significantly contributed to the refinement and improvement of this study.

To our beloved parents, friends, and classmates, for their unwavering support, encouragement, and understanding during the most challenging times of this journey.

Lastly, to all members of the research group, for their teamwork, commitment, and shared determination in bringing this project to its successful completion.

Group ByteWorks

DEDICATION

This work is wholeheartedly dedicated to all those who have provided unwavering support, guidance, and encouragement throughout the completion of this study.

Above all, to the Almighty God, the source of all wisdom and strength, this achievement is humbly offered. Every success and challenge overcome was made possible through His grace and guidance.

To our beloved parents, whose sacrifices, unconditional love, and continuous support have been our greatest foundation. This accomplishment is not only ours but also a reflection of your faith in us and your dedication to our future.

To Cebu Technological University, for shaping us into capable and responsible individuals, and for providing an environment that fosters growth, learning, and excellence.

To the faculty of the College of Computer, Information and Communications Technology, for their dedication to teaching and for equipping us with the knowledge and skills necessary to face real-world challenges.

To our classmates and friends, who shared this journey with us, thank you for the support, laughter, and encouragement that made even the most difficult moments more manageable. The memories and experiences we shared will always be cherished.

And to everyone who believed in us and supported us in any way, this work is dedicated to you. Your encouragement has been an essential part of this achievement.

Renz Kayne, Christ Jhon Denver, Ulric, Jannalyn

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CHAPTER 1

PROJECT CONTEXT AND RESEARCH DESIGN

This section outlines the background and framework of the study. The project context describes the setting, key issues, and rationale that led to the research, providing readers with a clear understanding of why the study is important. It highlights relevant conditions, challenges, or gaps that the research aims to address.

The research design, on the other hand, explains how the study is structured to achieve its objectives. It details the methods, approaches, and procedures used for data collection and analysis, ensuring that the research is systematic, valid, and reliable. Together, the project context and research design establish the foundation of the study, guiding its direction and ensuring that the findings are meaningful and credible.

PROJECT CONTEXT

The project context introduces the background, purpose, objectives, significance, scope and limitations of the study, providing an overview of the situation that led to the research. It explains the key issues, challenges, or gaps that the project aims to address, helping readers understand its purpose and relevance. By outlining the environment and conditions surrounding the study, the project context establishes a strong foundation for the research and guides its overall direction.

Background of the Study

The Lapu-Lapu Arnis de Abanico Organization manages member records, training attendance, event participation, player progress, and organizational funds as part of its regular operations. These records are important because they support training monitoring, event preparation, decision-making, and financial transparency. However, when records are handled manually through notebooks, paper logs, spreadsheets, or disconnected files, the organization experiences delayed updates, duplicate or inconsistent records, weak monitoring, and difficulty in producing organized reports. This issue becomes more complex because the organization handles both public players and private players. Public players are commonly connected to school-based or group training arrangements, while private players may attend paid, private, or customized

sessions. Because attendance, event preparation, belt progression, and financial recording vary depending on these arrangements, the organization needs a management system that follows its actual workflow and not only a generic sports application (Hambali et al., 2024; Jafridin & Salamat, 2023).

The need for a digital Arnis management system is timely because sports and community-based organizations increasingly depend on accurate information, fast communication, and reliable monitoring to manage activities effectively. In information systems research, system quality, information quality, service quality, user satisfaction, and net benefits are treated as important factors in determining whether a system truly supports users and organizations (DeLone & McLean, 2003; Jeyaraj, 2020). In the Philippine setting, Arnis is also significant because Republic Act No. 9850 declares Arnis as the national martial art and sport of the Philippines, while Republic Act No. 10588 institutionalizes the Palarong Pambansa and supports sports participation among learners (Republic Act No. 9850, 2009; Republic Act No. 10588, 2013). This makes the organized management of Arnis programs relevant not only for record keeping but also for cultural, educational, and sports development.

Recent studies and systems show that digital platforms can improve membership management, coordination, real-time access, and operational efficiency. Web-based membership systems centralize member information and reduce administrative burden, while web-based sports centre systems help reduce problems caused by paper-based records and disconnected data (Hambali et al., 2024; Jafridin & Salamat, 2023). Studies on digital transformation in sports also emphasize that technology can improve communication, monitoring, and organizational responsiveness when it is properly aligned with the actual needs of the users (Christodimitropoulou, 2025; Haffner et al., 2025; Santomier, 2024; Zhang, 2024). However, these studies and systems are not designed specifically for an Arnis organization that needs to manage member profiles, automated attendance verification, event documentation, fund monitoring, registration approval, player suggestions, belt progression, and public-private player arrangements in one platform.

This gap supports the development of Arnis Connect: An Integrated Web and Mobile-Based Management System for Arnis Sports Program. The system is proposed as a browser-accessible platform that centralizes member management, registration approval, automated attendance, event management, player suggestion, belt progression, finance, and reports. Unlike general sports team platforms, Arnis Connect is designed around the workflow of the Lapu-Lapu Arnis de Abanico Organization and includes features that directly support Arnis training and player

progress. The study contributes mainly to SDG 9, Industry, Innovation, and Infrastructure, by applying digital innovation to a local sports organization. It also supports SDG 4, Quality Education, by helping organize training participation, learner involvement, and progress monitoring in a structured Arnis program (United Nations, 2015; United Nations Department of Economic and Social Affairs, n.d.-a, n.d.-b).

Purpose and Description of the Project

The primary purpose of Arnis Connect is to replace manual and fragmented administrative practices with a centralized, role-based, and browser-accessible management system. The system is designed to help the Lapu-Lapu Arnis de Abanico Organization manage members, attendance, events, funds, player progress, registration requests, and reports in one platform. This follows the principle that information systems should collect, process, store, and distribute useful information to support operational control and decision-making (Laudon & Laudon, 2020; Valacich & George, 2020).

Arnis Connect is built using PHP, MySQL, HTML, CSS, and JavaScript. The system is web-based, but it is also mobile-accessible because users can open it through standard mobile browsers on smartphones and tablets. This means that the system does not require a fully native mobile application in the current version. It is intended for the Admin or Lead Coach, Coach, Finance Officer, Member or Player, and Applicant. Each role receives access only to functions connected to its responsibility, following role-based access control principles that assign permissions through roles rather than individual manual permissions (Ferraiolo et al., 2001; National Institute of Standards and Technology, 2016; Sandhu et al., 1996).

The system includes member management, registration management, automated attendance management, training and belt progression, event management, player suggestion, finance management, reports and monitoring, and security control. The attendance module is improved through facial recognition and geolocation verification. The system checks whether the player is physically near the training location and verifies the identity of the user before the attendance record is stored. Because no system can absolutely prevent all forms of spoofing or misuse, suspicious attendance records are flagged for coach review instead of being automatically accepted without validation. This makes the feature more realistic, privacy-aware, and defensible.

The player suggestion feature is also revised so that it does not depend only on attendance or rank. The system may consider attendance percentage, belt or rank eligibility, event type, public

or private player category, past event participation, achievements, availability, and coach remarks. The final selection of players remains under the authority of the coach or lead coach. The belt progression feature presents player development in a structured progress interface, where the player can see the current belt, completed requirements, locked next belt level, and coach assessment status. This supports motivation and progress monitoring while keeping promotion decisions under coach validation.

Key features of the application include:

Table 1
Key Features of Arnis Connect

Feature	Description
Login and Session Management	Provides secure user authentication, login, logout, session control, and password verification to ensure that only authorized users can access the system
User Role Management	Restricts system access based on assigned user roles such as Administrator, Coach, and Finance User to protect data integrity and sensitive organizational records
Member Management	Stores, updates, searches, and monitors member records including personal information, public or private player classification, rank, member type, status, and training preference
Registration Management	Allows applicants to submit registration requests through a public form and enables the Administrator or Lead Coach to review, approve, or reject applications before converting them into official member accounts
Automated Attendance Management	Records training attendance using face verification, geolocation checking, timestamp logging, and coach review for flagged records.
Training and Belt Progression	Tracks the current belt, completed requirements, next-belt unlock status, coach assessment, and belt history.
Event Management	Creates events, assigns participants, stores event documents, and tracks participation and outcomes.
Player Suggestion	Suggests possible participants using attendance, belt eligibility, event type, player category, achievements, availability, and coach validation.
Finance Management	Records income and expenses, links transactions to events when needed, and supports fund transparency.
Reports and Monitoring	Generates member, attendance, event, finance, and dashboard summary reports.

Multi-Platform Browser Access	Allows users to access the system through desktops, laptops, tablets, and smartphones using modern browsers.

Objectives of the Study

This study aims to design, develop, and evaluate Arnis Connect: An Integrated Web and Mobile-Based Management System for Arnis Sports Program. Specifically, it seeks to accomplish the following objectives arranged according to the SDLC flow:

1. To analyze the current administrative practices of the Lapu-Lapu Arnis de Abanico Organization in managing members, training sessions, attendance, event participation, funds, player progress, and reports.
2. To identify the key problems, user needs, functional requirements, and non-functional requirements needed to improve the organization's manual and disconnected workflow.
3. To design the system architecture, database structure, user interface, functional decomposition diagram, use case diagrams with include and extend relationships, and swimlane activity diagram of Arnis Connect based on the actual workflow of the organization.
4. To develop the member management and registration modules with features for member profiling, public and private player classification, member access, and approval or rejection of registration requests.
5. To develop an automated attendance module using facial recognition, geolocation verification, timestamp logging, and coach review of flagged records to support more reliable training attendance monitoring.
6. To develop the training and belt progression module to monitor current belt level, completed requirements, locked/unlocked progression status, coach assessment, and belt history.
7. To develop the event management and player suggestion modules to support event documentation, participant assignment, and event preparation using attendance, belt eligibility, event type, player category, achievements, availability, and coach validation.
8. To develop the finance, reports, and role-based access control modules for organized fund recording, report generation, secure system access, and data integrity.

9. To evaluate Arnis Connect in terms of functional suitability, usability, reliability, performance efficiency, security, and compatibility using selected ISO/IEC 25010 software quality characteristics administered to selected respondents.

Scope and Limitations

This section will discuss the in-scope and out-scope of the study.

Scope

This study covers the design, development, and evaluation of Arnis Connect: An Integrated Web and Mobile-Based Management System for Arnis Sports Program for the Lapu-Lapu Arnis de Abanico Organization. The system centralizes the administrative and operational functions of the organization, including member records, registration requests, training attendance, events, player suggestions, belt progression, fund records, audit logs, and reports. This is aligned with the purpose of information systems, which is to collect, process, store, and distribute information to support organizational operations and decision-making (Laudon & Laudon, 2020; Valacich & George, 2020). The system is intended for the Administrator or Lead Coach, Coach, Finance Officer, Member or Player, and Applicant.

The system allows applicants to create an account, complete verification, and submit a member application form. The Administrator or Lead Coach can review submitted applications and approve or reject requests. Once approved, the applicant is stored as an official member in the database. Members can access the system to view their profile, application status, attendance history, upcoming trainings, event participation, and belt progression. Coaches can create training sessions, manage attendance, review flagged attendance records, assess belt progression, and assist in event preparation. The Finance Officer can record income and expenses, monitor balances, and generate financial summaries. The Administrator or Lead Coach can manage users, members, events, reports, audit logs, and overall system settings. Access to these functions is controlled through role-based access, which limits user permissions based on assigned responsibilities (Ferraiolo et al., 2001; Sandhu et al., 1996).

The attendance module includes facial recognition and geolocation verification to improve the accuracy of attendance recording during training sessions. The system checks the member's identity and location before attendance is accepted or marked for coach review. The belt progression module provides a structured progress interface where the next belt level remains locked until the required training milestones and coach assessment are completed. The player

suggestion feature provides decision support for event preparation by considering attendance, belt level, player category, event type, achievements, availability, and past participation. However, the final selection of players remains under the decision of the coach or lead coach.

The system is designed to be accessible through standard web browsers on desktops, laptops, tablets, and smartphones. Since the system is browser-based, users do not need to install a separate desktop or mobile application. The system supports multi-device access as long as the device has a supported browser and internet connection.

Limitations

The current version of Arnis Connect is web-based and mobile-accessible through a browser, but it is not a fully native mobile application. It depends on the availability of an internet connection, browser compatibility, and the device used by the user. Some features, especially automated attendance verification, may not function properly if the member's device does not allow camera access, location access, or if the internet connection is weak or unstable during attendance taking.

The automated attendance process is designed to improve attendance accuracy, but it still requires proper user cooperation and device readiness. If the device cannot access the camera or location service, or if the connection fails during submission, the attendance record may need to be reviewed or validated by the coach. For this reason, the system includes flagged attendance review to allow authorized personnel to check records that could not be fully verified by the system. Since the system involves personal information and attendance-related data, proper notice, consent, access control, and responsible data handling must be observed in accordance with the Data Privacy Act of 2012 and related privacy guidelines (National Privacy Commission, 2016; Republic Act No. 10173, 2012).

The player suggestion feature is rule-based in the current version and does not yet use advanced machine learning. It only provides decision support based on stored records such as attendance, belt level, category, achievements, availability, and participation history. The final decision in selecting players for events still depends on the coach or lead coach. The belt progression feature also monitors progress and requirements, but actual promotion to the next belt level still requires coach assessment and approval.

The system is limited to the Lapu-Lapu Arnis de Abanico Organization and does not yet support multi-organization deployment. Its effectiveness depends on accurate data entry, proper role assignment, responsible user behavior, stable internet connectivity, and proper maintenance of the database and hosting environment.

Significance of the Study

The study is significant because it provides a centralized and practical system for improving Arnis organizational management. By replacing scattered files and paper-based records with a role-based digital platform, Arnis Connect supports better data organization, faster retrieval, improved monitoring, and more transparent reporting. The study is also aligned with design science research because it creates and evaluates a purposeful system artifact intended to address a real organizational problem (Hevner et al., 2004).

Lead Coach / Administrator. The Lead Coach or Administrator benefits from centralized access to members, registration approvals, attendance, events, player suggestions, belt progression, funds, and reports. This improves monitoring and decision-making.

Coach. Coaches benefit from faster attendance recording, access to member progress, event participant suggestions, and a structured belt progression view that supports training assessment.

Finance Officer. The Finance Officer benefits from a more organized structure for recording income and expenses, tracking event-related funds, and producing financial summaries.

Members / Players. Members benefit from access to their attendance records, event participation, and belt progression status. This helps them understand their development and responsibilities in the organization.

Applicants. Applicants benefit from a more organized registration process where their information can be submitted, reviewed, approved, or rejected through the system.

Organization. The organization benefits from a system that matches its actual workflow, including public and private player arrangements, training records, event documentation, and fund monitoring.

Filipino Community. The study supports the modernization of Arnis administration, which is relevant because Arnis is formally recognized as the national martial art and sport of the Philippines (Republic Act No. 9850, 2009; Yap, 2017).

Future Researchers and Developers. The study may serve as a reference for future systems involving sports administration, cultural sports management, attendance verification, role-based access control, or community-based information systems.

RESEARCH DESIGN

The research design outlines the overall plan and structure of the study, ensuring a systematic and organized approach to achieving its objectives. It highlights the methods used, as well as the flow of the study from data collection to analysis. This section also describes the research environment, identifies the respondents involved, and explains the instruments utilized to gather data.

Methods

This study uses a descriptive-developmental research design. The descriptive component is used to examine the current practices, user roles, documents, and problems of the Lapu-Lapu Arnis de Abanico Organization. The developmental component guides the design, development, testing, and evaluation of the proposed Arnis Connect system. This approach is suitable because the study does not only describe a problem but also creates a working system to address that problem (Hevner et al., 2004; Pressman & Maxim, 2019).

The development process follows an iterative and incremental approach. The system is divided into manageable modules such as login, registration, members, attendance, belt progression, events, finance, and reports. Each module can be designed, developed, tested, revised, and improved based on feedback. Iterative and incremental development is appropriate for information systems projects because it supports repeated refinement and gradual improvement over time (Larman & Basili, 2003; Sommerville, 2016).

The study also uses mixed methods. Qualitative data from interviews and document analysis support the identification of user needs, workflows, and system requirements. Quantitative data from the questionnaire support the evaluation of system quality and acceptability using selected ISO/IEC 25010 quality characteristics. Purposeful sampling is used because the

respondents are selected based on their direct involvement in the organization and their ability to provide relevant feedback about the system (Creswell & Creswell, 2018; Palinkas et al., 2015).

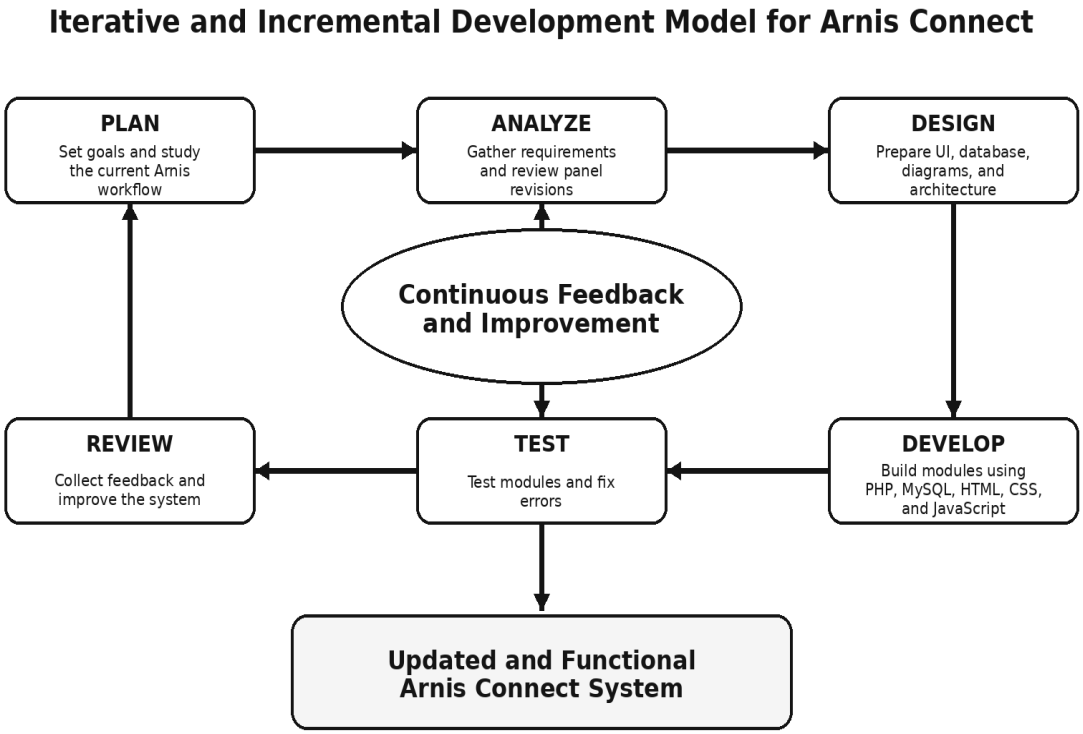


Figure 1
Iterative and Incremental Development Model for Arnis Connect

Flow of the Study

This section presents the logical progression of the study from data gathering up to the proposed system output.

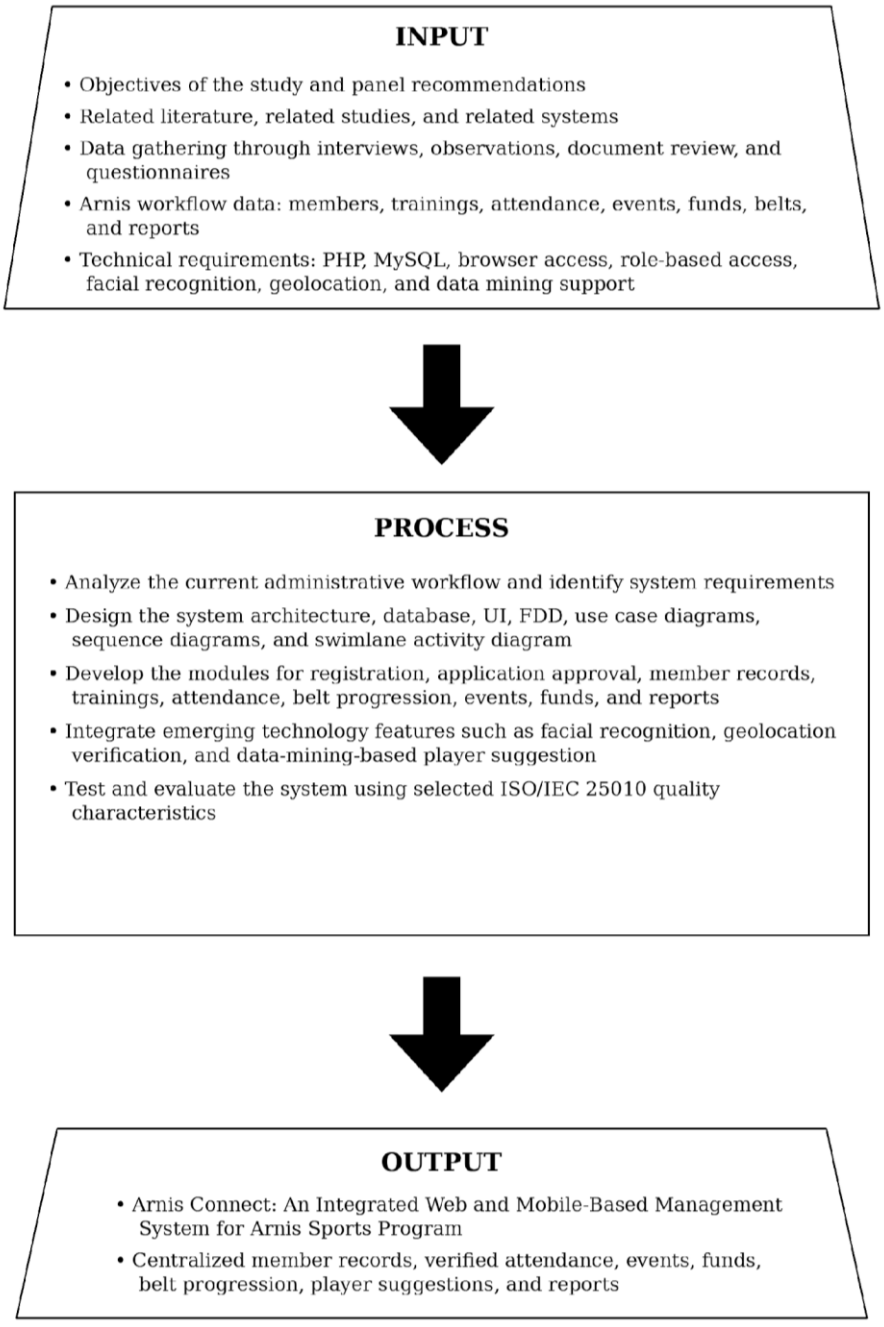


Figure 2
Flow of the Study

The flow of the study begins with the collection of inputs such as stakeholder requirements, existing records, related studies, related systems, and panel recommendations. These inputs are processed through requirements analysis, system design, iterative development, testing, and evaluation. The output is Arnis Connect, a centralized system for managing members, attendance, events, player progress, funds, and reports.

Environment of the Study

The study will be conducted within the operational setting of the Lapu-Lapu Arnis de Abanico organization in Lapu-Lapu City, Cebu, on Mactan Island. Lapu-Lapu City is a highly urbanized city in Metro Cebu and is officially described by the city government as a historic and strategic gateway area connected to mainland Cebu by major bridges and served by the Mactan-Cebu International Airport. This physical setting is appropriate for the study because it places the proposed system in the actual local environment where Arnis-related training, coordination, and public activities are carried out.

As for the institutional and organizational background, publicly available records identify Lapu-Lapu Arnis de Abanico as an active Arnis/Eskrima group in Lapu-Lapu City that participates in heritage, instructional, and community-centered activities. News coverage of the 503rd Kadaugan sa Mactan commemoration in April 2024 identified internationally acclaimed eskrimador Jonel Espinosa Pepito as the father of the group, while public event materials and tourism-linked programming also show the organization's involvement in Arnis workshops and cultural demonstrations in Lapu-Lapu City. These records indicate that the organization is not merely symbolic but functions as an active local Arnis group engaged in training and public representation of Filipino martial arts.

In terms of operational context, the study takes place in an environment where Arnis is practiced not only as a martial art but also as a structured cultural and sports activity. Public reports show that Arnis demonstrations involving large groups of eskrimadors are conducted in major city commemorations, while related educational and workshop initiatives in Lapu-Lapu City support the teaching, practice, and promotion of Arnis among students, teachers, and community participants. This environment is relevant to the study because it reflects a workflow that may involve member organization, training participation, event preparation, public performances, and coordination among coaches, officers, and participants.

From the academic and technical perspective, the environment supports the development of a browser-based management system because the organization operates in a city context that is connected to education, tourism, and public cultural programming, while the proposed Arnis Connect platform is intended to run through standard web technologies and internet-accessible devices. The legal and cultural recognition of Arnis under Republic Act No. 9850 also strengthens the relevance of designing a system specifically for Arnis-related administration, since the sport is formally recognized by the State as a national martial art and sport. In practical terms, a web-based platform is appropriate for this environment because it can support centralized records, multi-device access, and easier management of organizational activities without requiring separate local installations on each device.

This environment is appropriate for the present study because it directly supports the objectives of developing and evaluating a management system for an actual Arnis organization. The physical location in Lapu-Lapu City places the study within a historically and culturally relevant Arnis setting, the organization itself demonstrates active participation in training and heritage-related events, and the operational context reflects the type of administrative workflow that a system like Arnis Connect is meant to improve. For these reasons, the selected environment provides both practical relevance and contextual grounding for the proposed research.

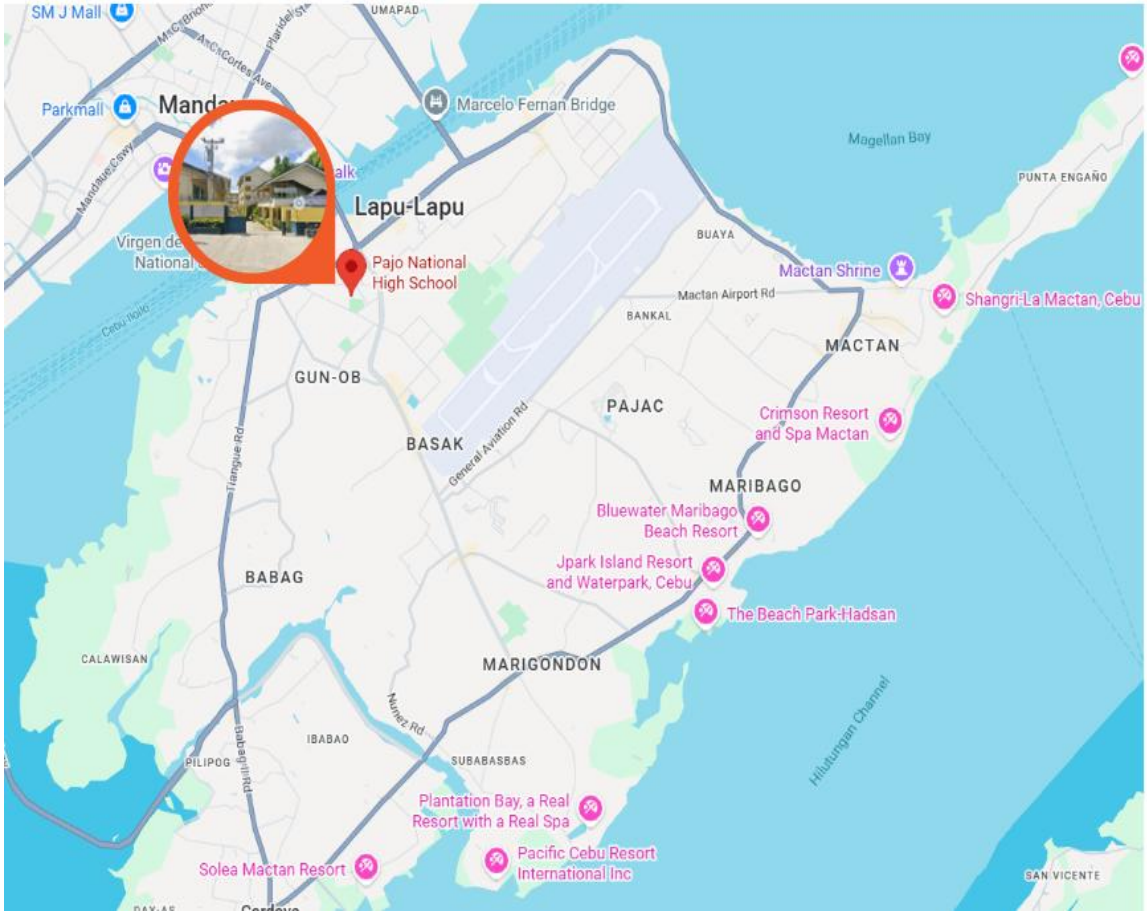


Figure 3
Locale of the Study

Respondents

The respondents of this study consisted of forty-one (41) individuals who had direct relevance to the operational context and evaluation of the proposed system. Since Arnis Connect is intended for the Lapu-Lapu Arnis de Abanico Organization, the respondents were selected from among those who are directly involved in the organization's administrative, training, coordination, and participation processes. Purposive sampling was used in selecting the respondents because it is appropriate when participants are intentionally chosen based on their ability to provide rich and relevant information for the study (Palinkas et al., 2015)

Lead Coach / Administrator. One (1) Lead Coach / Administrator was included as a respondent in this study. This respondent was considered essential because the role involves overall supervision of the organization, including the monitoring of member records, approvals, activities, and other major administrative functions. As the primary authority within the organization, the Lead Coach / Administrator is in a strong position to evaluate the usefulness, relevance, and practicality of the proposed system in relation to actual organizational workflow.

Coaches / Trainers. Five (5) Coaches / Trainers were included as respondents. They are directly involved in the conduct of trainings, attendance monitoring, event preparation, and coordination with players. Their inclusion is important because they represent core operational users who can assess whether the system supports day-to-day training-related and event-related tasks in an organized and practical manner.

Officers / Committee Members. Five (5) Officers / Committee Members were included as respondents. These individuals are involved in organizational coordination and support functions related to activities, records, and internal processes. Their participation is necessary because they can help evaluate how the proposed system supports administrative coordination, documentation, and other organization-level tasks connected to the management of the Arnis sports program.

Active Members / Players. Thirty (30) Active Members / Players were included as respondents. As individuals whose profiles, attendance, and event participation are managed within the organization, they provide an important user-side perspective regarding the relevance, usability, and acceptability of the proposed system. Their responses help determine whether the system is understandable and beneficial from the viewpoint of those who are directly affected by the organization's administrative processes.

The total respondent pool consisted of forty-one (41) participants, composed of one (1) Lead Coach / Administrator, five (5) Coaches / Trainers, five (5) Officers / Committee Members, and thirty (30) Active Members / Players. These respondents were considered appropriate because they represent the actual organizational context in which Arnis Connect will be used and evaluated. Their responses are expected to provide relevant insights into the practicality, usability, and acceptability of the proposed system in the context of the Arnis organization.

Table 2
Distribution of Respondents

Respondent Category	Number of Respondents	Percentage	ISO25010 Characteristics
Lead Coach / Administrator	1	2.4%	Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, Portability
Coaches / Trainers	5	12.2%	Functional Suitability, Usability, Performance Efficiency
Officers / Committee Members	5	12.2%	Functional Suitability, Usability, Performance Efficiency
Active Members / Players	30	73.2%	Usability, Reliability, Functional Suitability
Total	**41**	**100%**	

Instruments

This study uses three primary data-gathering instruments: a semi-structured interview guide, a documentary analysis checklist, and a structured evaluation questionnaire. These instruments were selected to support both the requirements analysis and the system evaluation phases of the study. Their combined use is appropriate because Arnis Connect is intended not only to be developed as a system artifact but also to be evaluated in terms of its acceptability and quality in the actual organizational context of the Lapu-Lapu Arnis de Abanico Organization.

The semi-structured interview guide is used during the requirements elicitation stage to gather qualitative information from the Lead Coach / Administrator and other relevant organization personnel. The interview focuses on the current workflow of the organization in relation to member record handling, training attendance, event preparation, financial recording, reporting practices, and the differing arrangements involving public and private players. Because the interview is semi-structured, it ensures that the major topics needed by the study are consistently covered while still

allowing the respondents to explain their actual practices, difficulties, and expectations in detail. This makes the gathered requirements more grounded in the real operational context of the organization.

The documentary analysis checklist is used to review project-related and system-related documents relevant to the development of Arnis Connect. These include the manuscript draft, project planning documents, software design materials, diagrams, and the actual SQL database structure. The checklist helps ensure that the description of the system remains aligned with the real project files, particularly the implemented records and relationships reflected in the database. Based on the uploaded files, the actual database structure includes tables such as users, members, trainings, attendance, events, event_documents, event_participants, funds, and achievements. The use of documentary analysis is important in this study because it helps maintain consistency between the manuscript, the technical design, and the actual implemented system.

The main evaluation instrument of the study is a structured questionnaire based on selected ISO/IEC 25010 software quality characteristics. In line with the objectives of the study, the questionnaire is designed to assess Arnis Connect in terms of functional suitability, usability, reliability, performance efficiency, and security. The questionnaire items are written in a context-specific manner so that they reflect the actual functions of the system, such as user access, member management, training and attendance recording, event handling, fund recording, and report-related functions. To make the evaluation more appropriate to the respondents, the questionnaire may use common core items for all respondent groups together with role-sensitive items that match their actual interaction with the system.

Table 3
ISO/IEC 25010 Quality Characteristics and Sample Questionnaire Item

Characteristic	Description	Sample Questionnaire Item
Functional Suitability	The degree to which the system provides functions that meet user needs.	Arnis Connect provides the needed features for managing members, attendance, events, funds, and reports.
Usability	The degree to which the system is easy to learn, understand, and use.	The system interface is easy to navigate even with minimal training.
Reliability	The degree to which the system performs consistently without failure.	The system remains stable while recording attendance and managing records.
Performance Efficiency	The degree to which the system responds quickly and uses resources properly.	The system loads records and reports within an acceptable time.
Security	The degree to which the system protects data and controls access.	The system protects personal, attendance, and financial records through proper access control.
Compatibility	The degree to which the system works properly across browsers and devices.	The system can be accessed using common desktop and mobile browsers.

The structured questionnaire will be administered through Google Forms for accessibility and convenience. Google Forms is appropriate because it allows the researchers to distribute the instrument online and to collect responses efficiently using computers, tablets, or smartphones. The questionnaire uses a five-point Likert scale ranging from Strongly Disagree to Strongly Agree in order to quantify the respondents' assessment of the system. This format supports easier tabulation and statistical treatment of responses during the evaluation stage.

To strengthen the quality of the instrument, the questionnaire shall be reviewed first for clarity, relevance, and alignment with the objectives and actual system functions before full administration. Its internal consistency shall also be tested using Cronbach's alpha, with a coefficient of 0.70 or higher considered acceptable for reliability. In this way, the instrument supports a more valid and reliable evaluation of Arnis Connect from the perspective of its intended respondents.

Treatment of Data

The study treated the gathered data using both qualitative and quantitative approaches. The qualitative data collected through the semi-structured interviews and documentary review were analyzed using thematic analysis. The responses and document findings were organized, coded, and grouped into meaningful themes such as workflow problems, user needs, role-based access

requirements, attendance management issues, and event-related concerns. Thematic analysis is appropriate because it helps identify recurring patterns and convert them into interpretable categories that can support system requirements and design decisions (Braun & Clarke, 2006).

The quantitative data collected through the structured questionnaire were intended to be analyzed using frequency count, percentage, and weighted mean. Frequency count would determine how many respondents selected each response option, percentage would describe the proportion of those responses, and weighted mean would summarize the overall ratings given for the selected ISO/IEC 25010 criteria. These statistical treatments are appropriate for describing evaluation results in system assessment studies and for supporting interpretation of overall user acceptability. The results would then be presented in tabular form for the evaluation chapter of the manuscript.

Scoring Procedures

The responses in the structured questionnaire were measured using a 5-point Likert scale. This scale was used to determine the respondents' level of agreement regarding the quality and acceptability of Arnis Connect in terms of functional suitability, usability, reliability, and performance efficiency. Likert's original scaling method remains a widely accepted approach for measuring attitudes and perceptions in survey-based research (Likert, 1932).

Table 4
Likert Scale Interpretation

Scale	Interpretation
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

To interpret the weighted mean of the responses, the following level of acceptability was used:

Table 5
Level of Acceptability of Arnis Connect

Weight	Scale Range	Category	Verbal Description
5	4.20 – 5.00	Strongly Agree	Indicates that the respondents strongly agree that the system is effective and acceptable.
4	3.40 – 4.19	Agree	Indicates that the respondents agree that the system is useful and functional.
3	2.60 – 3.39	Neutral	Indicates that the respondents are undecided about the effectiveness of the system.
2	1.80 – 2.59	Disagree	Indicates that the respondents believe that some parts of the system still need improvement.
1	1.00 – 1.79	Strongly Disagree	Indicates that the respondents believe that the system is not yet effective or acceptable.

Definition of Terms

Arnis Connect – the proposed integrated web and mobile-based management system for the Arnis sports program.

Administrator / Lead Coach – the authorized user with the highest level of access, responsible for system management, approval of requests, and overall monitoring of records.

Coach – an authorized user who manages day-to-day operational tasks such as attendance recording and event preparation.

Finance Officer – an authorized user responsible for recording and monitoring income, expenses, and related summaries.

Member – an individual who belongs to the Arnis organization and participates in trainings and events.

Public Player – a member associated with public or group-based training arrangements.

Private Player – a member associated with private, customized, or fee-based training arrangements.

Registration Approval Workflow – the process in which a prospective member submits an application and is reviewed for approval or rejection by the administrator.

Attendance Tracking – the digital recording of a member's training participation, including attendance status and timestamps.

Event Documentation – the structured recording of event details such as date, location, type, and participants.

Player Suggestion Feature – a feature that recommends eligible members for event participation based on defined criteria such as attendance and related records.

Functional Suitability – the degree to which the system provides functions that meet specified user needs.
Usability – the degree to which the system is easy to learn, understand, and use.

Reliability – the degree to which the system performs consistently under stated conditions.

Performance Efficiency – the degree to which the system provides appropriate response time and resource use during operation.

Web-Based System – a system that is accessed through a web browser rather than as a standalone installed application.

CHAPTER 2

RELATED LITERATURE AND TECHNICAL BACKGROUND

This chapter presents the review of existing studies, systems, and technologies related to the development of the Arnis Connect system. It highlights relevant studies and literature on scheduling systems, database management, and mobile and web-based development that provide insights and support for the study. This chapter also discusses the technical tools, frameworks, and methodologies used in the system's development, serving as the foundation for the design and implementation of Arnis Connect.

RELATED LITERATURE

The related literature section presents a review of existing studies, theories, and published works that are relevant to the research topic. It provides a foundation for understanding the study by highlighting what has already been discovered, as well as identifying gaps that the current research aims to address. By analyzing and synthesizing previous findings, this section helps establish the significance of the study and supports its direction and framework.

Related Studies

The reviewed studies collectively emphasize the growing importance of digital transformation in sports and organizational management. Existing literature shows that many organizations still experience inefficiencies caused by manual, paper-based, and fragmented record-keeping systems, leading to problems in attendance monitoring, membership management, event coordination, financial tracking, and reporting. Previous studies and existing sports management platforms demonstrated that web-based systems improve operational efficiency, centralize records, reduce administrative errors, and enhance accessibility and organizational control. Related works also highlighted the significance of attendance monitoring, participation tracking, integrated financial management, and role-based access control in maintaining organized and secure administrative operations. These studies support the development of Arnis Connect because the proposed system integrates membership management, attendance tracking, event documentation, financial recording, reporting, and player suggestion features into a centralized platform specifically tailored to the workflow and operational needs of an Arnis organization.

Current Process of Administrative Management in Sports Organizations

Sports organizations increasingly rely on digital tools to improve administration, communication, and record management. Thompson, Naraine, and Parent (2024) found that sport organizations use digital technologies to support communication, stakeholder engagement, and organizational change, although many organizations still remain at the level of small-scale digital improvements rather than full transformation. In a related review, Stegmann and Lang (2025) reported that digital transformation in voluntary sports organizations is generally driven by the expectation of improved service quality, operational efficiency, and better handling of organizational tasks. These findings are relevant to Arnis Connect because the proposed system is also intended to move the Arnis organization from fragmented manual processes toward a more structured digital management platform.

Despite the growing role of digital technologies, many organizations still depend on manual records, paper-based logs, and scattered spreadsheets. In the case of the Lapu-Lapu Arnis de Abanico Organization, the project documents describe existing difficulties in managing member profiles, attendance, event records, and funds through disconnected records. These problems include slow retrieval of information, inconsistent updates, weak reporting, and difficulty in monitoring organizational performance. Arnis Connect was proposed precisely because the organization needed a centralized, role-based, and more realistic system that reflects its actual workflow, including the distinction between public and private players.

Web-Based Membership and Record Management Systems

One of the strongest related studies for Arnis Connect is the work of Hambali, Alsamman, Osman, and Mohammed (2024), who developed and evaluated a web-based membership management system. Their study emphasized that as organizations grow, manual management of many members becomes impractical and prone to errors. They found that a web-based system can centralize member data, automate administrative work, and improve usability for both administrators and members. This is directly relevant to Arnis Connect because one of its main functions is the centralized handling of member records, registration requests, and participation history.

A related study by Jafridin and Salamat (2023) developed a web-based management system for a sports centre. Their paper identified the paper-based method as a source of duplicate bookings and lost customer information, and their proposed solution used a PHP- and MySQL-based web

system to organize records more systematically. Although their context involved facility and booking management rather than Arnis administration, the underlying issue is highly similar: manual and fragmented records make operations slower, less reliable, and more difficult to track. Their study supports the argument that web-based systems are effective in improving record organization, operational control, and administrative efficiency.

These studies support the development of Arnis Connect because they show that membership- and record-focused information systems are effective when they centralize data and reduce manual administrative work. In Arnis Connect, this concept is extended by combining member management with attendance tracking, event documentation, funds recording, and reporting in one integrated platform.

Attendance, Participation, and Event Monitoring

Attendance and participation records are among the most important operational data in sports organizations because they affect training monitoring, performance visibility, and decision-making for future activities. The product features defined in the Arnis Connect SRS specifically include training attendance tracking, event and performance documentation, and reporting and analytics. These functions are designed to create reliable participation histories and complete activity trails for both members and events.

Related systems literature also supports these features. Team management platforms and scheduling tools used in sports commonly emphasize schedules, member availability, attendance tracking, event organization, and communication as their core functions. For example, TeamSnap highlights schedules, member availability, assignments, messaging, and money management as part of its all-in-one team management solution, while TeamLinkt presents registration, scheduling, communication, reporting, and attendance-related app features for leagues, clubs, and associations. These examples show that effective sports administration platforms typically integrate attendance, planning, and communication rather than treating them as isolated features.

Arnis Connect adopts the same integrated principle but customizes it to the workflow of an Arnis organization. Unlike many general team apps, it is designed not only for practices and member visibility, but also for official member registration approval, event documentation, organization-specific reports, and player suggestion based on attendance and participation history.

Financial Monitoring and Reporting in Organizational Systems

Financial tracking is also an important part of organizational management systems because administrative transparency depends not only on activity records but also on clear documentation of income and expenses. The Arnis Connect SRS and project charter specifically state that the system must log and categorize organizational financial transactions, including income from training fees, private trainings, and talent fees, and must support financial summaries and reports.

In existing sports team management platforms, financial tools are increasingly integrated with schedules and administration. TeamSnap includes invoicing and sponsorship support as part of its money management features, while Spond includes payment requests, event planning, attendance tracking, and reminders for groups and clubs. These existing systems show that financial recording is most useful when connected with events, members, and operational activities rather than treated as a separate spreadsheet-based process. This supports the decision to include a finance module in Arnis Connect instead of leaving fund management outside the system.

Role-Based Access and Centralized Information Systems

The proposed system also relies on the principle of role-based access. The Arnis Connect project documents define different user roles such as administrator or lead coach, coach, and finance officer, each with specific access privileges and responsibilities. This is consistent with the NIST role-based access control model, which explains that access is more effectively administered when permissions are assigned through roles rather than individually. RBAC helps simplify administration while protecting data by ensuring that users access only the functions relevant to their role. This is highly relevant for Arnis Connect because attendance, member records, events, and funds are not meant to be managed by all users at the same level.

Overall, the related studies show that digital transformation in sports organizations, centralized membership systems, web-based administrative platforms, attendance and event tracking, financial monitoring, and role-based control all support the need for a system like Arnis Connect. What distinguishes Arnis Connect is that it combines these ideas in a way that is specific to the needs of an Arnis organization, including public and private player arrangements, registration approval, and smart player suggestion.

Related Systems

Several existing systems provide features that are related to the proposed Arnis Connect system. These platforms focus on sports team coordination, scheduling, attendance, communication, member handling, and in some cases, financial tools. However, while they provide valuable reference points, they are not fully tailored to the specific needs of an Arnis organization.

TeamSnap

TeamSnap is a team management platform that integrates communication, rosters, schedules, member availability, assignments, lineups, statistics, and financial management into a unified system. It is widely used in sports organizations to streamline coordination among team members, coaches, and administrators by providing a centralized interface for managing team activities and information.

TeamSnap is similar to Arnis Connect in terms of combining scheduling, attendance visibility, and communication features within a single platform. However, it differs in scope because it is designed for general sports team management, particularly youth and recreational teams. In contrast, Arnis Connect focuses on organization-specific workflows such as formal registration approval, structured member management, event documentation, and role-based financial recording tailored to the needs of the Lapu-Lapu Arnis de Abanico Organization.

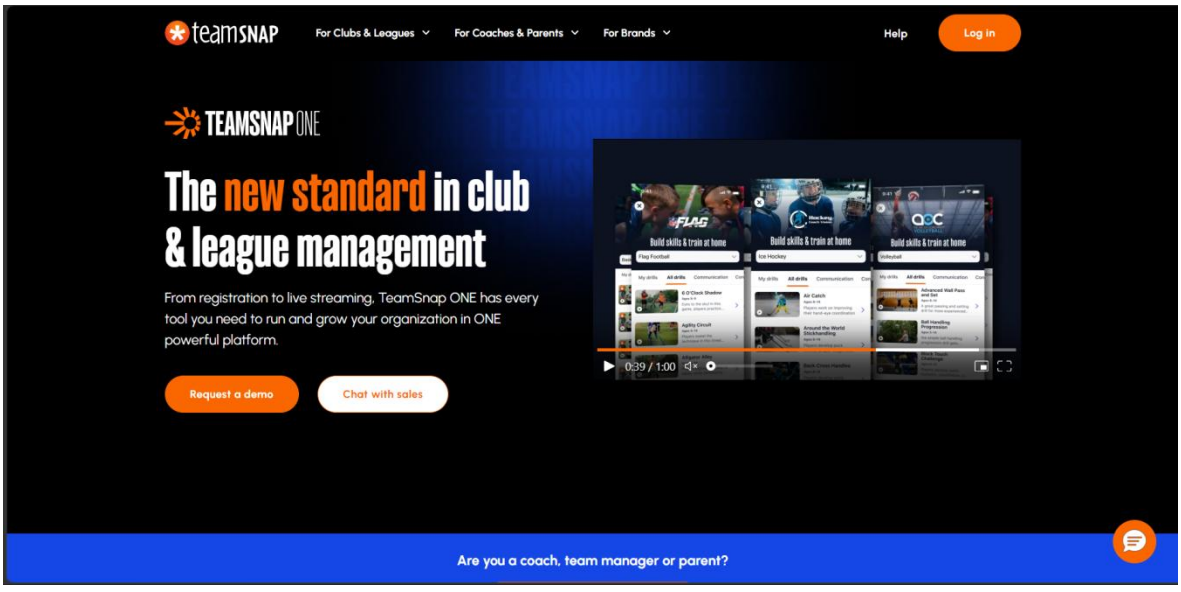


Figure 4
Homepage of TeamSnap

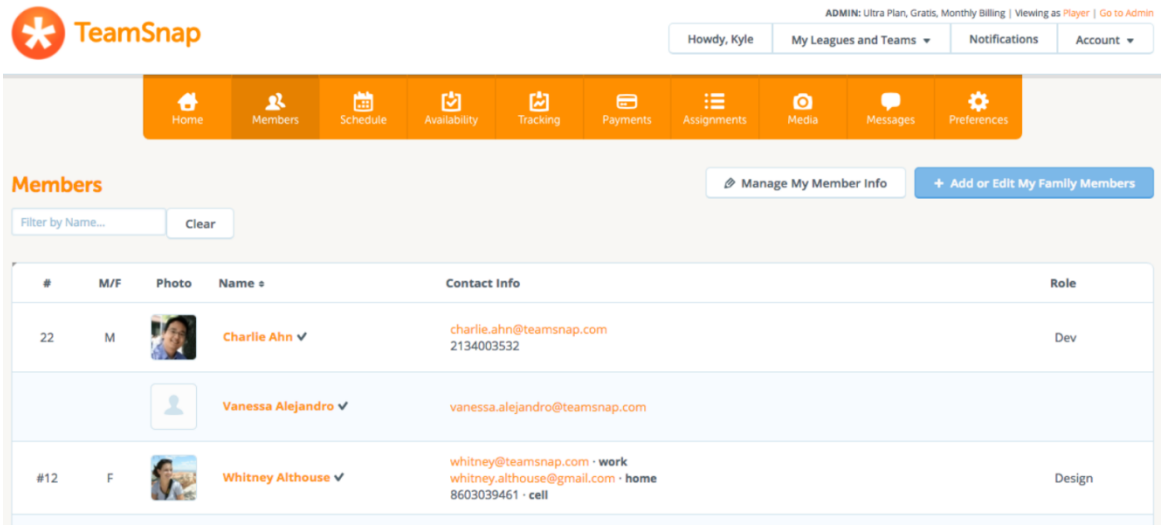


Figure 5
TeamSnap Features/ Dashboard Sample

TeamLinkt

TeamLinkt is an all-in-one sports management platform that supports registration, scheduling, website building, tournaments, communications, scorekeeping, and team app functions. It also advertises attendance tracking and organization-wide reporting features. This makes it a strong related system because it demonstrates a broader sports management approach that combines operations, communication, and reporting.

Nevertheless, TeamLinkt is primarily designed for leagues, clubs, associations, and tournaments at a broad organizational level, while Arnis Connect is more specifically designed for the operational workflow of the Lapu-Lapu Arnis de Abanico Organization.

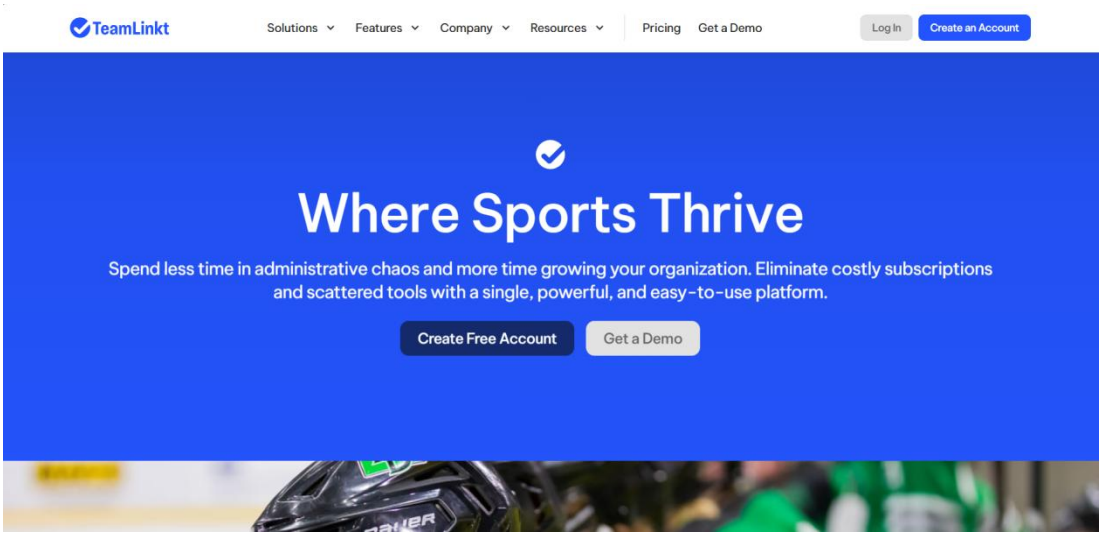


Figure 6
Homepage of TeamLinkt

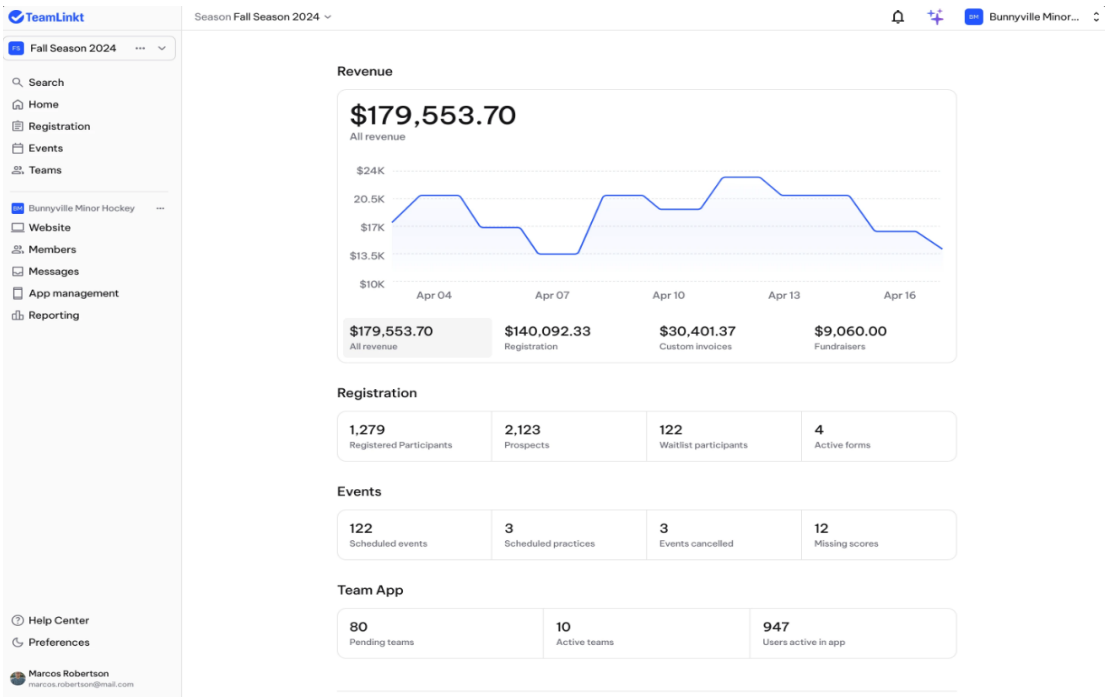


Figure 7
TeamLinkt App/ Dashboard Sample

Teamup

Teamup is a customizable shared calendar platform that enables users to organize schedules, manage resources, and coordinate team activities across multiple devices. It is commonly used for scheduling events and ensuring that all participants have access to updated timelines and activity plans.

Teamup is similar to Arnis Connect in its ability to manage and share schedules among users. However, it differs significantly in functionality because it is primarily a scheduling tool rather than a full management system. Unlike Arnis Connect, Teamup does not provide integrated features such as member profile management, attendance tracking, financial recording, or registration approval processes, making Arnis Connect a more comprehensive solution.

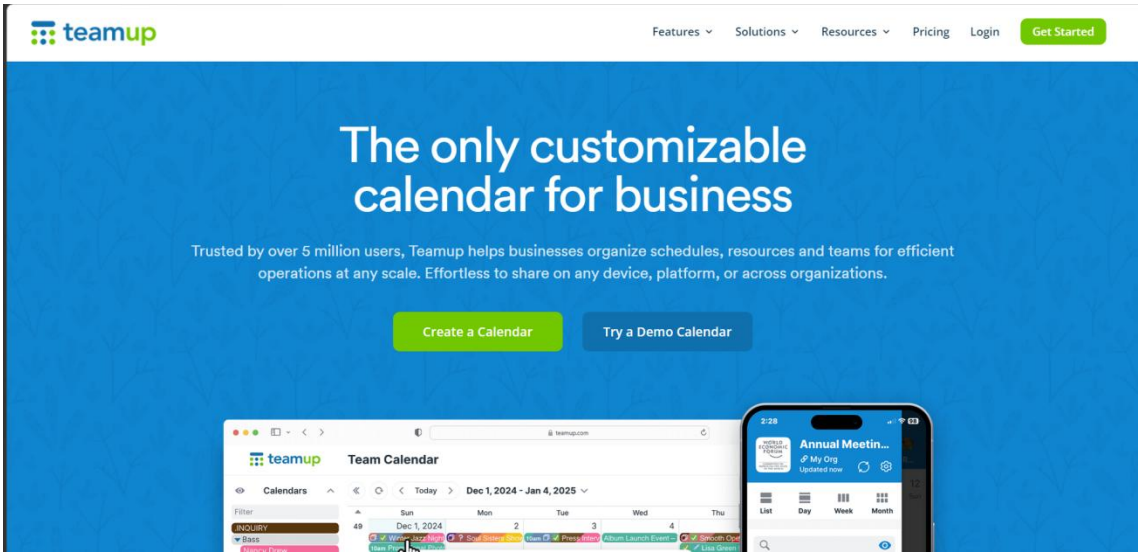


Figure 8
Homepage of TeamUp

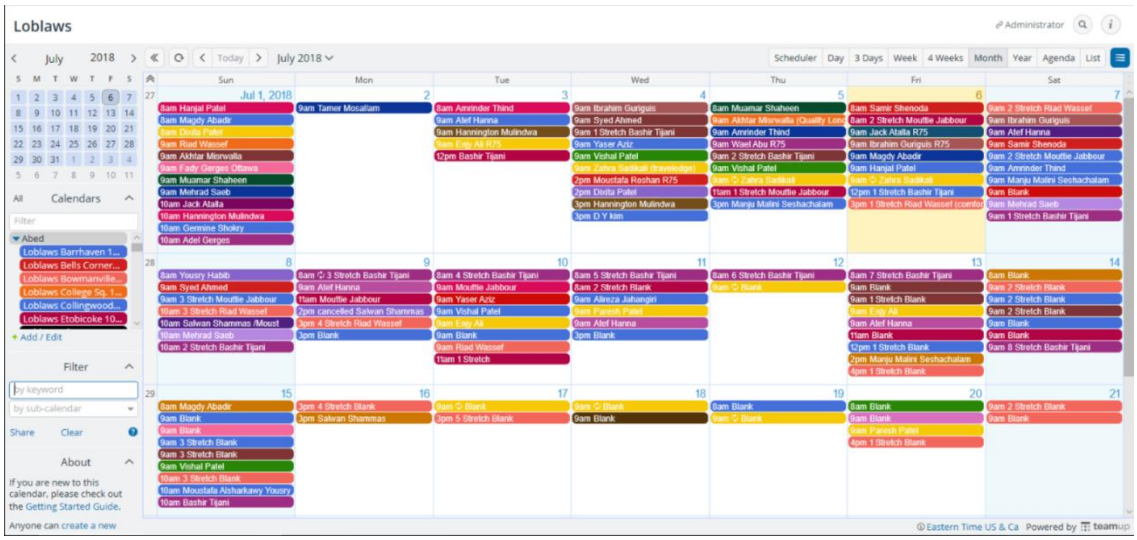


Figure 9
TeamUp Calendar / Schedule Sample

Spond

Spond is a sports team and club management application that provides tools for event planning, attendance tracking, reminders, communication, and payment requests. It is designed to simplify coordination within teams and clubs by offering an easy-to-use platform for managing group activities and interactions.

Spond is similar to Arnis Connect in terms of integrating event scheduling, attendance tracking, communication, and financial-related features within one system. However, Spond primarily focuses on group coordination and communication, whereas Arnis Connect extends beyond this by incorporating administrative reporting, structured member records, formal registration workflows, and organization-specific processes. This makes Arnis Connect more suitable for managing the operational and administrative

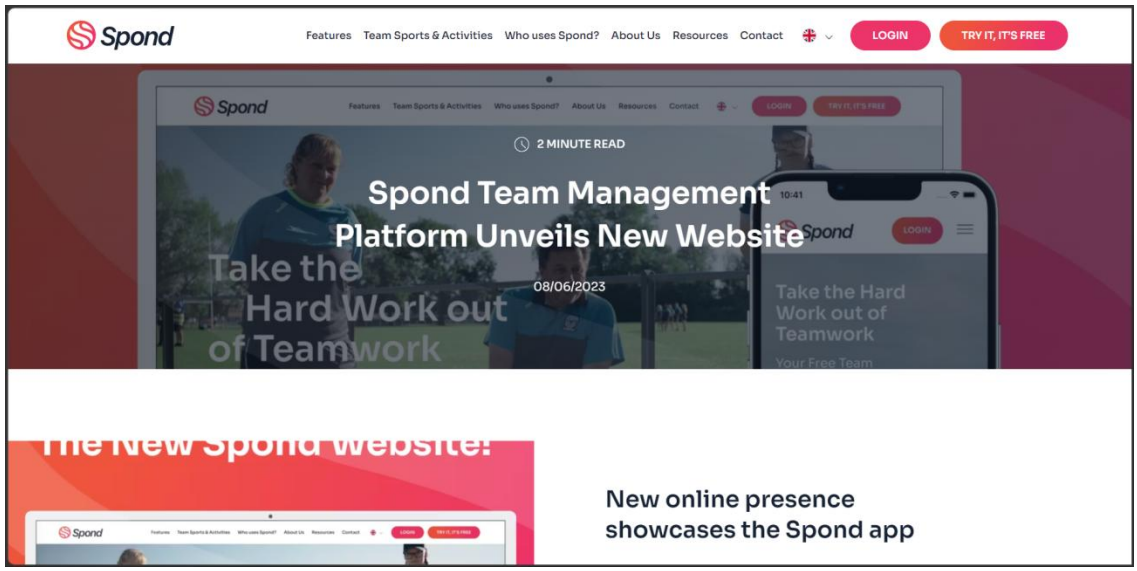


Figure 10
Homepage of Spond

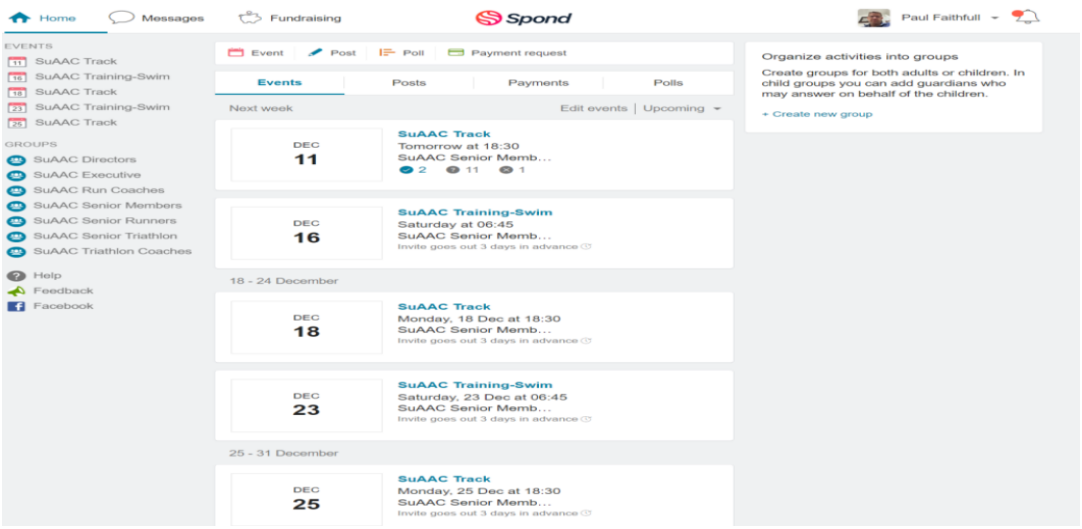


Figure 11
Spond App / Attendance Sample

Comparative Matrix

The comparative matrix below presents how Arnis Connect compares with selected related systems based on features that are important to the proposed study.

Table 6
Comparative Matrix

Features	TeamSnap	TeamLinkt	Teamup	Spond	Arnis Connect
Member profile management	✓	✓	✗	✓	✓
Attendance monitoring	✓	✓	Limited	✓	✓
Event scheduling / planning	✓	✓	✓	✓	✓
Communication / notifications	✓	✓	✓	✓	✓
Financial tools / payments	✓	✓	✗	✓	✓
Role-based access	Limited	✓	Limited	Limited	✓
Registration approval workflow	✗	Partial	✗	✗	✓
Event documentation with participants	Limited	Partial	✗	Limited	✓
Reports generation	Partial	✓	Limited	Limited	✓
Player suggestion feature	✗	✗	✗	✗	✓
Total	62.5%	75%	35%	65%	100%

The matrix shows that the selected existing systems provide useful combinations of sports management features, especially in scheduling, attendance, communication, and payments. However, Arnis Connect differs because it is designed to integrate features that are specifically needed by the Arnis organization, including a registration approval workflow, event documentation linked to members, structured reporting, and a player suggestion feature. These combinations make the proposed system more specialized and more aligned with the actual organizational context

described in the project documents.

Competitor Analysis

Competitor analysis examines how existing systems respond to the same general management needs addressed by Arnis Connect. This analysis helps identify which existing features are already available in the market and which gaps remain unaddressed.

Table 7
Competitor Analysis

Related System	Description	Features / Strengths
TeamSnap	A sports team management app used by coaches, parents, and clubs to coordinate team activities.	Strong in schedules, member availability, messaging, assignments, and money management. (TeamSnap)
TeamLinkt	An all-in-one sports management platform for leagues, clubs, tournaments, and associations.	Strong in registration, scheduling, communication, scorekeeping, reporting, and attendance-related app tools. (TeamLinkt)
Teamup	A customizable shared calendar platform used to organize schedules, teams, and resources.	Strong in scheduling, sharing, and calendar visibility across devices. (Teamup.com)
Spond	A sports team and club management app focused on events, attendance, reminders, and payments.	Strong in event planning, attendance tracking, reminders, communication, and payment requests. (Spond)
Arnis Connect	A proposed integrated web and mobile-based management system for the Arnis sports program.	Strong in member management, attendance, event documentation, fund recording, reports, registration approval, and player suggestion specific to Arnis organizational workflow.

The competitor analysis shows that while existing platforms already support general sports team management, none of them are designed specifically for the administrative and organizational realities of an Arnis program. This supports the relevance of Arnis Connect as a specialized management information system rather than a simple imitation of existing apps.

TECHNICAL BACKGROUND

This section presents the technical foundation underlying the development of Arnis Connect. It discusses the core technologies utilized in the development of the system, the theoretical concepts that guide its structure and functionality, and the related technologies integrated into its overall development environment. The section also explains the programming languages, development tools, frameworks, libraries, database technologies, and analytical components used to support the implementation of the system's features, including member management, attendance monitoring, event documentation, financial recording, reporting, and player suggestion analytics. Furthermore, it describes how these technologies collectively contribute to the development of a centralized, role-based, and web-based management platform for the organization.

Introduction to the Technology

Arnis Connect is designed as a web-based management information system, meaning it is accessed through web browsers instead of requiring a separately installed desktop application. This approach is practical because it allows multiple users to access the system through desktops, laptops, tablets, and smartphones using the same centralized platform. The Arnis Connect project documents consistently describe the system as web- and mobile-accessible, with PHP, MySQL, HTML, CSS, JavaScript, and Apache as the planned technology stack.

The software design documentation further explains that Arnis Connect follows a client-server architecture with three layers: the presentation layer, which handles the user interface; the application layer, which processes the system's business logic; and the data layer, which stores and retrieves records using a relational database. It also identifies the major subsystems of the project as Member Management, Attendance Management, Event Management, Financial Management, and User Authentication. This architecture was chosen because centralized storage and processing make it easier to maintain data consistency, support multiple users, and manage updates and backups.

Theoretical Concepts

One important concept underlying Arnis Connect is the Management Information System (MIS) perspective, which emphasizes the collection, processing, storage, and presentation of data to support organizational operations and decision-making. This is relevant to Arnis Connect because the system is not limited to data storage; it is intended to provide organized records, usable reports, and decision-support features such as player suggestion and participation monitoring. In information systems literature, the DeLone and McLean model explains that the success of an information system is closely related to its system quality, information quality, service quality, use, user satisfaction, and net benefits. This is relevant to Arnis Connect because the system's value depends on whether it is accurate, usable, reliable, and beneficial to the organization.

A second important concept is database management, particularly the use of a relational database. According to the MySQL reference manual, MySQL is an open-source SQL database management system in which data is stored in separate but related tables. This structure is useful for Arnis Connect because the project involves multiple related data entities such as users, members, trainings, attendance, events, and funds, all of which must be linked and managed systematically.

A third important concept is Role-Based Access Control (RBAC). The NIST RBAC model explains that permissions are assigned through roles, which simplifies administration and makes access control more consistent. This is directly relevant to Arnis Connect because the system is designed for different roles such as administrator or lead coach, coach, and finance officer, each with different access needs and responsibilities. Role-based access ensures that users interact only with the modules and records appropriate to their role.

Related Technologies

HTML (HyperText Markup Language) is the standard markup language used to structure content on web pages. In Arnis Connect, HTML is responsible for organizing the content of pages such as forms, dashboards, tables, and module views. CSS (Cascading Style Sheets) is the styling language used to control layout, colors, spacing, and the visual presentation of those pages. JavaScript adds interactivity to the system, such as dynamic page behavior, search, filtering, validation, and responsive interface actions. The MDN Web Docs identify HTML, CSS, and JavaScript as the core technologies of the web, which makes them appropriate foundational tools for a browser-based system like Arnis Connect.

PHP is a widely used server-side scripting language suited for web development. In Arnis Connect, PHP handles server-side processing such as login authentication, form submission handling, record creation and updating, and communication with the database. It is therefore central to the system's backend logic and dynamic page generation.

MySQL is the relational database management system used to store, organize, and retrieve the system's records. It supports structured tables and relationships, which are essential for storing interconnected records such as members, attendance, events, and funds. Because Arnis Connect is a data-driven system, MySQL functions as the core persistence layer of the application.

Apache HTTP Server is the web server that handles HTTP requests and serves the web application to the client browser. The Apache project describes the server as a robust, freely available, and feature-rich implementation of an HTTP web server. In typical PHP-MySQL application stacks such as XAMPP, Apache serves as the runtime environment that connects browser requests to PHP processing and database operations.

Together, these technologies form a practical and commonly used web stack for small and medium-sized organizational systems. Their combination supports accessible interfaces, interactive behavior, centralized data management, backend processing, and browser-based deployment, all of which are necessary for the implementation of Arnis Connect.

CHAPTER 3

METHODOLOGY

This chapter presents the research methodology used in the development of Arnis Connect. It discusses the research design, system development methodology, data gathering procedures, system architecture, software and hardware requirements, database design, testing procedures, and evaluation methods applied throughout the study. The chapter also explains the processes involved in analyzing the existing organizational workflow, identifying system requirements, designing the proposed solution, and implementing the web-based management system.

The study utilized a descriptive-developmental research approach to examine the current administrative practices of the organization and develop a system that addresses the identified operational problems. Agile methodology was adopted to support iterative system development, continuous feedback, and incremental improvement of system features. Through this methodology, the researchers were able to design, develop, test, and refine the system based on the needs of administrators, coaches, finance users, and members.

Furthermore, this chapter describes the technologies, programming languages, frameworks, and analytical features integrated into the system, including attendance monitoring, financial management, event documentation, player suggestion analytics, and role-based access control. The chapter concludes with the procedures used to evaluate the functionality, usability, reliability, and overall acceptability of the developed system.

PLANNING PHASE

This phase establishes the foundation of the study by defining the system requirements, identifying the key stakeholders, and outlining the overall design approach for the Arnis Connect system. It focuses on analyzing the current challenges faced by the Lapu-Lapu Arnis de Abanico Organization, particularly the use of manual and scattered record-keeping in managing members, attendance, events, and financial transactions, which often leads to inefficiencies, data inconsistency, and delayed reporting. During this phase, the needs of the organization’s lead coach, administrators, coaches, finance-related users, and prospective members are identified and translated into structured system components and functional modules. These include login and session management, member management, registration approval, attendance tracking, event documentation, financial monitoring, and reports generation. By organizing these requirements and

aligning them with the objectives of the study, this phase ensures a clear and coherent direction for system development and guides the succeeding phases toward a centralized, efficient, and user-oriented solution.

Feasibility Study Model

This section presents the feasibility analysis of Arnis Connect, an integrated web-based management system designed to support the administrative operations of an Arnis sports program. The purpose of this analysis is to evaluate whether the proposed system is practical, achievable, and beneficial for implementation. It examines key aspects including technical, economic, operational, schedule, and legal feasibility to determine the overall viability of the system.

Through this evaluation, the study aims to ensure that the required resources, technologies, and processes are available and suitable for the successful development and deployment of Arnis Connect. The results of this analysis provide a structured basis for decision-making and confirm whether the system can effectively address the identified problems within the organization.

Technical Feasibility

Technical feasibility determines whether the development team possesses the necessary tools, technologies, frameworks, and technical skills to successfully develop and deploy Arnis Connect within the defined project scope. The proposed system is designed as a web-based management platform developed using PHP, MySQL, HTML, CSS, and JavaScript, operating through a centralized role-based environment accessible through modern web browsers. The system includes modules for member management, attendance monitoring, event management, financial recording, reporting, player suggestion analytics, geolocation-based attendance validation, and facial recognition-based attendance verification.

The development environment utilizes XAMPP as the local server platform, phpMyAdmin for database administration, and Visual Studio Code as the primary development environment. Frontend responsiveness and interface consistency are supported through Bootstrap, while JavaScript libraries such as DataTables, SweetAlert, and Chart.js are integrated to improve interactivity, data presentation, and reporting visualization. The system may also integrate Python-based analytics and facial recognition modules to support attendance analysis, player suggestion processing, attendance verification, and organizational insights generation.

Table 8
Technical Feasibility Assessment

Component	Assessment	Rating
Web-Based System Platform (PHP, HTML, CSS, JavaScript)	PHP supports backend processing, authentication, role-based access control, attendance management, event management, and financial recording. HTML, CSS, and JavaScript provide responsive and interactive user interfaces accessible through standard web browsers.	Feasible
Database Management (MySQL)	MySQL provides centralized storage for users, members, attendance, events, achievements, financial records, reports, and attendance verification records. phpMyAdmin simplifies database administration and management during development and testing.	Feasible
Local Development Environment (XAMPP)	XAMPP provides Apache and MySQL services required for local deployment, testing, and debugging of the web-based system before final deployment.	Feasible
Development Tools (Visual Studio Code)	Visual Studio Code supports multi-language web development with extensions for PHP, JavaScript, SQL, and frontend technologies, improving coding efficiency and debugging capabilities.	Feasible
Frontend Framework (Bootstrap)	Bootstrap enables responsive layouts and consistent interface design across desktop and mobile browsers without requiring separate application versions.	Feasible
Data Presentation and Reporting (Chart.js, DataTables)	Chart.js supports graphical reports and dashboard visualization, while DataTables improves table searching, sorting, filtering, and pagination for large organizational records.	Feasible
Notification and User Interaction (SweetAlert)	SweetAlert enhances user interaction through modern confirmation dialogs, alerts, and validation prompts, improving overall usability and user experience.	Feasible
Geolocation Attendance Validation	Geolocation services may be integrated through browser-based GPS APIs to validate whether members are within authorized training or event locations during attendance recording. This improves attendance authenticity and monitoring accuracy.	Feasible
Facial Recognition Attendance Verification	Facial recognition modules using Python and OpenCV may be integrated to automate attendance verification through member facial identification. This reduces manual attendance errors and strengthens attendance	Feasible with Constraints

	validation, although it introduces moderate implementation complexity and training requirements.	
Player Suggestion and Analytics Module (Python Analytics Integration)	Python analytics modules may be integrated to analyze attendance, participation history, achievements, availability, and event records for player suggestion and organizational insights. This introduces additional processing complexity but remains achievable within the project scope.	Feasible with Constraints
Web Browser Compatibility	The system is compatible with commonly used browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox without requiring additional software installation.	Feasible

The primary technical considerations of the proposed system involve the integration of analytics processing, geolocation validation, and facial recognition technologies alongside the PHP-based web platform. Facial recognition functionality requires image processing libraries, camera integration, and proper facial dataset preparation, while analytics processing introduces additional computational and integration requirements. Geolocation features also depend on device GPS permissions and internet connectivity for accurate coordinate validation. Despite these considerations, the selected technologies, frameworks, and development tools are widely supported, compatible with one another, and manageable within the technical capabilities of the development team. Overall, the technical feasibility of the proposed system is assessed as high.

Economic Feasibility

Economic feasibility evaluates whether the expected benefits of Arnis Connect justify the estimated development and operational costs. The proposed system primarily utilizes open-source and freely available technologies such as PHP, MySQL, XAMPP, Bootstrap, JavaScript libraries, and Python-based analytical tools, minimizing software acquisition expenses. The development process mainly requires existing computer hardware, internet connectivity, and development tools already accessible to the researchers.

The implementation of the system is expected to reduce administrative inefficiencies caused by manual record keeping, paper-based attendance monitoring, fragmented spreadsheets, and repetitive reporting tasks. By centralizing records and automating administrative processes, the system may reduce operational delays, improve reporting accuracy, and decrease the time required for attendance monitoring, event management, and financial tracking.

Table 9
Economic Feasibility Assessment

Component	Assessment	Rating
Development Software and Tools	Most technologies and development tools used are open-source or freely available, reducing software acquisition costs.	Feasible
Hardware Requirements	The system can operate using existing computers, laptops, webcams, and internet-enabled devices available to the organization.	Feasible
Maintenance Costs	System maintenance mainly involves database backup, software updates, and minor technical support, resulting in manageable operational costs.	Feasible
Administrative Efficiency	The system reduces manual paperwork, duplicate records, and repetitive administrative tasks, improving productivity and reducing operational workload.	Feasible
Reporting and Record Management	Automated report generation and centralized records reduce time spent on manual computations and data retrieval.	Feasible
Analytics and Attendance Validation Features	Additional implementation costs for facial recognition and analytics integration may increase development complexity but remain manageable within the project scope.	Feasible with Constraints

The proposed system is considered economically feasible because it relies primarily on low-cost technologies while providing long-term operational benefits through improved efficiency, centralized information management, and reduced administrative workload.

Operational Feasibility

Operational feasibility determines whether the proposed system can function effectively within the operational environment of the organization and whether intended users can adapt to the system. Arnis Connect is designed to support the actual workflow of administrators, coaches, finance officers, members, and applicants through role-based dashboards and centralized management functions.

The system simplifies attendance monitoring, event coordination, member management, registration approval, and financial recording through an organized web-based interface. By

replacing fragmented records and manual tracking methods, the system improves operational coordination and accessibility of organizational data.

Table 10
Operational Feasibility Assessment

Component	Assessment	Rating
Role-Based Access Control	Different user roles ensure that users only access functions relevant to their responsibilities, improving security and operational organization.	Feasible
User Interface and Accessibility	The web-based interface is designed to be understandable and accessible through common web browsers.	Feasible
Attendance and Event Management	Automated attendance recording and centralized event monitoring improve operational coordination and record accuracy.	Feasible
Financial Recording Operations	Finance users can manage transactions and summaries more efficiently compared to manual spreadsheet recording.	Feasible
Member and Registration Management	Centralized member records and applicant approval workflows improve monitoring and reduce administrative confusion.	Feasible
User Adaptability	Users may require initial orientation and familiarization with the system, particularly for analytics and facial recognition features.	Feasible with Constraints

The proposed system is operationally feasible because its functions align with the existing workflow and administrative needs of the organization while improving efficiency, accessibility, and coordination.

Schedule Feasibility

Schedule feasibility evaluates whether the development and implementation of the proposed system can be completed within the available academic timeline and project constraints. The development of Arnis Connect follows an Agile-based methodology, allowing iterative development, continuous testing, and incremental feature implementation.

Core modules such as login authentication, member management, attendance recording, event management, and financial monitoring are prioritized during early development phases, while advanced features such as analytics processing, geolocation validation, and facial recognition may be implemented progressively.

Table 11
Schedule Feasibility Assessment

Component	Assessment	Rating
Requirements Gathering and Planning	System requirements and organizational workflows are clearly identified through interviews, observations, and documentation review.	Feasible
Core System Development	Fundamental system modules can be developed within the academic development period using available technologies and tools.	Feasible
Testing and Debugging	Incremental testing and Agile iteration support manageable debugging and refinement processes.	Feasible
Analytics and Facial Recognition Integration	Advanced features may require additional development and testing time due to integration complexity.	Feasible with Constraints
Deployment and User Orientation	The web-based deployment process is relatively simple and can be conducted within the project timeline.	Feasible

Although advanced integrations introduce moderate development complexity, the project timeline remains achievable through phased implementation and prioritized feature development. Overall, the proposed system is considered schedule feasible.

Legal Feasibility

Legal feasibility evaluates whether the proposed system complies with applicable legal, ethical, and data privacy considerations. Arnis Connect handles organizational records including member information, attendance logs, financial records, and facial recognition data; therefore, proper data protection and controlled system access are necessary.

The system implements authentication, role-based access control, and secured record management procedures to minimize unauthorized access and misuse of information. Personal information collected from members and applicants is intended solely for organizational and administrative purposes.

Table 12
Legal Feasibility Assessment

Component	Assessment	Rating
Data Privacy and Protection	User and member information is protected through authentication and restricted system access.	Feasible
Role-Based Access Control	System permissions ensure that sensitive records are accessible only to authorized users.	Feasible
Organizational Record Usage	Collected information is used only for attendance, membership, event, and financial management purposes.	Feasible
Facial Recognition Data Handling	Facial recognition implementation requires responsible handling of image data and user consent procedures.	Feasible with Constraints
Software Licensing	Most development tools, frameworks, and libraries used are open-source and legally accessible for academic development purposes.	Feasible

The proposed system is legally feasible provided that user information, attendance data, financial records, and facial recognition images are handled responsibly and used strictly for authorized organizational operations.

System Context Diagram

The system context diagram presents the high-level view of Arnis Connect and its interaction with the external entities involved in the management of the Arnis sports program. It defines the system boundary and shows how data flows between the proposed system and the main stakeholders. In Arnis Connect, the major external entities are the Administrator or Lead Coach, Coach, Finance Officer or finance-related user, and Applicant or Prospective Member. The central system receives data such as registration requests, member information, attendance entries, event details, and financial transactions, then processes and stores them in a centralized platform for viewing, updating, and reporting.

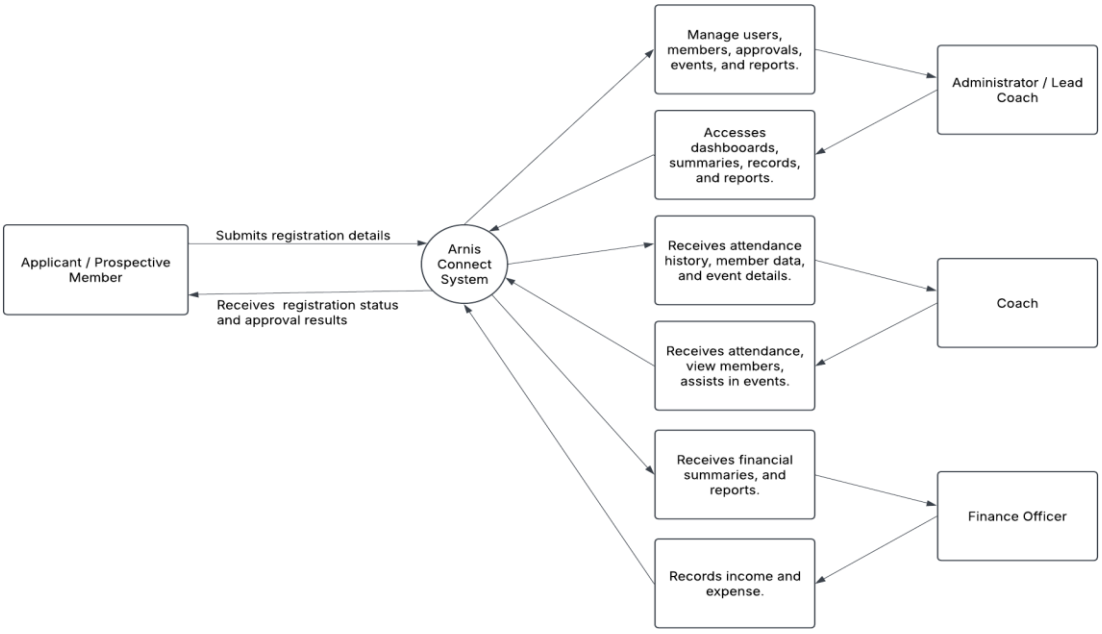


Figure 12
System Context Diagram (Level 0 DFD)

FIgure 12 illustrates Arnis Connect as the central process that facilitates the exchange of information among the external entities of the organization. The administrator or lead coach manages members, approvals, users, reports, and event records. Coaches update attendance and view training or event-related information. Finance-related users record income and expenses and retrieve financial summaries. Applicants submit registration data for review and approval. Through this context diagram, the system is shown as a centralized platform that improves coordination, accessibility of records, and efficiency in performing administrative tasks.

User Flow Diagrams

This section presents the workflow of Arnis Connect, which illustrates the step-by-step processes and decision flows experienced by the primary user types within the system. The workflow diagrams serve as a detailed representation of how each user interacts with the platform from initial entry through the completion of their respective tasks and goals. These diagrams complement the System Context Diagram by moving beyond system boundaries to depict the internal logic, conditional flows, decision points, and sequential actions that govern user interaction with Arnis Connect (Sommerville, 2016; Pressman & Maxim, 2019).

Five separate workflow diagrams are presented to accurately reflect the distinct roles, access levels, and interaction patterns of the following users: Applicant or Public Player, Member or Player, Administrator or Lead Coach, Coach, and Finance Manager or Finance Officer.

Applicant / Public Player

The applicant workflow begins when a prospective public or private player accesses Arnis Connect and selects the member registration option. The applicant creates an account by entering a username, password, email or Gmail address, and mobile number. The system then sends verification codes to validate the email and mobile number before the applicant can proceed. After successful verification, the applicant is redirected to the login page and can access the Member Application Form. The submitted application is stored as pending until reviewed by the Administrator or Lead Coach. If approved, the applicant is stored as an official member in the database and can access the member dashboard. If rejected, the applicant may view the application status or revise the submitted information depending on the organization's decision.

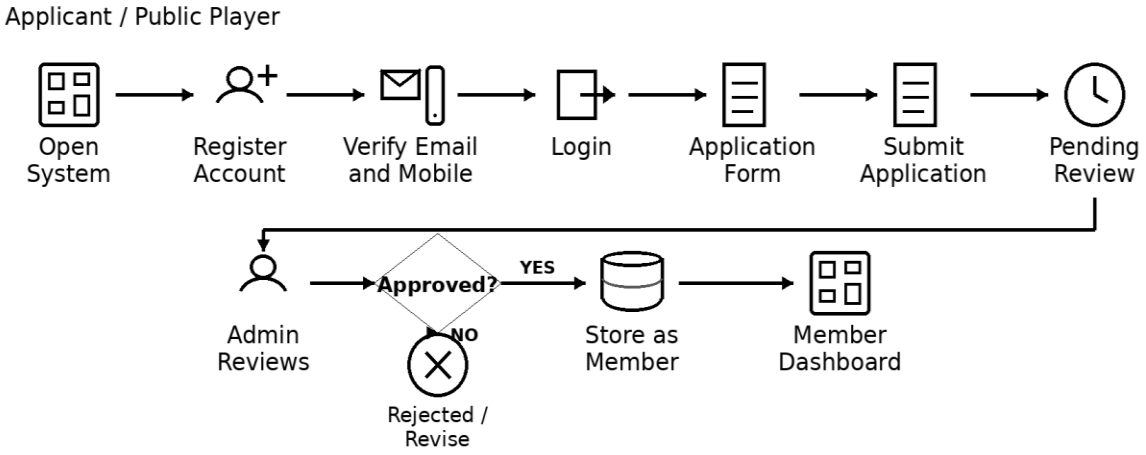


Figure 13
User Flow Diagram - Applicant / Public Player

Member / Player

The member workflow begins after the applicant has been approved and stored as an official member. The member logs in to Arnis Connect and accesses the member dashboard, where personal application status, upcoming trainings, attendance history, event participation, and belt progression can be viewed. For automated attendance, the member may enroll a face profile and use the Take Attendance feature during scheduled training sessions. The system checks facial recognition and geolocation to verify whether the member is physically present at the training location. Valid attendance is recorded in the database, while incomplete or suspicious attendance is flagged for coach review. The member can also monitor belt progress through a locked and unlocked progression display.

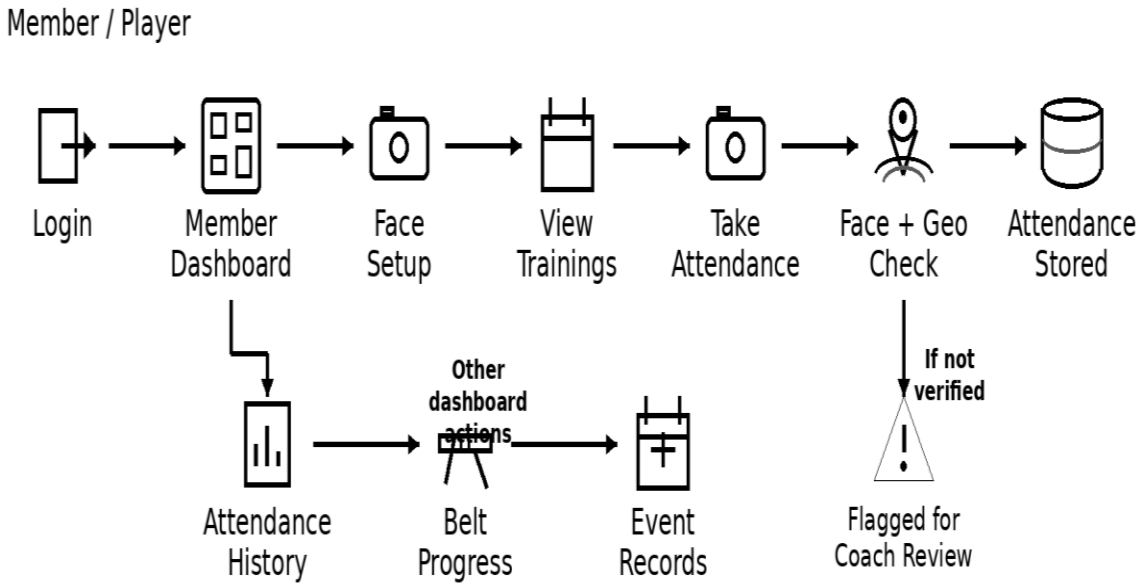


Figure 14
User Flow Diagram - Member / Player

Administrator / Lead Coach

The Administrator or Lead Coach workflow begins with secure login to the role-based dashboard. From the dashboard, the Administrator or Lead Coach reviews pending applications and approves or rejects applicant records. Approved applications are converted into official member records and stored in the database. The Administrator or Lead Coach can also manage user accounts, assign system roles, monitor member records, oversee trainings and events, and generate reports and audit logs. This workflow ensures that major administrative actions remain controlled by authorized users and that important organizational data are properly monitored.

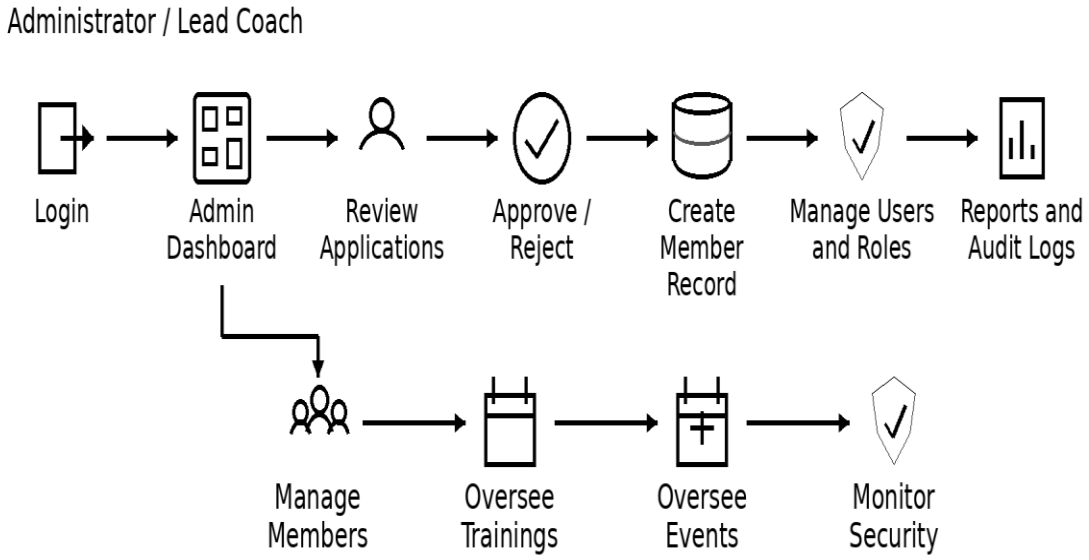


Figure 15
User Flow Diagram - Administrator / Lead Coach

Coach

The coach workflow begins with login to the coach dashboard. The Coach can schedule training sessions, set the training location, and define the geofence radius used for attendance verification. During training, members submit attendance through facial recognition and geolocation, while the Coach monitors recorded and flagged attendance entries. The Coach can review suspicious records, validate attendance when needed, update belt progression after assessment, create or manage events, generate player suggestions, and select final participants. The player suggestion feature supports the Coach by considering attendance, belt level, player category, event type, achievements, availability, and past participation, but the final decision remains under the Coach.

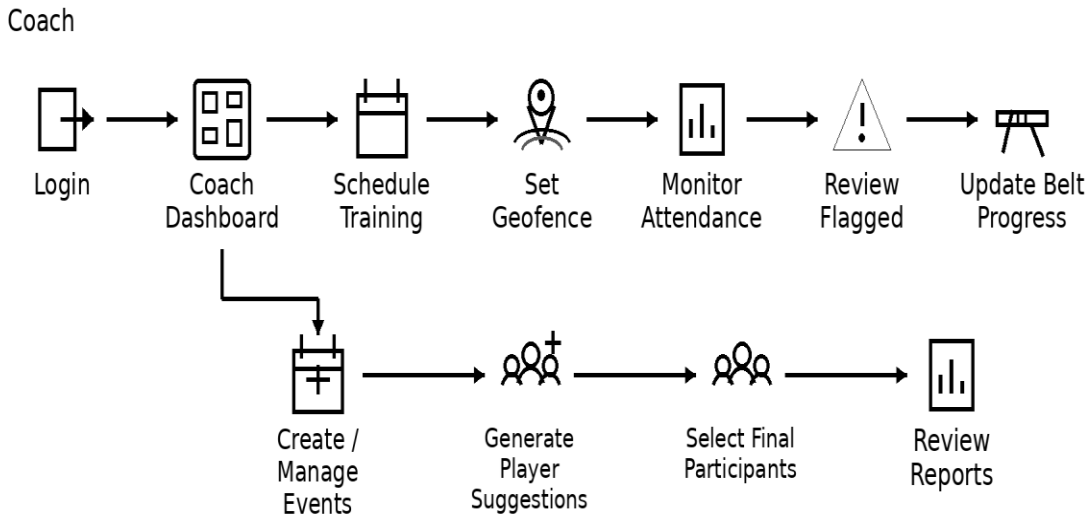


Figure 16
User Flow Diagram - Coach

Finance Manager / Finance Officer

The finance workflow begins when the Finance Manager or Finance Officer logs in to the finance dashboard. The user records income and expense transactions, categorizes financial data, links transactions to events or training-related activities when applicable, and updates the fund balance of the organization. The Finance Manager or Finance Officer may also view transaction history, check event-related financial records, generate financial reports, and prepare summaries for monitoring. All finance-related actions are stored with audit trails to support transparency and accountability within the organization.

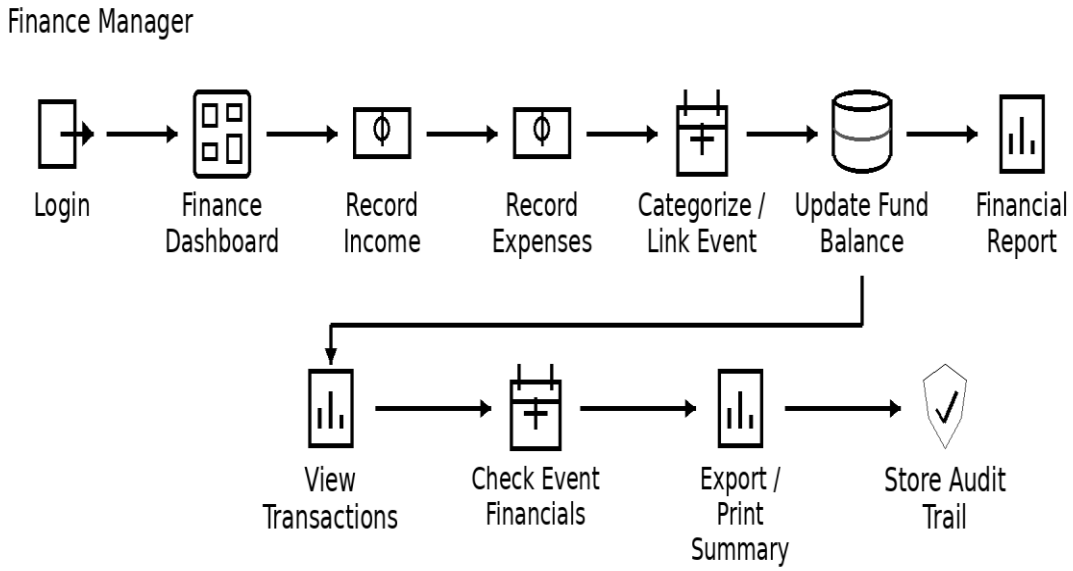


Figure 17
User Flow Diagram - Finance Officer

Functional Decomposition Diagram

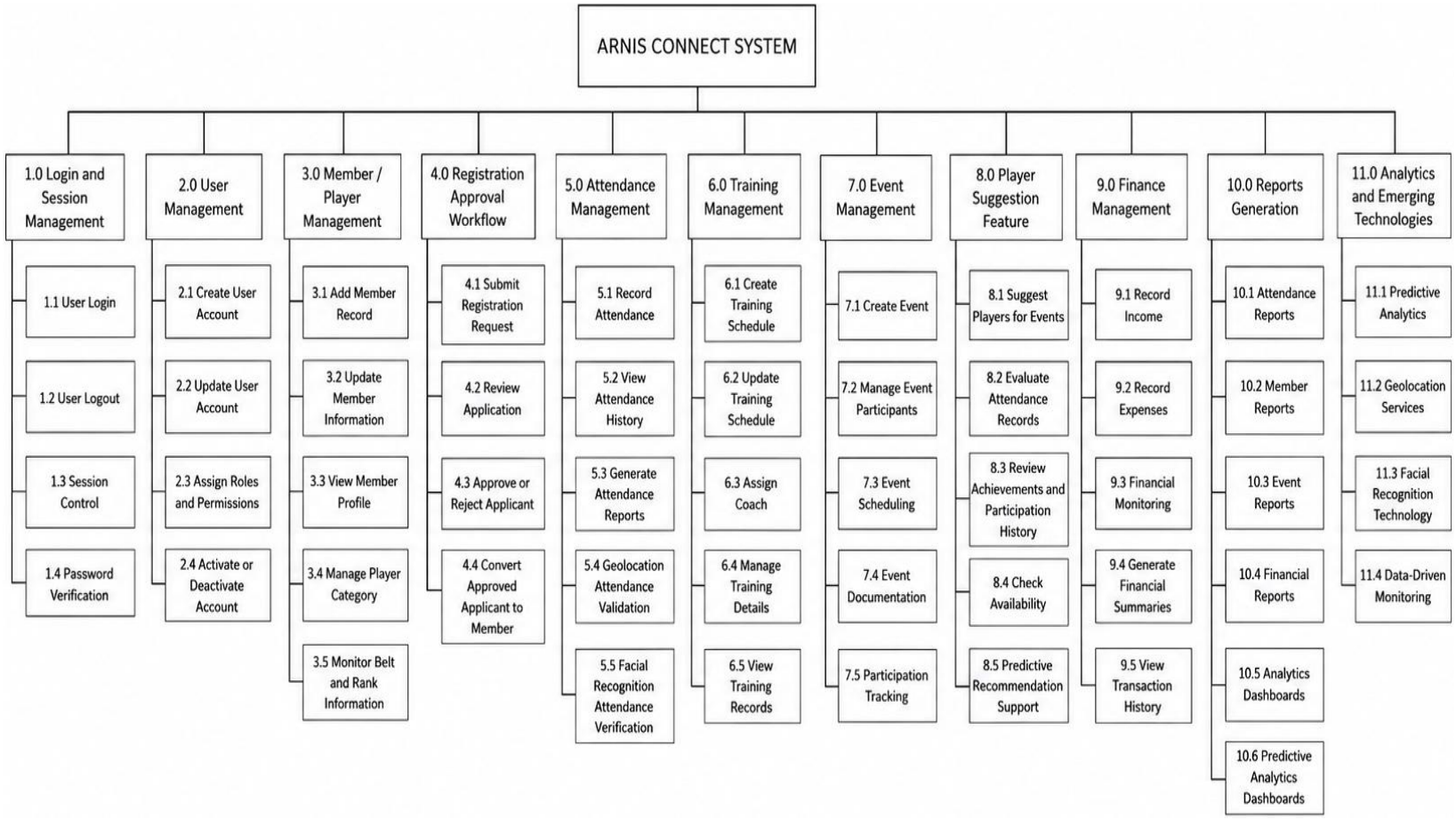


Figure 18
Functional Decomposition Diagram

The Functional Decomposition Diagram (FDD) of Arnis Connect presents the hierarchical breakdown of the system into its major functional modules and related sub-functions. This diagram shows how the proposed system is divided into smaller and more manageable components, making it easier to understand how each function supports the overall operation of the Arnis organization. Functional decomposition is useful in system development because it simplifies complex processes by organizing them into clear functional areas that can be designed, developed, tested, and maintained separately (Pressman & Maxim, 2019; Sommerville, 2016).

As shown in Figure 18, Arnis Connect is composed of the following major functional modules:

1.0 Login and Session Management – This module handles user authentication, login, logout, session control, and password verification. It ensures that only authorized users can access the system and that every user session is properly managed.

2.0 User Management – This module allows the administrator to create and update user accounts, assign roles, manage permissions, and activate or deactivate accounts. It supports role-based access for administrators, coaches, finance users, and members.

3.0 Member / Player Management – This module manages member and player records, including adding member records, updating member information, viewing member profiles, managing player categories, and monitoring belt and rank information. It supports the centralized storage of public and private player records.

4.0 Registration Approval Workflow – This module handles registration requests submitted by applicants. It includes submission of registration requests, review of applications, approval or rejection of applicants, and conversion of approved applicants into official members of the organization.

5.0 Attendance Management – This module manages the recording and monitoring of training attendance. It includes attendance recording, attendance history, attendance reports, geolocation attendance validation, and facial recognition attendance verification. This supports a more reliable attendance process by checking both the identity and location of the member during training.

6.0 Training Management – This module handles the creation and updating of training schedules. It also includes assigning coaches, managing training details, and viewing training records. This supports organized planning and monitoring of training sessions.

7.0 Event Management – This module manages organizational events and activities. It includes creating events, managing event participants, event scheduling, event documentation, and participation tracking. This helps the organization maintain proper records of performances, competitions, demonstrations, and other Arnis-related events.

8.0 Player Suggestion Feature – This module supports event preparation by providing player suggestions based on attendance, achievements, participation history, availability, and other relevant records. It also includes predictive player recommendation support, but the final decision remains with the coach or authorized user.

9.0 Finance Management – This module handles the financial recording functions of the system. It includes recording income, recording expenses, financial monitoring, financial summaries, and transaction history. This supports transparency and organized tracking of organizational funds.

10.0 Reports Generation – This module produces system reports such as attendance reports, member reports, event reports, financial reports, analytics dashboards, and predictive analytics dashboards. These reports help authorized users monitor activities and support decision-making.

11.0 Analytics and Emerging Technologies – This module represents the advanced features of Arnis Connect, including predictive analytics, geolocation services, and facial recognition technology. These features support improved attendance validation, player recommendation, and data-driven monitoring of organizational activities.

Overall, the Functional Decomposition Diagram shows that Arnis Connect is not a work breakdown structure but a system function breakdown. It focuses on what the system does and how its features are logically grouped. Through this structure, the system demonstrates a centralized and organized approach to managing members, registration, attendance, training, events, player suggestions, finances, reports, and emerging technology features for the Lapu-Lapu Arnis de Abanico Organization.

Work Breakdown Structure

The Work Breakdown Structure outlines the major project phases and tasks involved in the planning, analysis, design, development, testing, and deployment of Arnis Connect. It provides an organized sequence of work that ensures all necessary components of the project are addressed systematically.

Table 13
Work Breakdown Structure for Arnis Connect

Task No.	Description	Duration (Days)	Predecessor Tasks	Status
1.0	Arnis Connect Project			
1.1	Project Initiation			
1.1.1	Requirements Gathering (SRS)	10		Closed
1.1.2	UI/UX Wireframing & Design	8	1.1.1	Closed
1.1.3	Database Design (ERD)	5	1.1.2	Closed
1.1.4	Design Review & Approval	2	1.1.3	Open
1.2	Core & Member Module			
1.2.1	Database Setup	4	1.1.4	Open
1.2.2	Login/Authentication	6	1.2.1	Close
1.2.3	Member Management	8	1.2.2	Close
1.2.4	Iteration 1 Testing	3	1.2.3	Open
1.2.5	Iteration 1 Review	1	1.2.4	Open
1.3	Attendance & Event Module			
1.3.1	Attendance Tracking	9	1.2.5	Close
1.3.2	Event Documentation	9	1.3.1	Close
1.3.3	Module Integration	4	1.3.2	Open
1.3.4	Iteration 2 Testing	3	1.3.3	Open
1.3.5	Iteration 2 Review	1	1.3.4	Open

1.4	Finance & Reports Module			
1.4.1	Financial Tracking	8	1.3.5	Close
1.4.2	Basic Reports	7	1.4.1	Close
1.4.3	Data Mining Features	1	1.4.2	Close
1.4.4	Financial Integration	4	1.4.3	Open
1.4.5	Iteration 3 Testing	5	1.4.4	Open
1.4.6	Iteration 3 Review	1	1.4.5	Open
1.5	Finalization & Deployment			
1.5.1	System Integration Testing	5	1.4.6	Open
1.5.2	User Acceptance Testing	4	1.5.1	Open
1.5.3	Bug Fixing & Polishing	5	1.5.2	Open
1.5.4	Final Review & Contingency	4	1.5.3	Open
1.5.5	Server Deployment	2	1.5.4	Open
1.5.6	User Training & Documentation	2	1.5.5	Open
1.5.7	Project Handover	1	1.5.5	Open

Gantt Chart

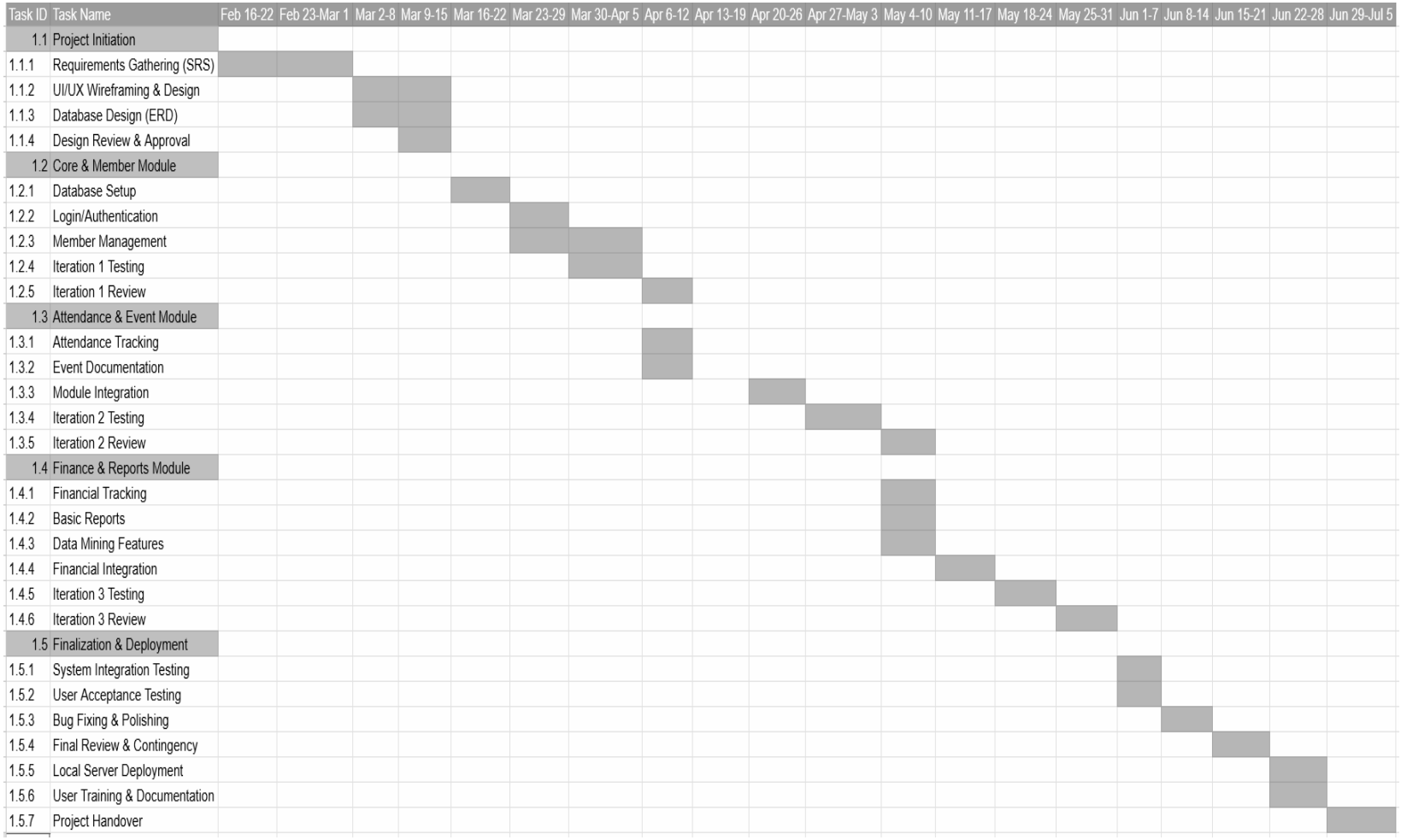


Figure 19
Project Gantt Chart

ANALYSIS PHASE

This phase focuses on examining the current workflow of the organization and identifying the functional and non-functional requirements of the system. The outputs of this phase serve as the basis for the detailed design and development of Arnis Connect. It ensures that the proposed solution reflects the actual operational activities of the organization and the expectations of its intended users.

Use Case Diagram

Use case identification presents the system actors and their interactions with Arnis Connect. It shows the actions that each role can perform within the system and helps define the functional requirements from the perspective of the users.

Administrator / Lead Coach Use Case

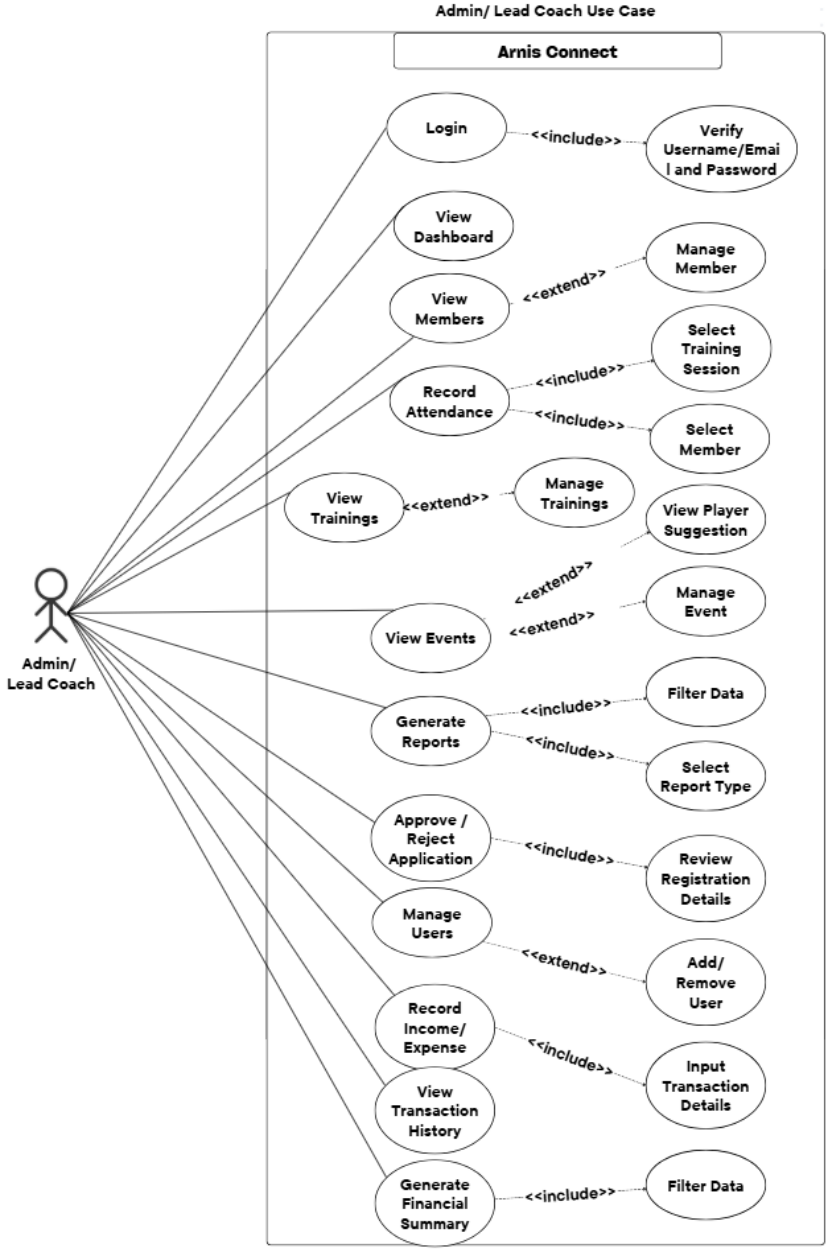


Figure 20
Administrator / Lead Coach Use Case Diagram

Figure 20 shows the use cases for the Administrator or Lead Coach. These include logging in, managing users, approving or rejecting registration requests, adding and updating member records, viewing reports, creating or managing events, and monitoring attendance and finances. This use case diagram demonstrates that the administrator acts as the main system controller responsible for coordinating and validating organizational data.

Coach Use Case

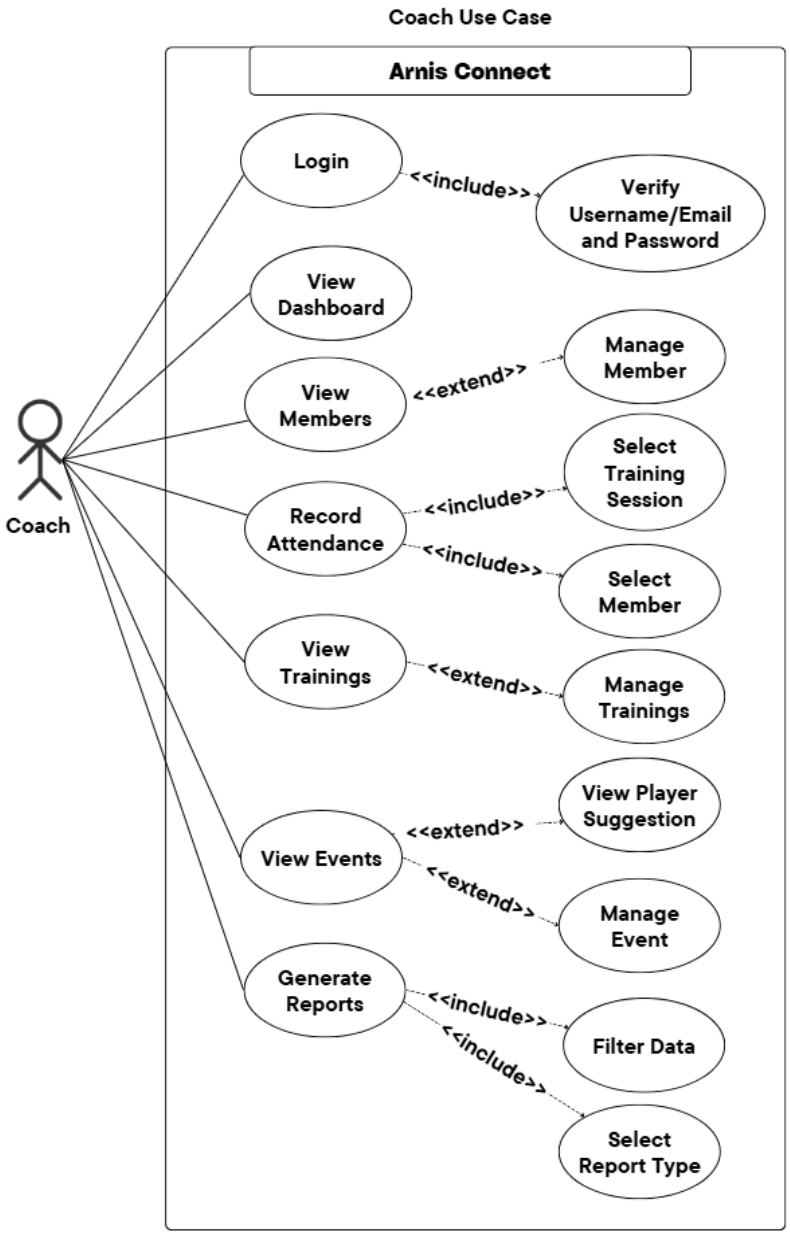


Figure 21
Coach Use Case Diagram

Figure 21 above shows the use cases for the Coach. These include logging in, viewing members, recording attendance, viewing training or event schedules, and checking suggested participants for events. This use case diagram emphasizes the operational role of the coach in day-to-day organization activities.

Finance User Use Case

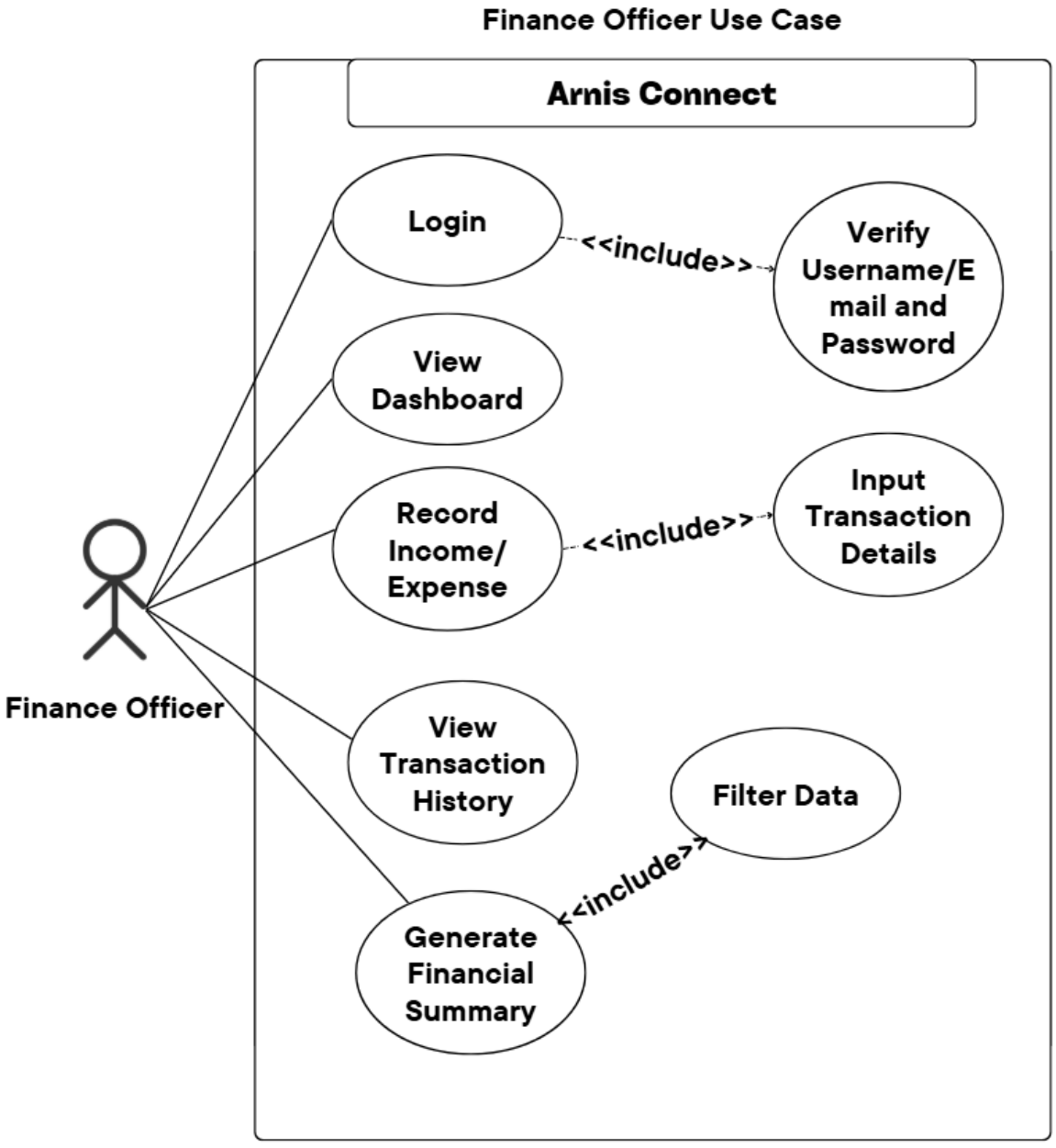


Figure 22
Finance User Use Case Diagram

Figure 22 above shows the use cases for the Finance User. These include logging in, recording income and expenses, viewing financial summaries, and generating financial reports. This diagram highlights the system's support for more organized and transparent financial record management.

Applicant Use Case

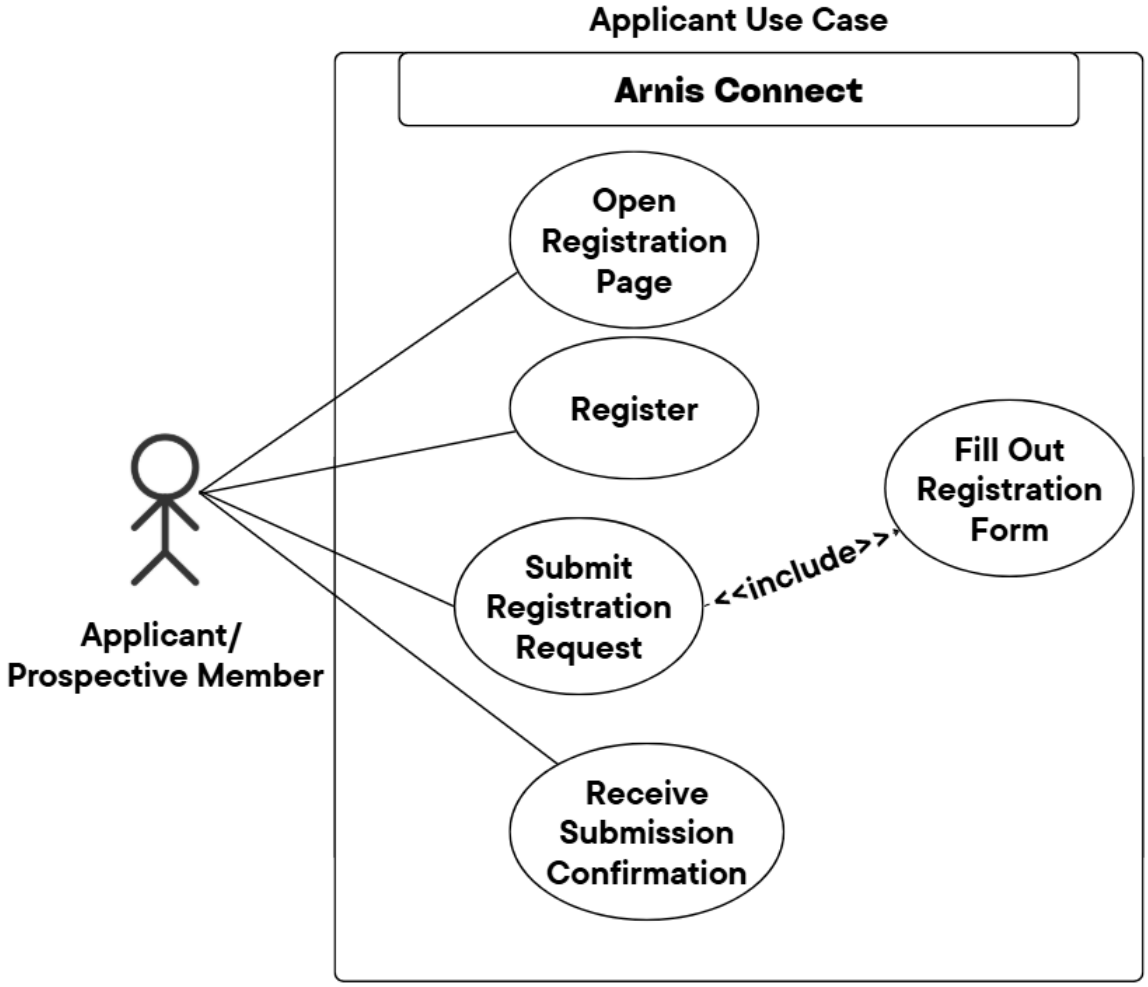


Figure 23
Applicant Use Case Diagram

Figure 23 presents the use cases for the Applicant or Prospective Member. These include viewing the registration page, submitting registration data, and waiting for the result of the approval process. This diagram demonstrates how the public-facing part of the system supports a structured intake of new members.

Member/ Player Use Case

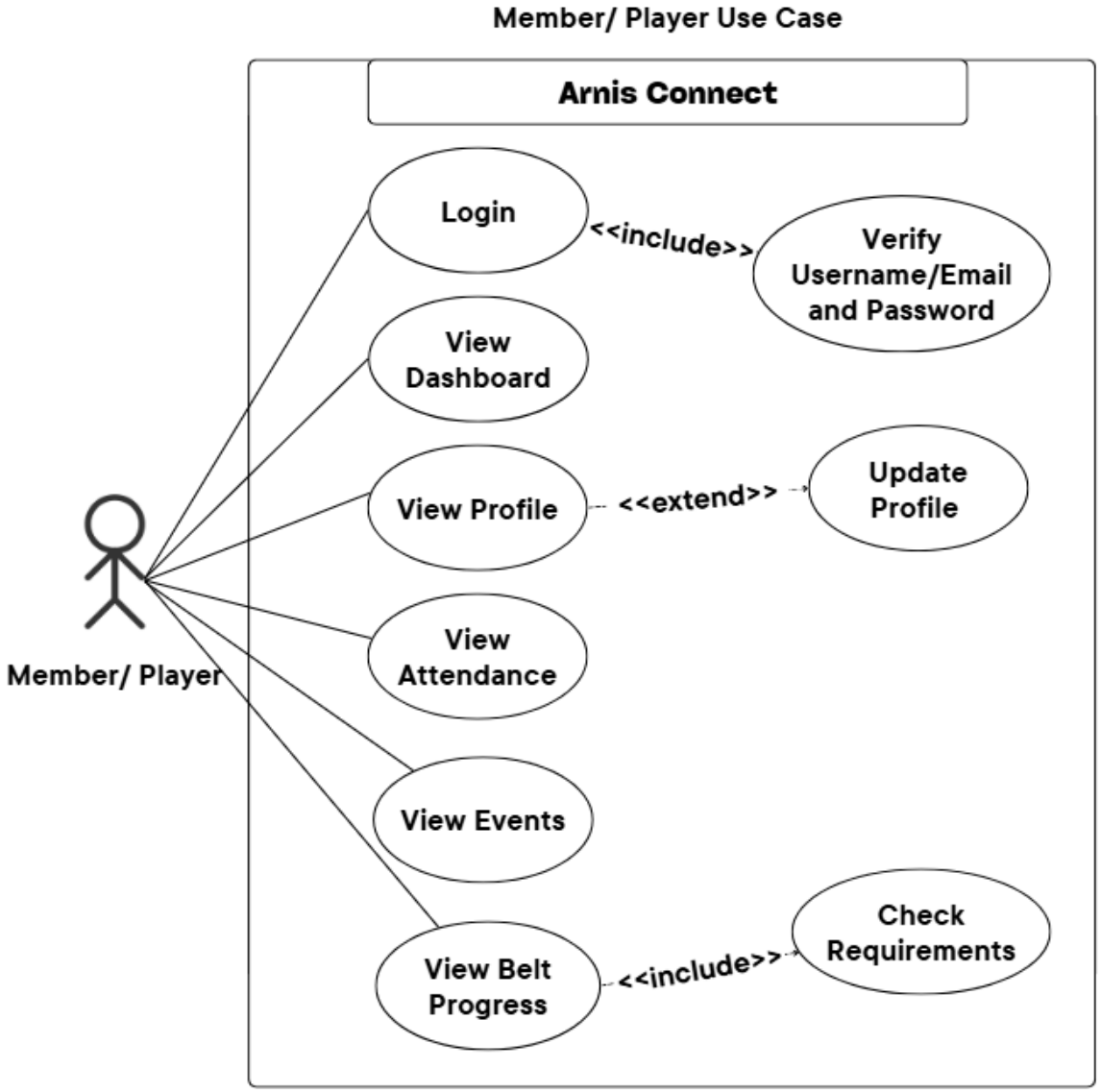


Figure 24
Member/Player Use Case Diagram

Figure 24 presents the use cases for the Applicant or Prospective Member. The Member or Player Panel allows approved members to access their profile, attendance records, event participation, and belt progress. This improves transparency and gives players better visibility of their participation history.

Sequence Diagram

This section presents the sequence diagrams that illustrate the step-by-step interaction among the user, interface, application logic, and database. These diagrams show how the major processes of Arnis Connect occur over time.

Login Authentication Sequence Diagram

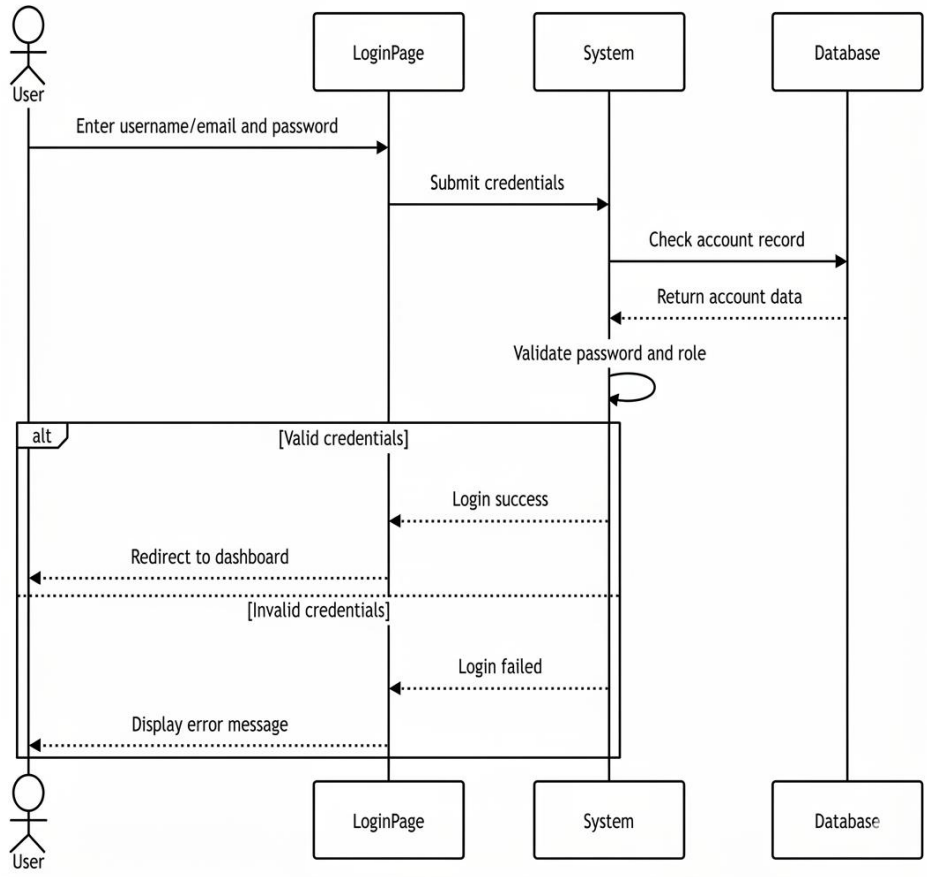


Figure 25
Login Authentication Sequence Diagram

Figure 25 illustrates the sequence of events during the login process. The user enters credentials, the system validates the input, checks the data in the database, verifies the account role, and then redirects the user to the appropriate dashboard. This process ensures secure and role-based access to the system.

Registration and Approval Sequence Diagram

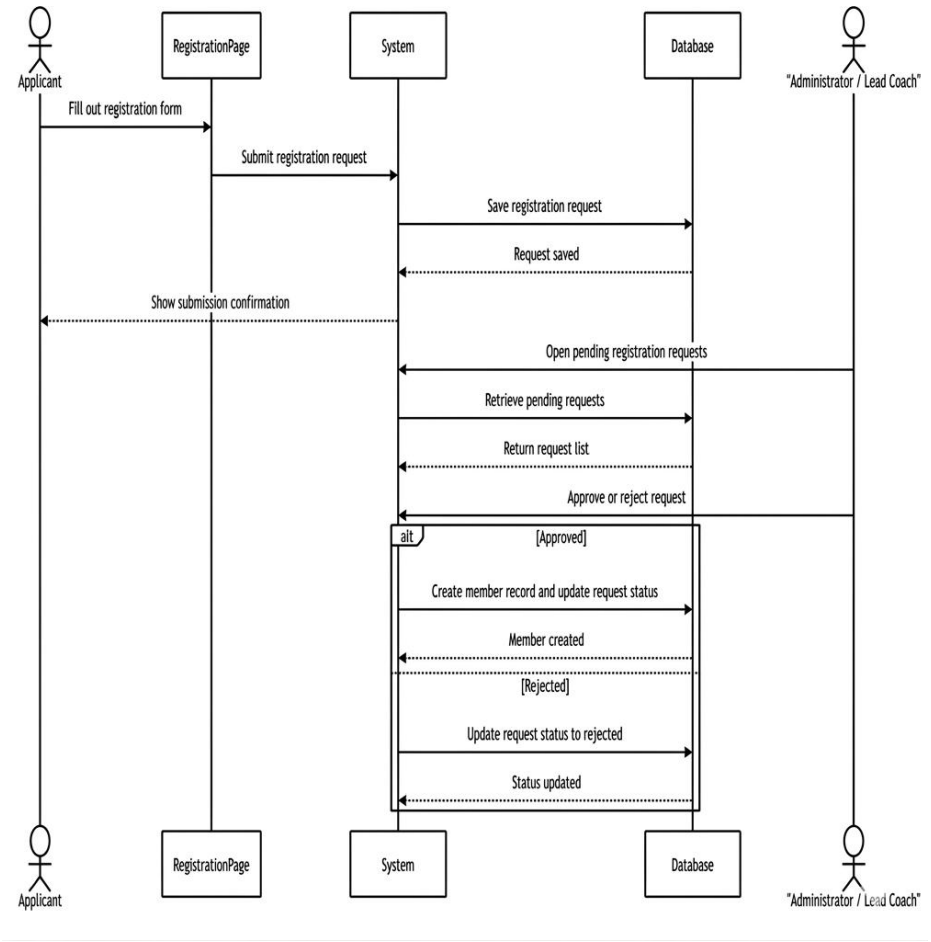


Figure 26
Registration and Approval Sequence Diagram

Figure 26 above shows the sequence of interactions when an applicant submits a registration request and the administrator reviews it. The applicant fills out the registration form, the system stores the request, and the administrator later approves or rejects the submission. If approved, the request is converted into an official member record.

Attendance Recording Sequence Diagram

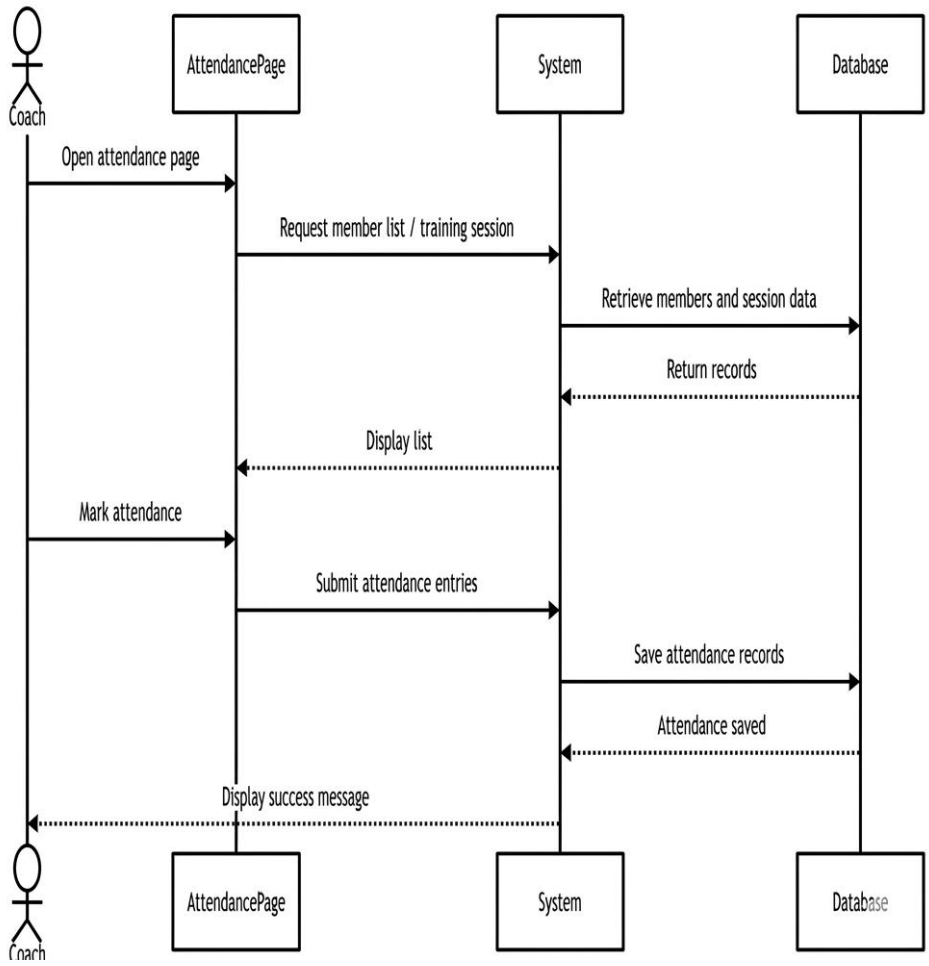


Figure 27
Attendance Recording Sequence Diagram

Figure 27 above illustrates the process of attendance recording. The coach selects a training session, marks member attendance, submits the record, and the system stores the data in the database. This sequence ensures that attendance records are encoded systematically and linked to the proper member and training record.

Event Creation and Player Suggestion Sequence Diagram

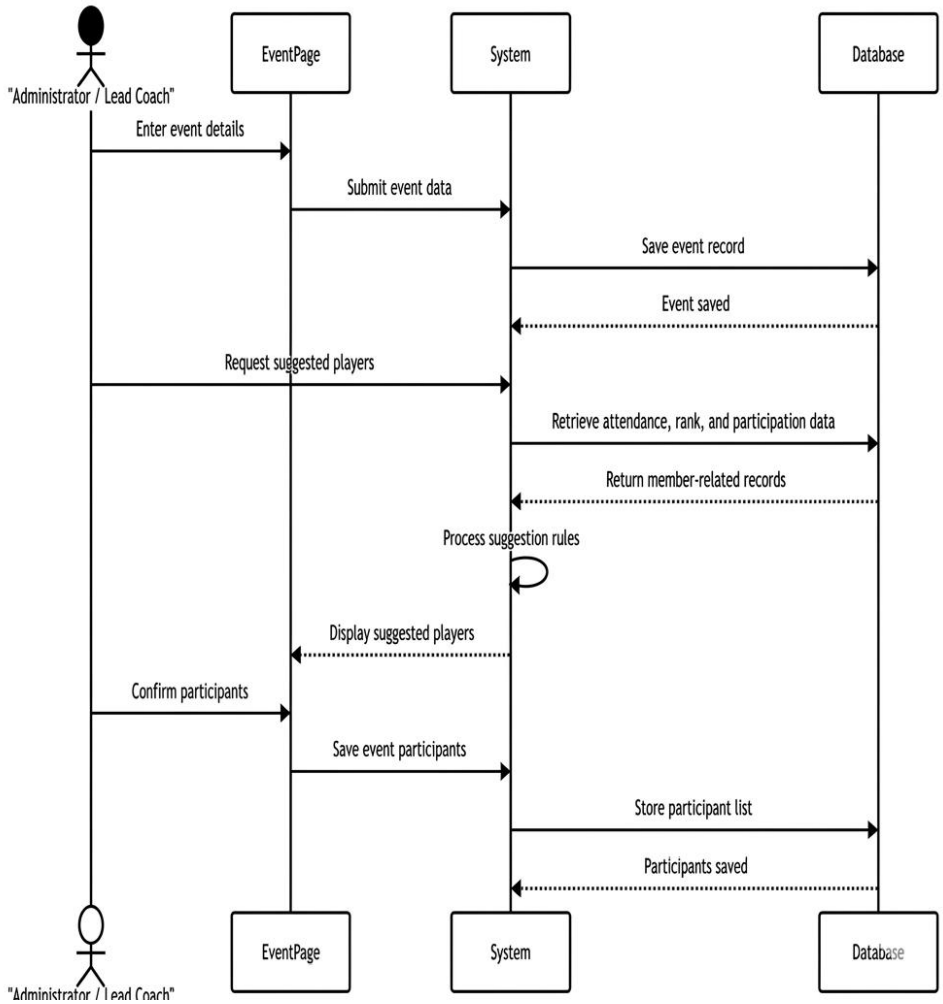


Figure 28
Event Creation and Player Suggestion Sequence Diagram

Figure 28 presents the sequence of actions involved in creating an event and suggesting players. The administrator or coach inputs event details, the system retrieves relevant member data such as attendance history and participation records, then suggests appropriate players based on the defined criteria. This helps improve event preparation and decision-making.

Financial Logging Sequence Diagram

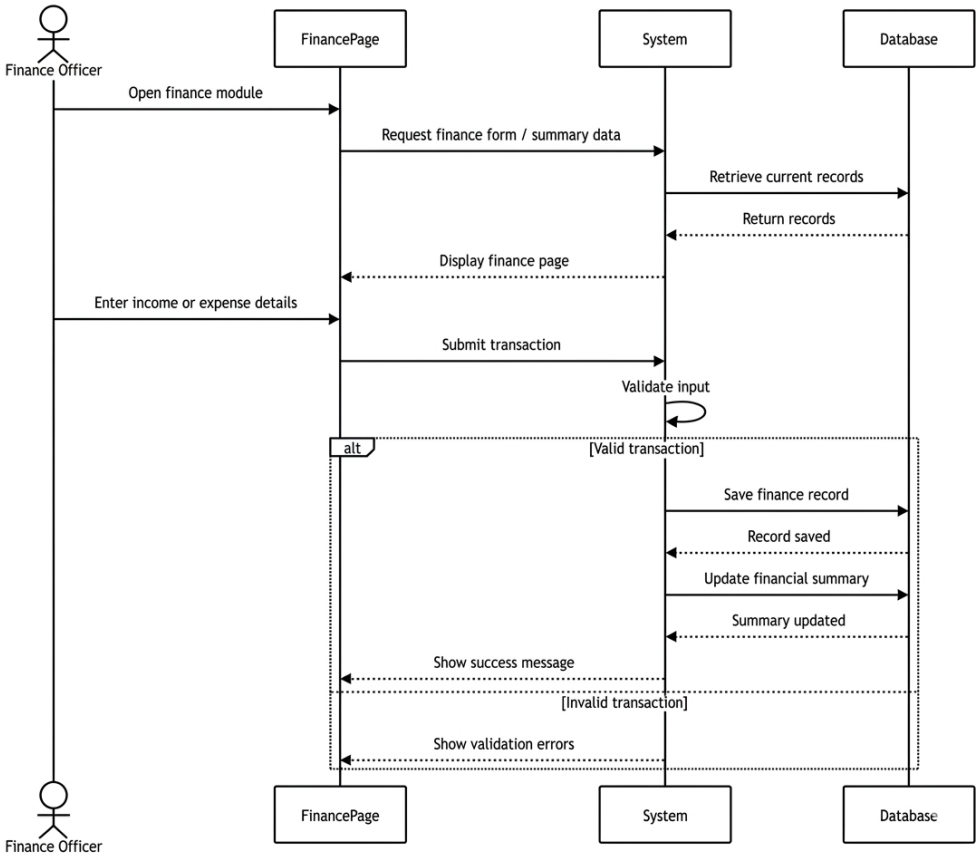


Figure 29
Financial Logging Sequence Diagram

Figure 29 shows the process of recording income or expense transactions. The finance user submits transaction details, the system validates the input, stores the record in the database, and updates the financial summary. This sequence ensures a more organized and centralized handling of financial data.

Activity Diagram

Activity diagrams illustrate how a system operates by mapping out the flow of actions involved in a process. They show the order of activities, decision points, conditions that influence the flow, concurrent operations, and the start and end of a process. These diagrams are useful for understanding how tasks progress within a system, especially when there are multiple possible paths or when different users or components interact. Compared to sequence diagrams, activity

diagrams focus more on the overall process flow rather than the interaction between objects, making them helpful for analyzing complex workflows.

Login and Session Management

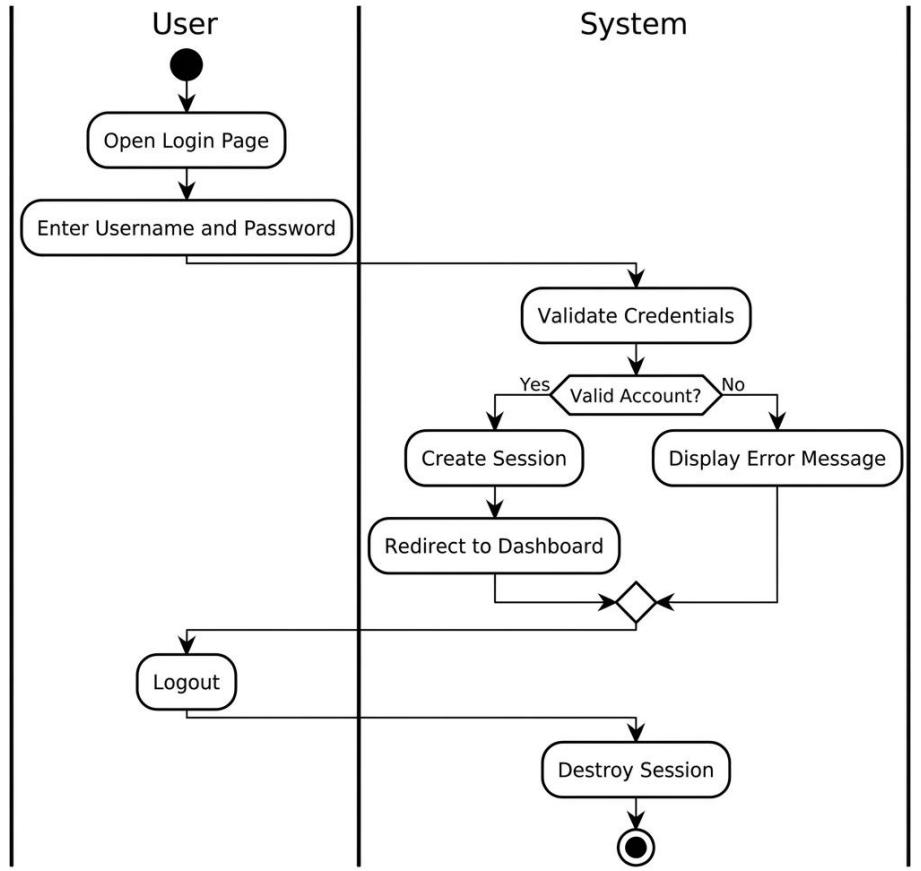


Figure 30
Login and Session Management Sequence Diagram

The Login and Session Management Module illustrates the process by which users securely access and exit the Arnis Connect system. Users enter their login credentials through the authentication page, and the system validates the provided username and password. If the credentials are valid, the system creates an active session and redirects the user to the appropriate dashboard based on their assigned role. If invalid credentials are entered, the system displays an error message and prompts the user to try again. Users may also log out at any time, which terminates the session and securely returns them to the login page.

User Management

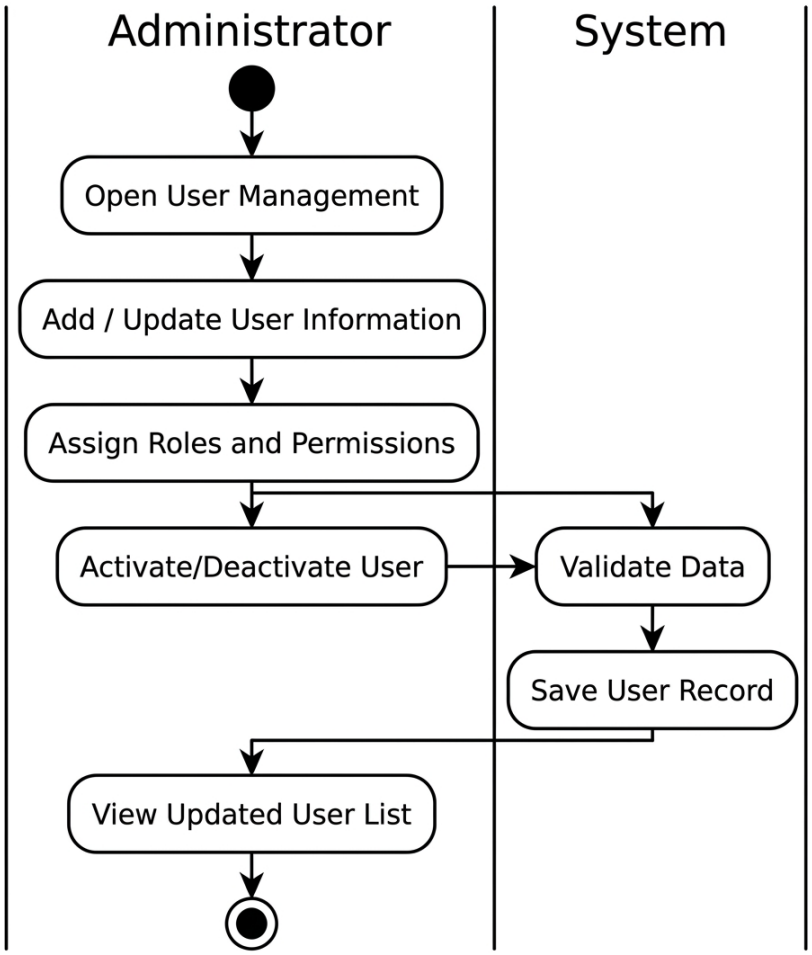


Figure 31
User Management Sequence Diagram

The User Management Module illustrates how administrators manage system users and their access privileges. Administrators can create new user accounts, update existing account information, assign user roles, and manage permissions according to organizational responsibilities. The system validates all submitted information before saving updates to the database. This module ensures controlled access to system functionalities and supports secure role-based access control throughout the platform.

Member/ Player Management

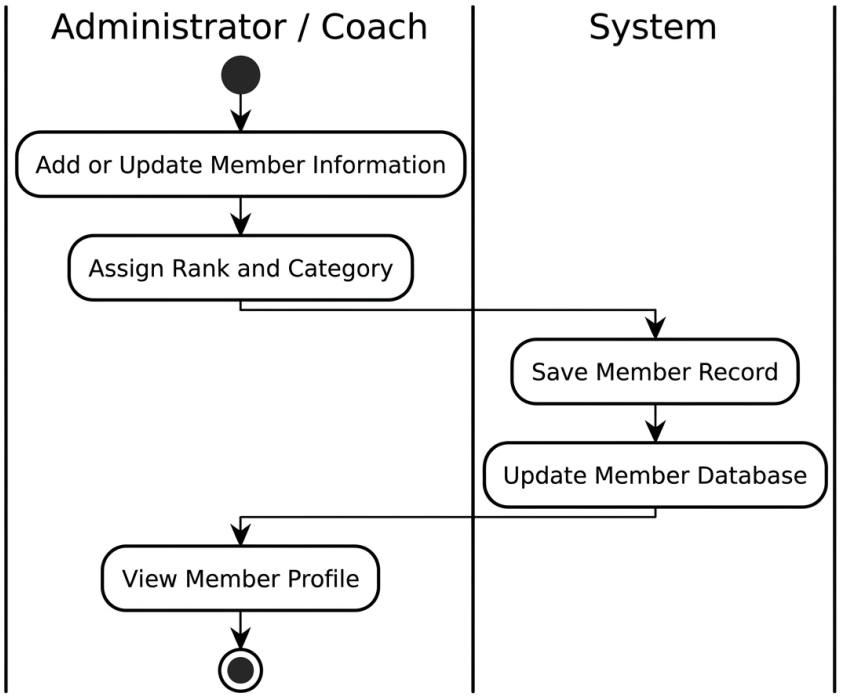


Figure 32
Member/ Player Sequence Diagram

The Member / Player Management Module illustrates the management of member records within Arnis Connect. Administrators and coaches may add new member profiles, update personal information, assign player categories, and monitor member status and rank progression. The system stores and updates all records in the centralized database, allowing authorized users to easily retrieve and review member information. This module supports organized tracking of player development and participation within the organization.

Registration Approval

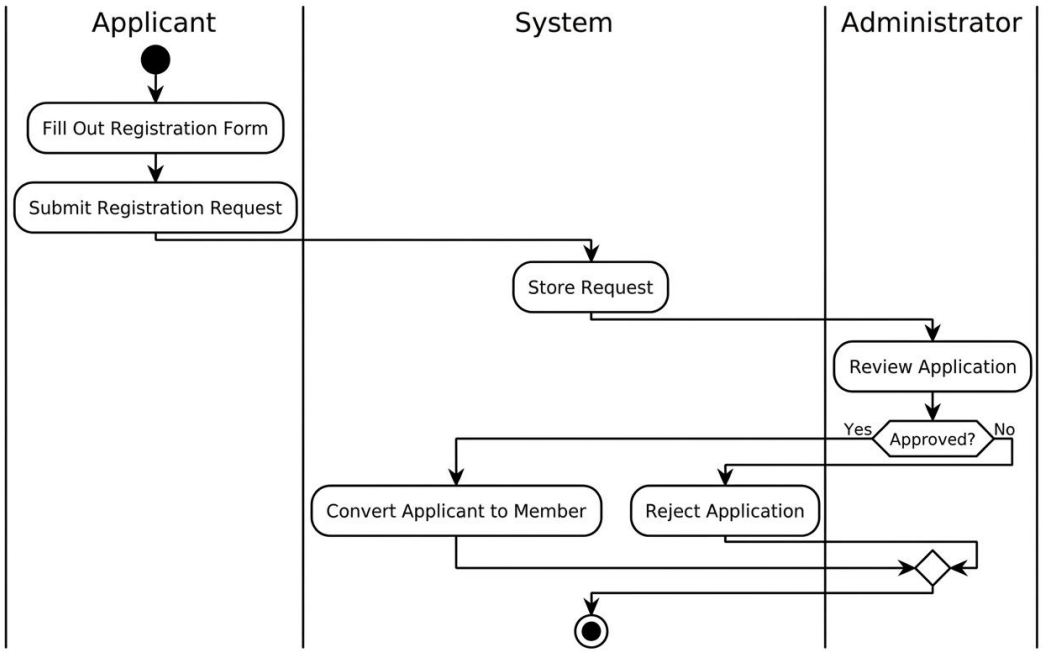


Figure 33
Registration Approval Sequence Diagram

The Registration Approval Workflow Module illustrates the process of handling applicant registrations. Applicants access the public registration form and submit their personal information and training preferences. The system stores the submitted request and notifies the administrator for review. The administrator evaluates the application and either approves or rejects the request. Approved applicants are automatically converted into official member records, while rejected applications remain archived for reference.

Attendance Management

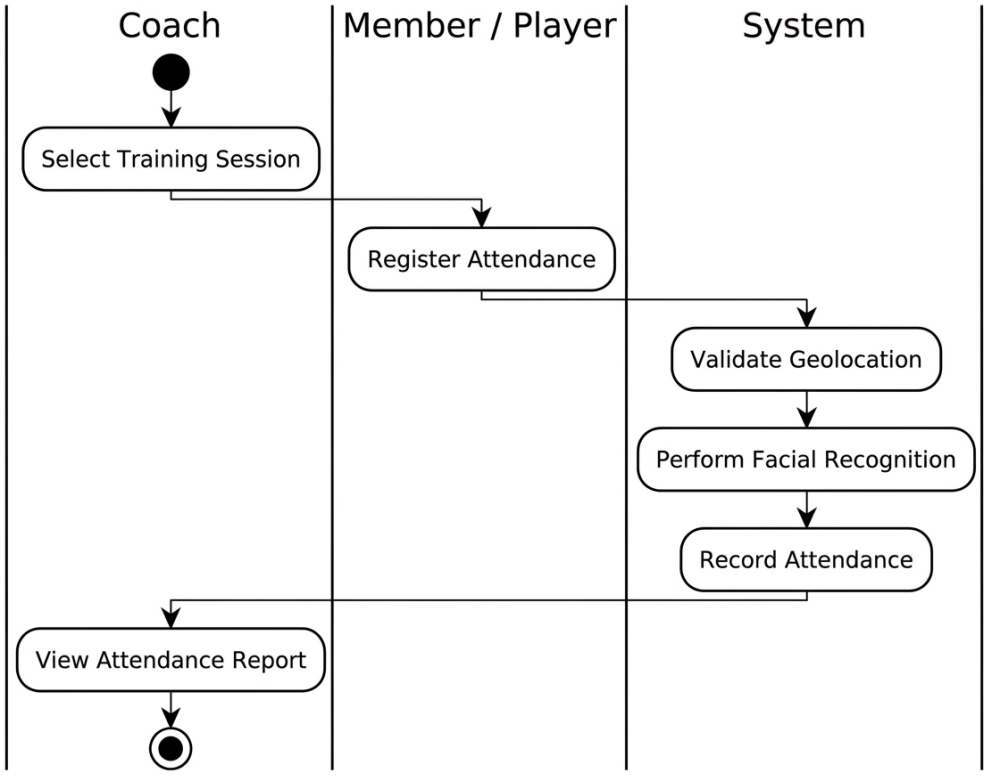


Figure 34
Attendance Management Sequence Diagram

The Attendance Management Module illustrates how attendance records are captured and monitored during training sessions. Coaches select the active training session while members or players register their attendance through the system. Attendance validation may include geolocation verification and facial recognition to improve accuracy and prevent fraudulent attendance entries. Once validated, the system records attendance information and updates attendance histories and reports for monitoring purposes.

Training Management

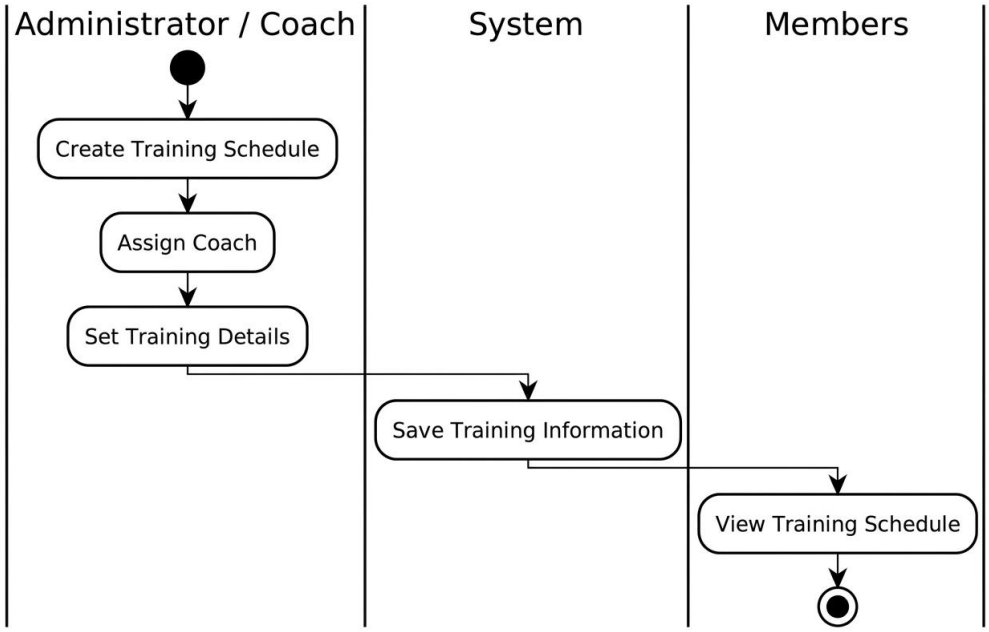


Figure 35
Training Management Sequence Diagram

The Training Management Module illustrates the process of organizing and maintaining training schedules and session information. Administrators or coaches create training sessions, assign coaches, specify schedules, and define training details such as location and session type. The system stores these records and makes them accessible to members and players for schedule viewing and participation tracking. This module supports structured coordination of training activities within the organization.

Event Management

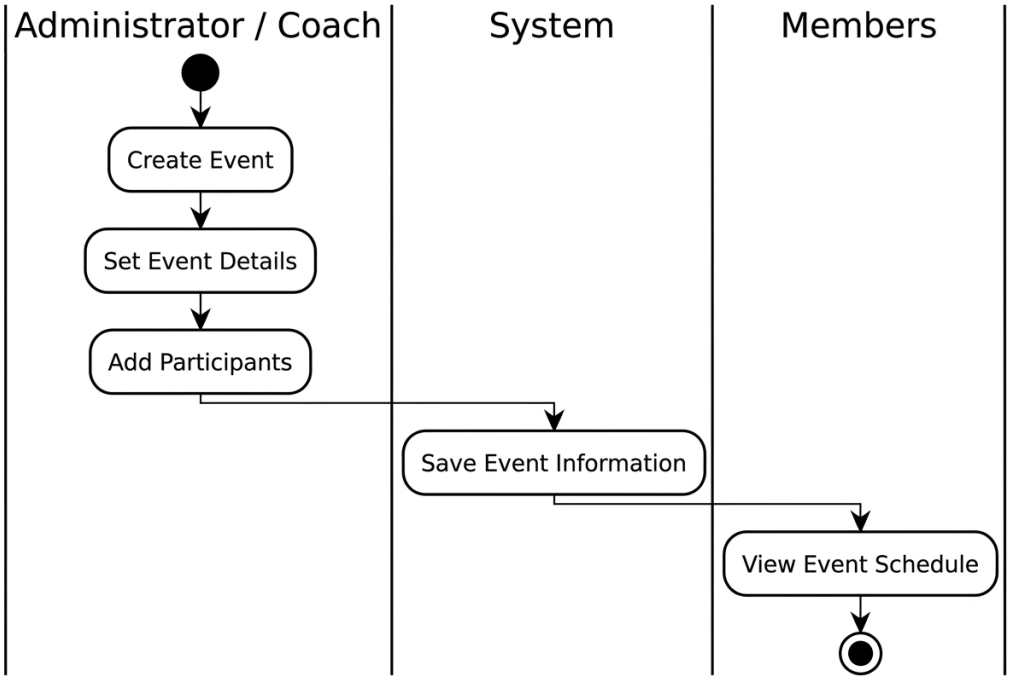


Figure 36
Event Management Sequence Diagram

The Event Management Module illustrates how organizational events are created, managed, and monitored. Administrators or coaches create event records, assign schedules, manage participant lists, and document event-related information. The system stores event details and participation records in the centralized database for future monitoring and reporting. Members and players may view upcoming events and their participation status through the system.

Player Suggestion

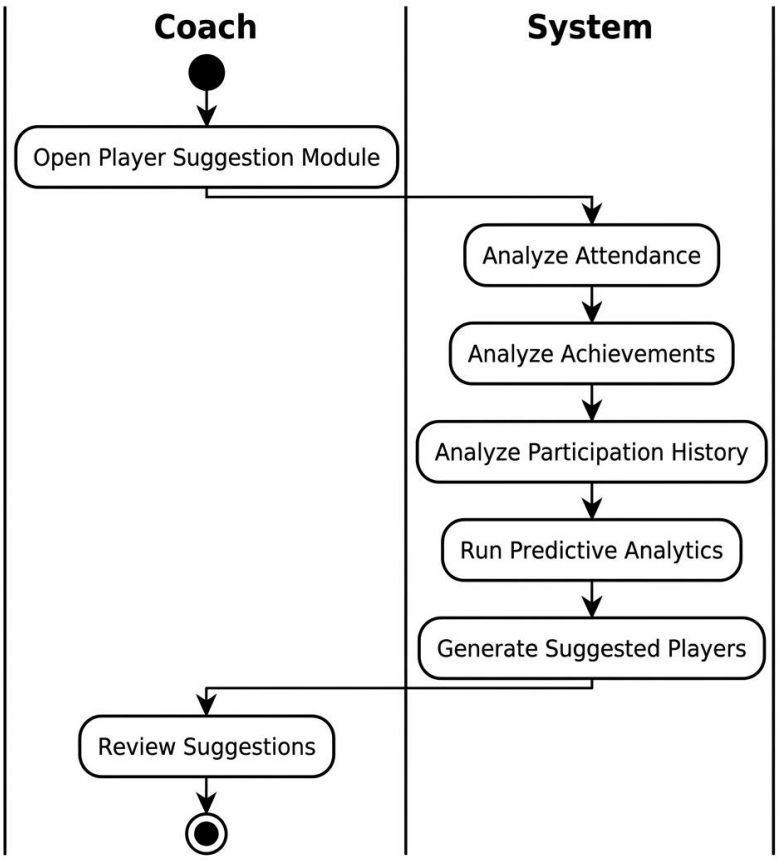


Figure 37
Player Suggestion Sequence Diagram

The Player Suggestion Feature Module illustrates the system's analytics-supported recommendation process for selecting event participants. The system analyzes attendance records, achievements, participation history, player category, and other relevant factors to generate suggested players for specific events. Predictive analytics techniques may also be used to identify suitable participants based on historical performance and attendance patterns. Final player selection decisions remain under the authority of the coach or lead coach.

Finance Management

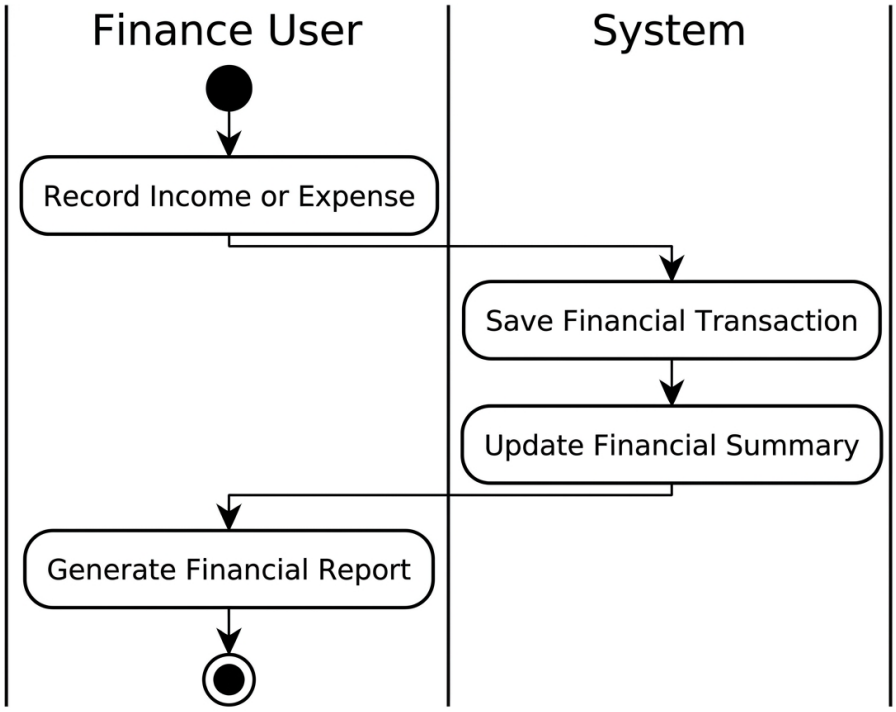


Figure 38
Finance Management Sequence Diagram

The Finance Management Module illustrates the recording and monitoring of organizational financial transactions. Finance users may encode income and expense records, update financial information, and review transaction histories. The system automatically updates financial summaries and generates reports that support transparency and monitoring of organizational funds. Access to this module is restricted to authorized finance personnel to maintain data confidentiality and integrity.

Reports Generation

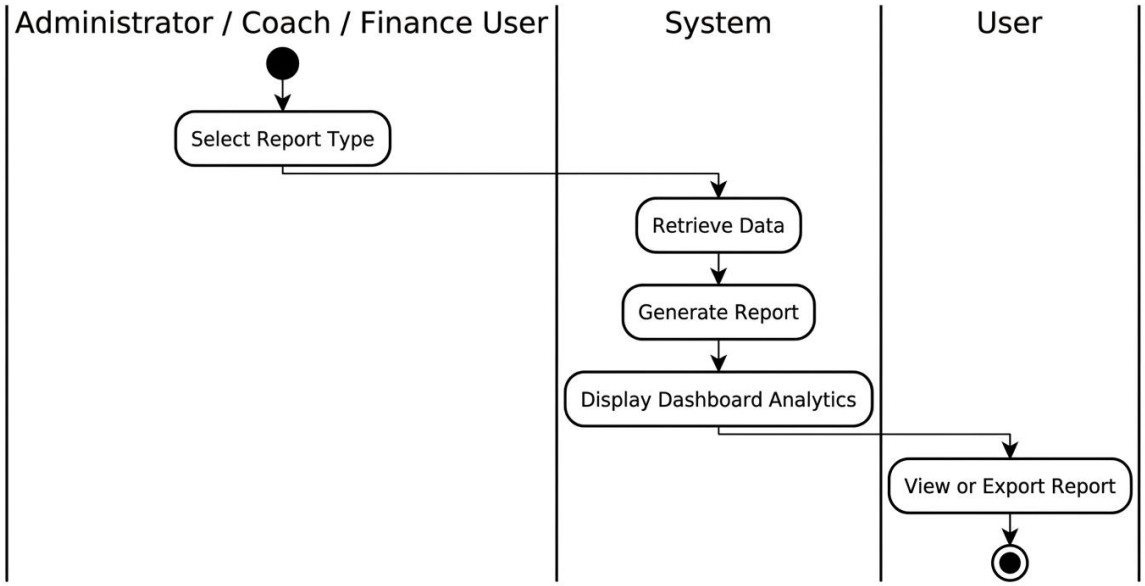


Figure 39
Reports Generation Sequence Diagram

The Reports Generation Module illustrates how the system produces operational and analytical reports for monitoring and decision-making. Authorized users may select report categories such as attendance, members, events, or finances. The system retrieves relevant records from the database, generates summaries and visual reports, and displays them through dashboards or printable formats. This module supports efficient evaluation of organizational activities and performance.

Analytics and Emerging Technologies

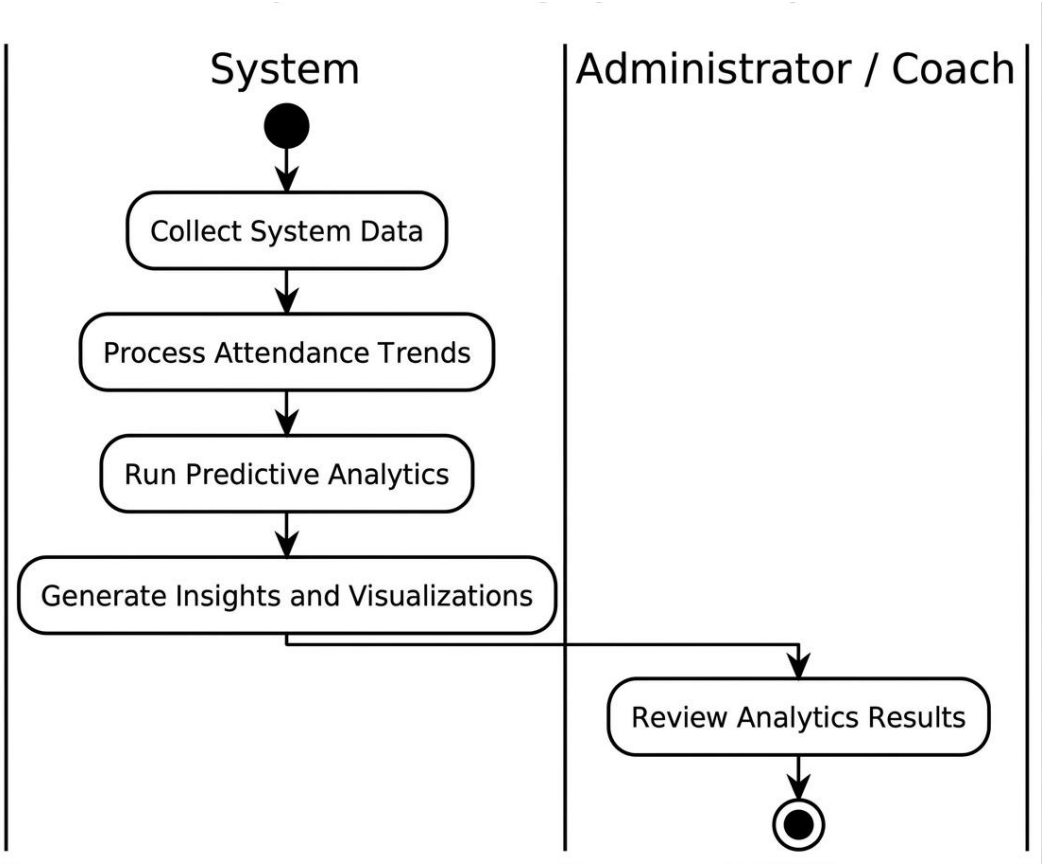


Figure 40
Analytics and Emerging Technologies Sequence Diagram

The Analytics and Emerging Technologies Module illustrates the integration of modern analytical and intelligent technologies within Arnis Connect. The system collects operational data from attendance, participation, events, and member activities to generate insights and trends through predictive analytics and data visualization techniques. Emerging technologies such as geolocation services, facial recognition, and machine learning-based recommendation features are incorporated to improve automation, monitoring accuracy, and decision support within the organization.

DESIGN PHASE

This phase focuses on translating the identified requirements into a structured technical and visual blueprint for the system. It includes the user interface design, database design, and system architecture design of Arnis Connect. This phase ensures that the final implementation of the system is aligned with the organizational workflow, user roles, and technical requirements identified in the earlier phases.

USER INTERFACE DESIGN

This section presents the updated user interface design of Arnis Connect based on the revised system requirements and latest prototype implementation. The updated interface uses a consistent dark-themed layout with purple accent colors to maintain a unified visual identity for the system. The screens show the complete flow from account registration, account verification, member application, member access, automated attendance support, belt progression, and administrative management.

The user interface is designed to support both administrative users and members. Administrative users can manage members, trainings, events, funds, users, and reports. Members can access their own dashboard, application status, face recognition setup, attendance-related training access, and belt progression. This supports the revised scope of Arnis Connect where members are no longer only recorded by the system but can also directly access system features.

Interface Screens

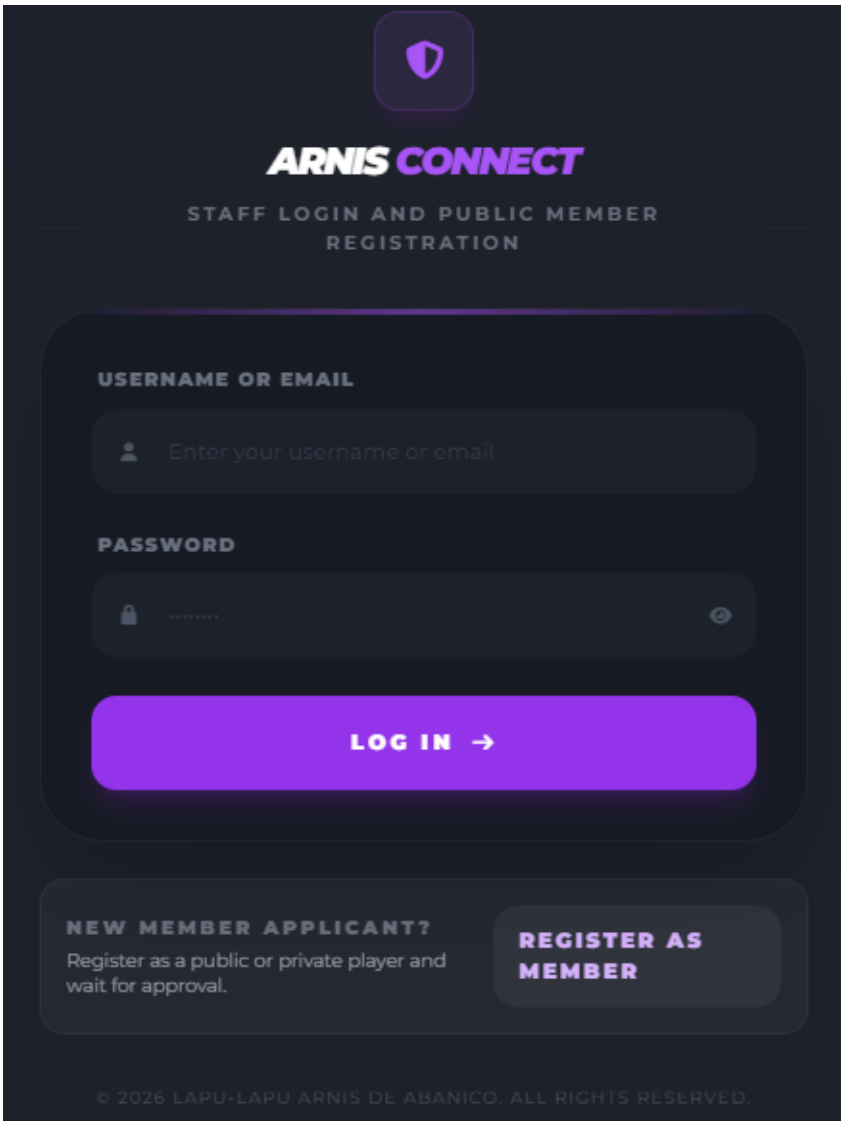


Figure 41
Login Page Interface

Figure 41 shows the login page of Arnis Connect. This interface allows existing users to enter their username or email and password to access the system. It also provides a registration option for new member applicants who need to create an account before submitting their membership application.

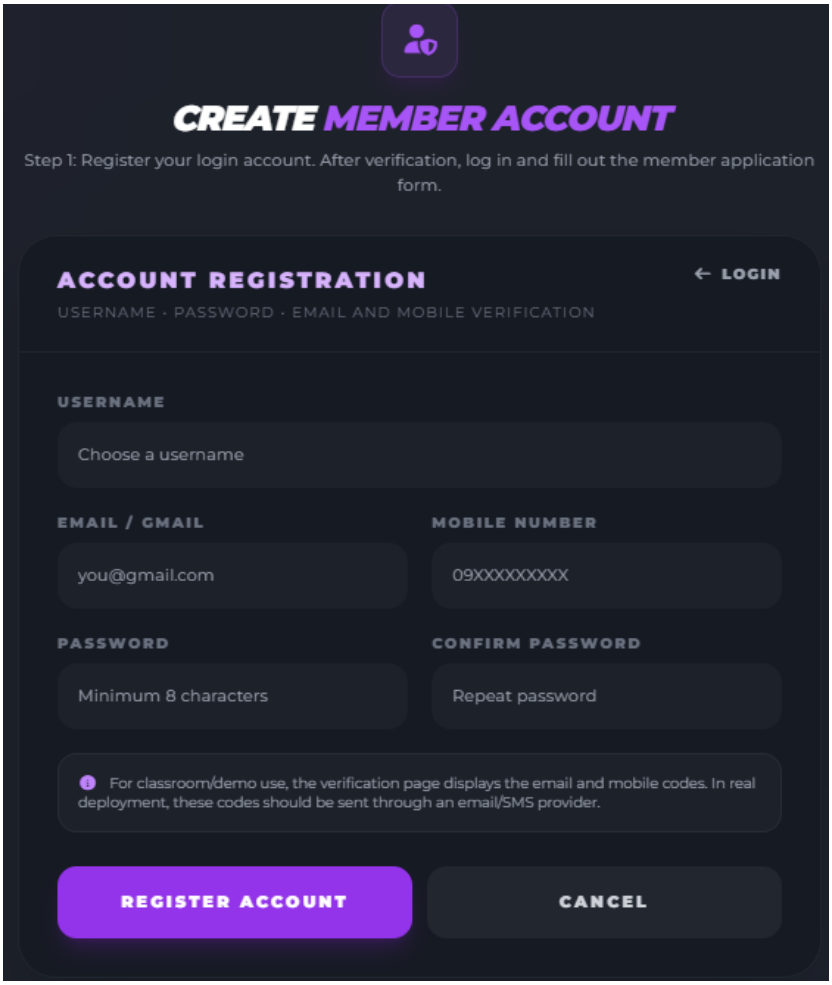


Figure 42
Member Account Registration Interface

Figure 42 shows the member account registration page. This page is used by new applicants to create their login account by providing a username, email or Gmail address, mobile number, password, and password confirmation. This supports the revised registration flow where account creation is done before completing the member application form.

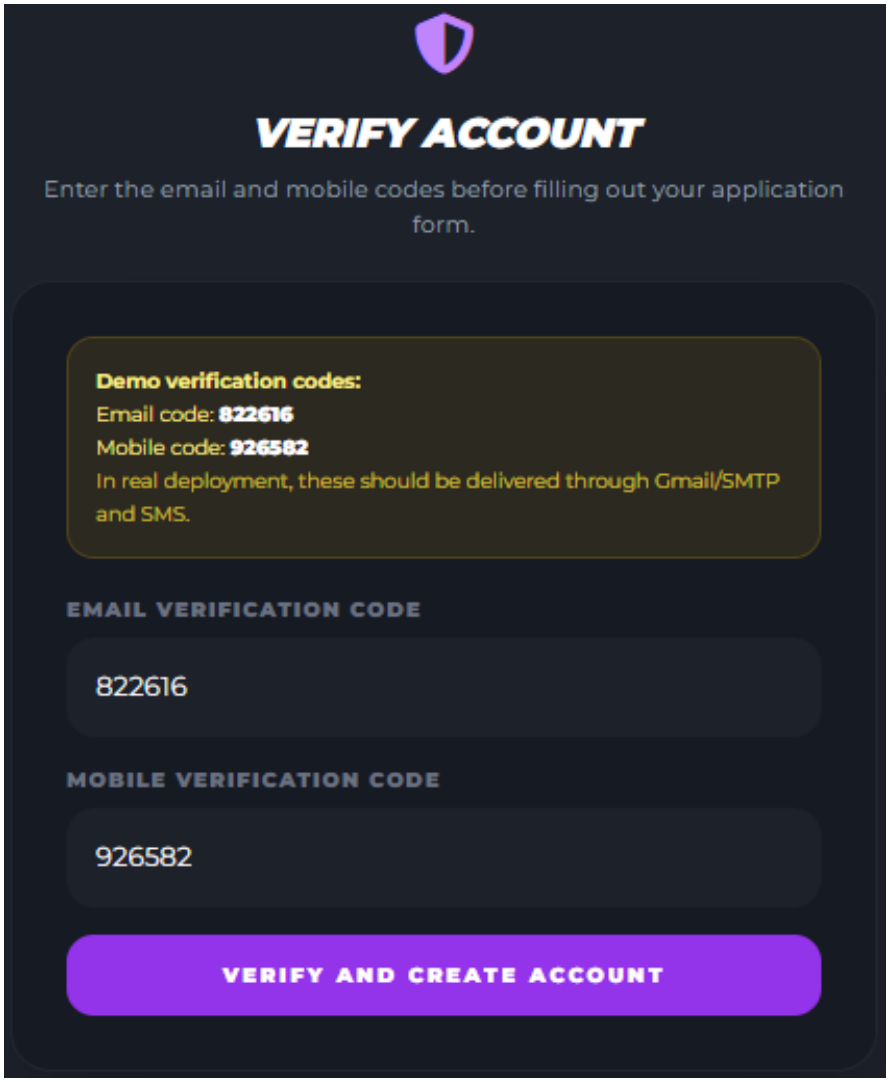


Figure 43
Account Verification Interface

Figure 43 shows the account verification page. After creating an account, the applicant enters the verification codes for email and mobile number. Once the verification is completed, the account can proceed to the next step, which is filling out the member application form.

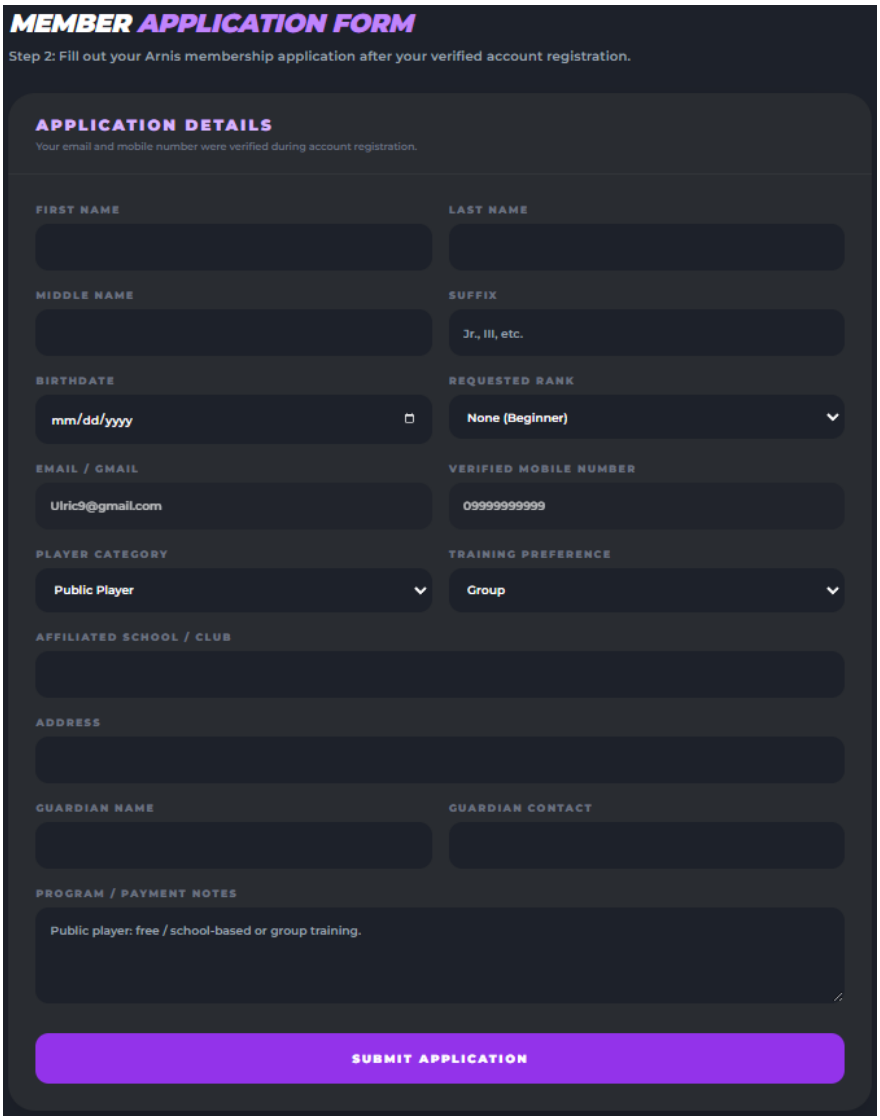


Figure 44
Member Application Form Interface

Figure 44 shows the member application form. This page collects the applicant details after account verification. The full name is separated into first name, middle name, last name, and suffix. The form also includes birthdate, requested rank, verified email, verified mobile number, player category, training preference, affiliated school or club, address, guardian details, and program or payment notes.

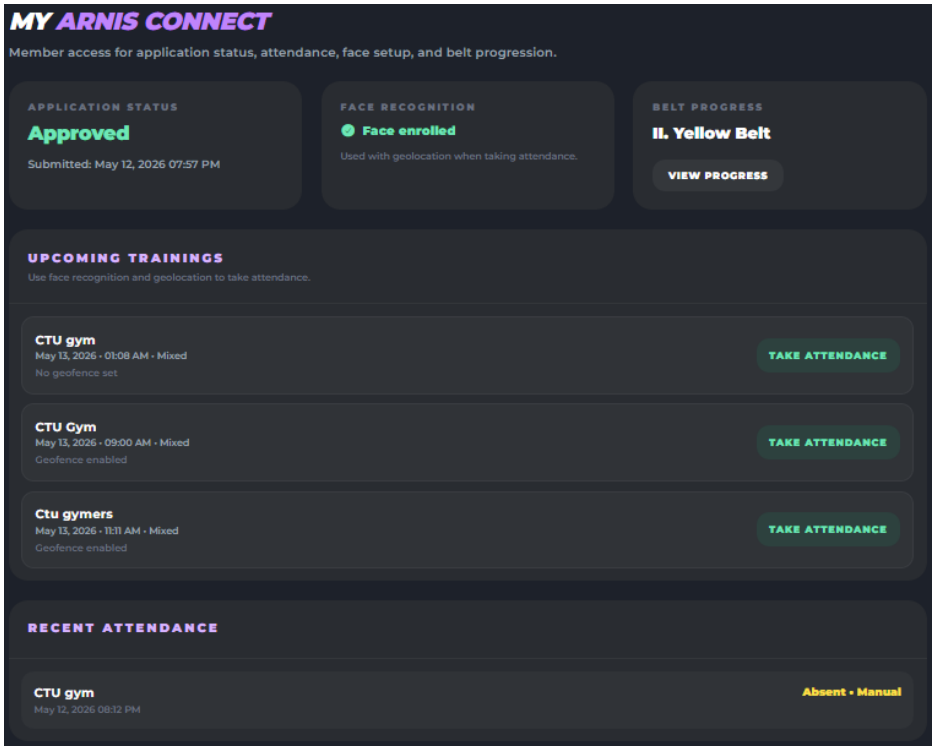


Figure 45
Member Dashboard Interface

Figure 45 shows the member dashboard. This interface gives members access to their application status, face recognition status, belt progress, upcoming training sessions, and recent attendance records. This supports the panel recommendation that members should be able to access the Arnis Connect system.

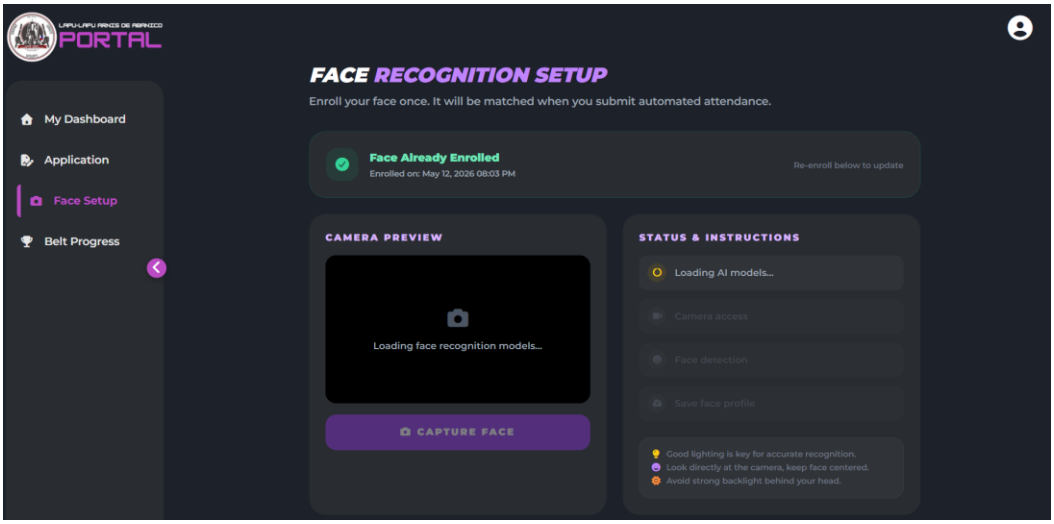


Figure 46
Face Recognition Setup Interface

Figure 46 shows the face recognition setup page. The member can enroll their face once through the camera preview. The captured face profile is used together with geolocation during automated attendance checking. This improves attendance monitoring by verifying that the correct player is taking attendance.

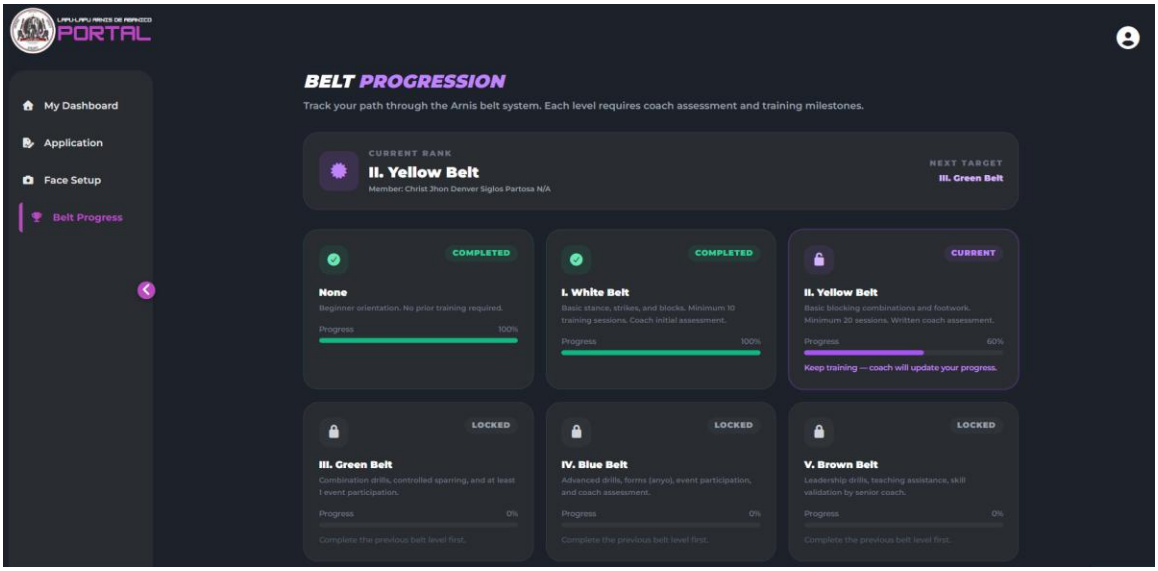


Figure 47
Belt Progression Interface

Figure 47 shows the belt progression page. This interface presents the player's current rank, next target rank, completed levels, current level, and locked levels. The design follows a progress-based layout where players need to complete the requirements of the current belt before unlocking the next belt level.

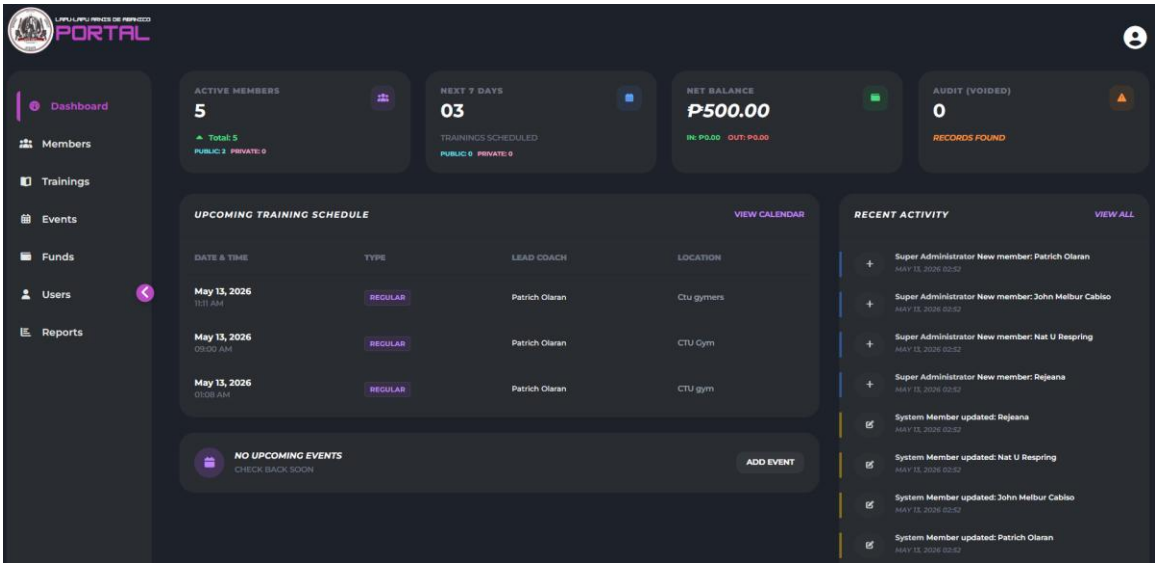


Figure 48
Administrator Dashboard Interface

Figure 48 shows the administrator dashboard. This page provides summary cards for active members, upcoming training schedules, fund balance, and audit records. It also displays upcoming training schedules and recent activities, allowing the administrator to monitor organizational operations in one view.

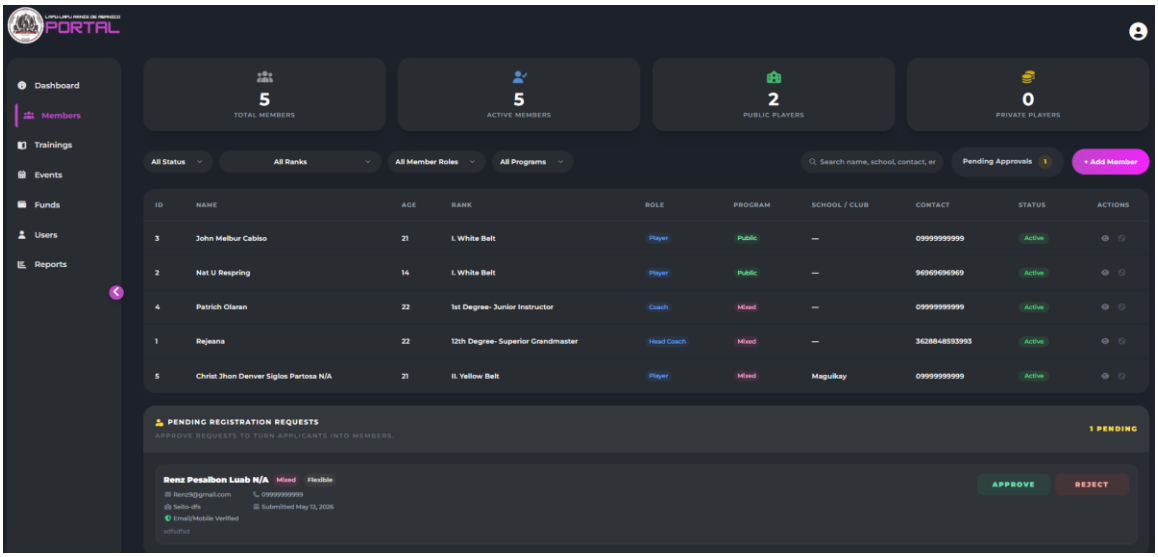


Figure 49
Members Management Interface

Figure 49 shows the members management page. This page displays member statistics, searchable member records, filters, and pending registration requests. The approve and reject

buttons are designed to process applications. When approved, the applicant is stored as an official member in the database and linked to the verified account.

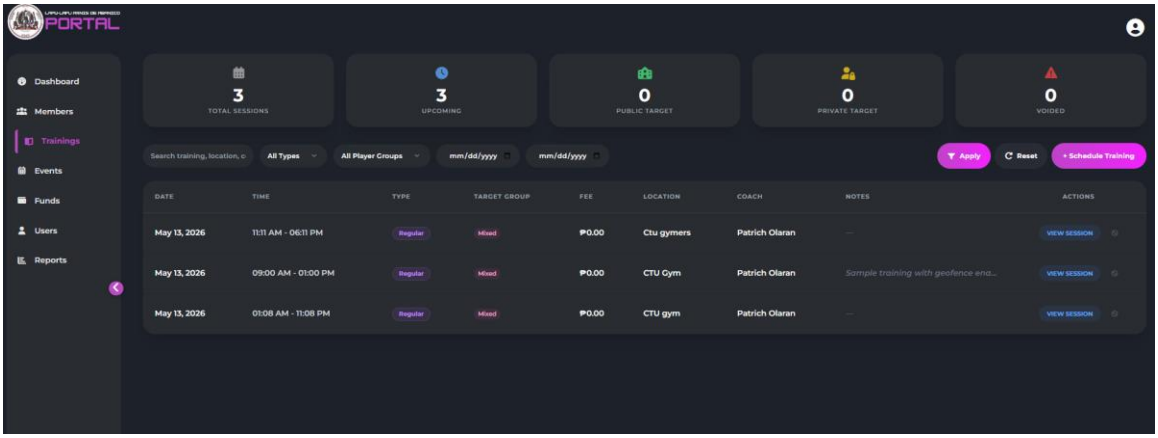


Figure 50
Trainings Management Interface

Figure 50 shows the trainings management page. The interface displays training statistics, filters, scheduled training sessions, training type, player group, fee, location, assigned coach, and actions. The training setup supports geofence information so that attendance can be verified based on the player's location.

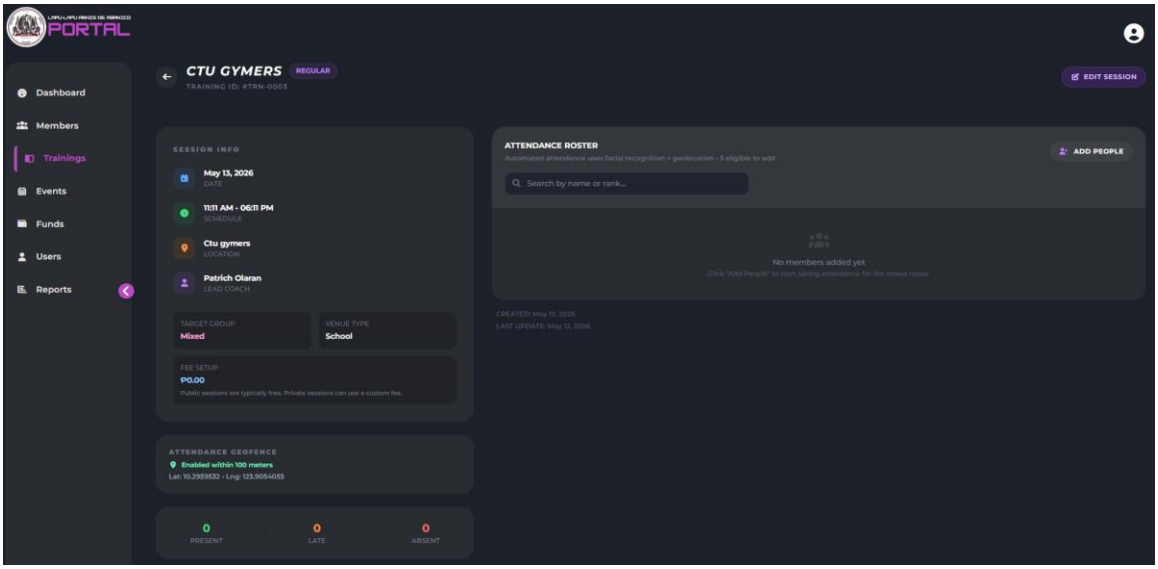


Figure 51
Training Session and Attendance Roster Interface

Figure 51 shows the training session detail and attendance roster. It displays the session information such as date, schedule, location, lead coach, target group, venue type, and geofence coordinates. The attendance roster supports automated attendance through face recognition and geolocation verification.

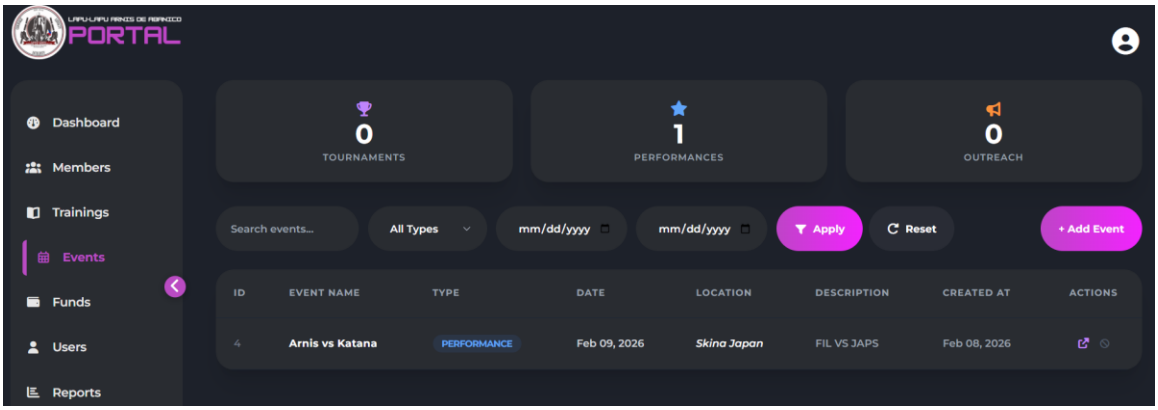


Figure 52
Events Management Interface

Figure 52 shows the events management page. This interface allows authorized users to view, search, filter, and add events. It also displays event counts by type such as tournaments, performances, and outreach activities.

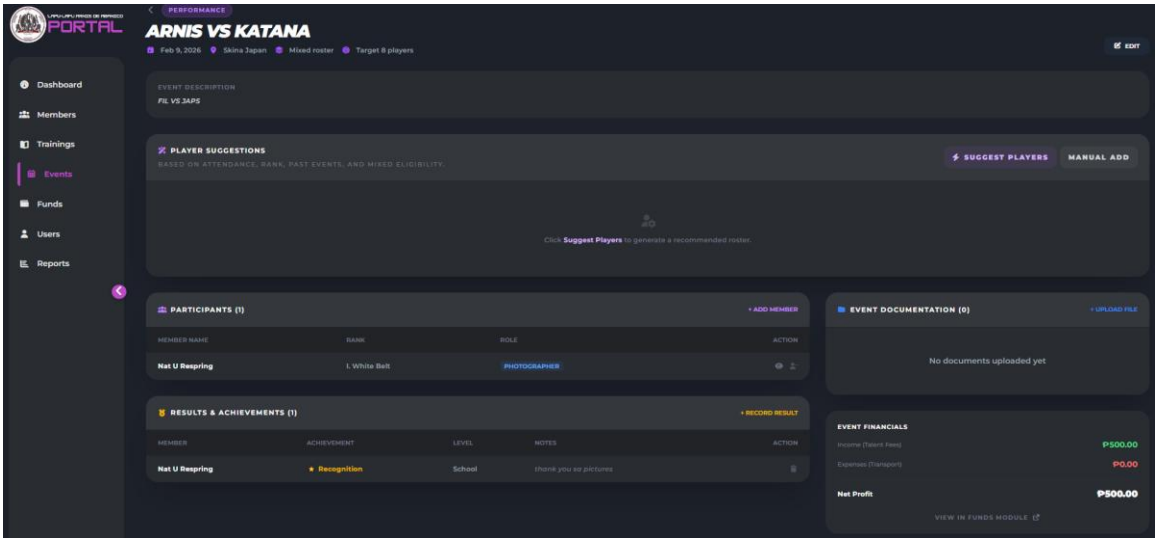


Figure 53
Event Detail Interface

Figure 53 shows the event detail page. This page displays the event name, date, location, target number of players, event description, player suggestions, participants, event documentation,

results and achievements, and event financial summary. This supports organized event preparation and documentation.

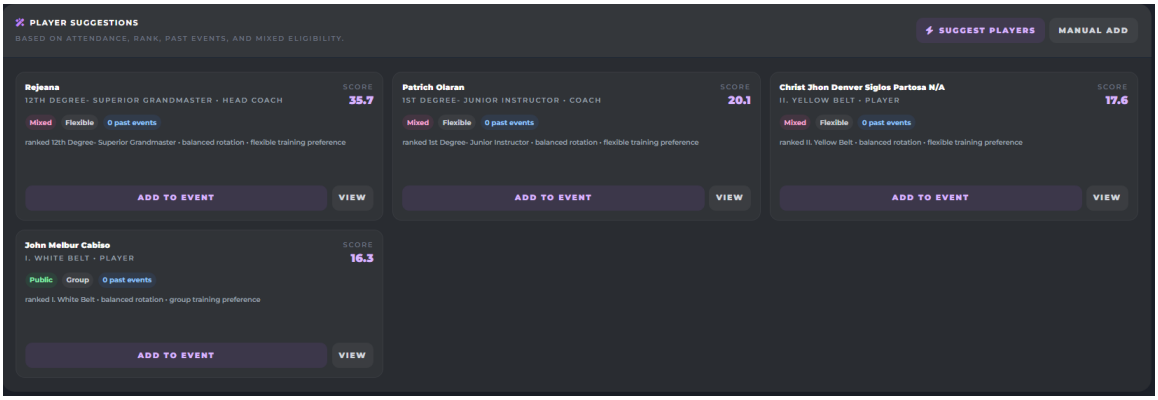


Figure 54
Player Suggestion Results Interface

Figure 54 shows the player suggestion results. The system suggests players based on a scoring method that considers attendance, rank, past events, player category, training preference, and balanced rotation. The coach can still make the final decision by adding or viewing suggested players.

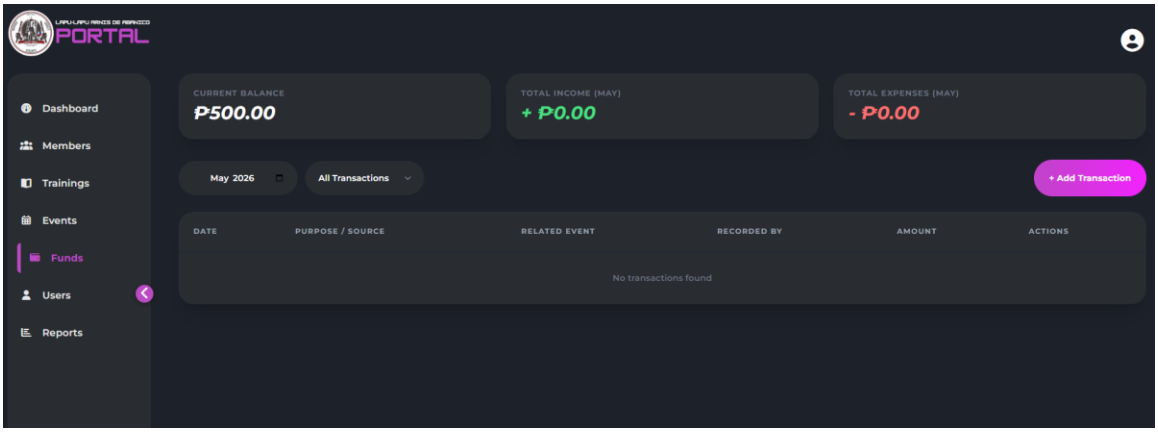


Figure 55
Funds Management Interface

Figure 55 shows the funds management page. This interface displays the current balance, total income, total expenses, transaction filters, transaction list, and add transaction button. It supports the recording and monitoring of organization funds.

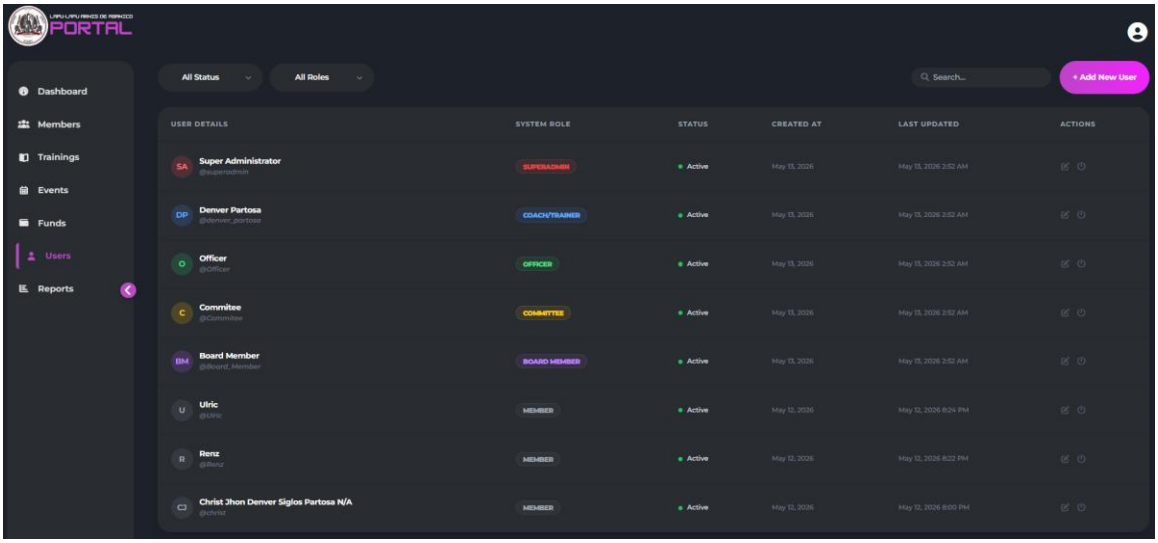


Figure 56
User Management Interface

Figure 56 shows the user management page. The page displays system users, roles, account status, creation date, last update, and actions. This supports role-based access control by managing user accounts such as superadmin, coach/trainer, officer, committee, board member, and member.

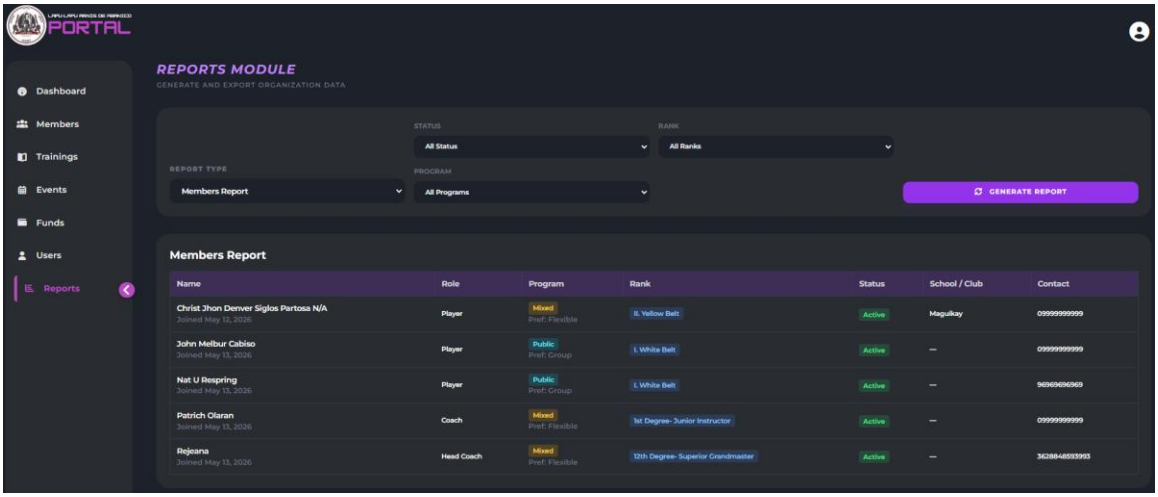


Figure 57
Reports Module Interface

Figure 57 shows the reports module. The interface allows authorized users to generate reports based on report type, status, rank, and program. The report output can display member data

including name, role, program, rank, status, school or club, and contact information. This supports organized reporting and decision-making.

Database Design

Database design defines the logical and physical structure of the data used by Arnis Connect. Since the system stores structured and highly related records such as users, members, trainings, attendance, events, registration requests, and financial transactions, a relational database is the most suitable core design for the project.

Capstone 1 - Relational Db (SQL DB): Conceptual Model (ERD, EERD)

The relational database design of Arnis Connect is represented through an Entity-Relationship Diagram (ERD), which identifies the major entities, attributes, and relationships of the system.

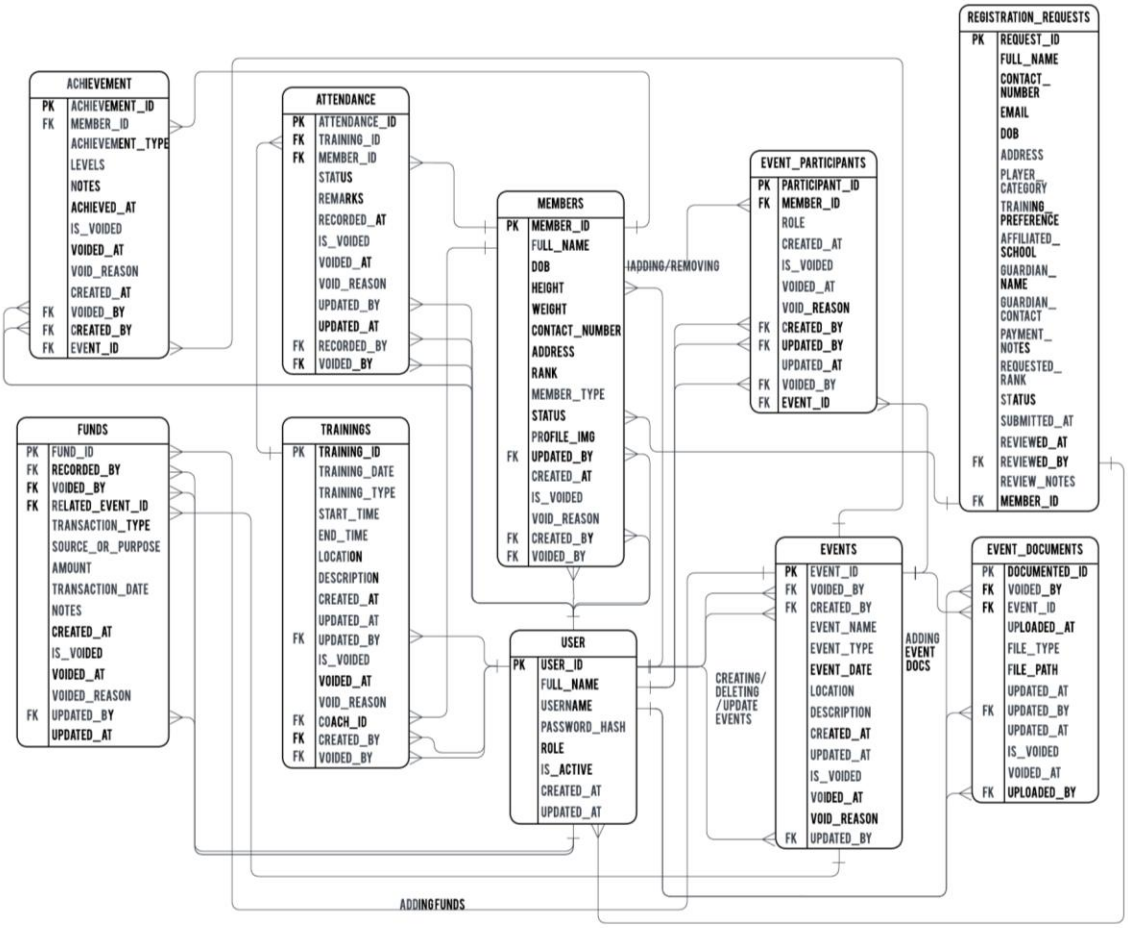


Figure 58
Conceptual Entity-Relationship Diagram of Arnis Connect

The database of Arnis Connect follows a centralized role-based model, wherein all system users are stored in a single USER table that manages authentication credentials, roles, and account status. The system distinguishes user responsibilities through predefined roles rather than separate account tables, enabling administrators, coaches, and finance users to access system functionalities based on role-based permissions. The core operational data of the system is centered on the USERS, MEMBERS, REGISTRATION_REQUESTS, TRAININGS, ATTENDANCE, EVENTS, EVENT_PARTICIPANTS, FUNDS, ACHIEVEMENTS, and EVENT_DOCUMENTS tables.

The USERS table serves as the root entity for system access and administrative actions, storing login credentials, role assignments, and account activity status. System users are responsible for creating, updating, and managing records across multiple modules. The MEMBERS table contains the official records of all registered individuals within the Arnis organization, including personal details, rank, membership type, and status. Each member record is linked to the user who created or last updated it, ensuring accountability and traceability of data modifications.

The REGISTRATION_REQUESTS table handles the intake of new applicants, capturing their personal information, training preferences, and requested rank. Each request is reviewed by a system user, recorded through the reviewed_by field, and may be linked to a corresponding member record upon approval via the member_id field. This establishes a clear lifecycle from applicant to official member while maintaining a record of all submitted requests regardless of outcome.

Training operations are managed through the TRAININGS table, which stores schedules, locations, and assigned coaches. Attendance tracking is implemented through the ATTENDANCE table, which serves as a transactional entity linking members to specific training sessions while recording their attendance status and remarks. This relationship ensures accurate monitoring of participation and performance over time.

Event management is handled through the EVENTS table, which stores details of organizational activities such as tournaments, performances, and outreach programs. The EVENT_PARTICIPANTS table acts as a junction entity that establishes a many-to-many relationship between members and events, allowing multiple members to participate in multiple events. Supporting documentation related to events, such as images, videos, and receipts, is stored in the EVENT_DOCUMENTS table, which maintains file references and upload metadata.

Financial transactions are managed through the FUNDS table, which records both income and expenses, including their source or purpose, associated events, and the user responsible for recording the transaction. This supports financial transparency and reporting within the organization. The ACHIEVEMENTS table captures recognitions earned by members during events, linking both the member and the event to maintain a structured record of accomplishments.

Data Dictionary

Table 14
Data Dictionary

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	FORMAT	RANGE *	REQUIRED	PK/FK	FK REF TABLE
USER	USER_ID	Unique identifier for each user	int(11)	Numeric	Auto increment	Yes	**PK**	
	FULL_NAME	User full name	varchar(100)	Text		Yes		
	USERNAME	Login username	varchar(50)	Text	Unique	Yes		
	PASSWORD_HASH	Encrypted password	varchar(255)	Text		Yes		
	ROLE	User role	enum	Text	Predifined roles	Yes		
	IS_ACTIVE	Account status	tinyint(1)	0/1	0-1	Yes		
	CREATED_AT	Date created	datetime	Timestamp		Yes		
	UPDATED_AT	Date updated	datetime	Timestamp		Yes		
MEMBERS	MEMBER_ID	Unique member id	int(11)	Numeric	Auto increment	Yes	**PK**	
	FULL_NAME	Member name	varchar(100)	Text		Yes		
	DOB	Date of birth	date	YYYY-MM-DD		No		

	HEIGHT	Height	decimal(5,2)	Numeric		No		
	WEIGHT	Weight	decimal(5,3)	Numeric		No		
	CONTACT_NUMBER	Contact number	varchar(20)	Text		No		
	ADDRESS	Address	varchar(255)	Text		No		
	RANK	Arnis rank	enum	Text	Predefined	Yes		
	MEMBER_TYPE	Role type	enum	Text	Predefined	Yes		
	STATUS	Member status	enum	Text	Active/ Inactive	Yes		
	IS_VOIDED	Void flag	tinyint(1)	0/1		Yes		
	CREATED_BY	Creator user	int(11)	Numeric		Yes	**FK**	USERS
	UPDATED_BY	Last updater	int(11)	Numeric		No	**FK**	USERS
	VOIDED_BY	User who voided	int(11)	Numeric		No	**FK**	USERS
REGISTRATION_REQUESTS	REQUEST_ID	Unique request ID	int(11)	Numeric	Auto Increment	Yes	**PK**	
	FULL_NAME	Applicant name	varchar(100)	Text		Yes		
	DOB	Date of birth of applicant	date	YYYY-MM-DD		No		
	EMAIL	Email address of applicant	varchar(100)	Text		No		
	CONTACT_NUMBER	Contact information of applicant	varchar(20)	Text		No		
	ADDRESS	Address of applicant	varchar(255)	Text		No		

	PLAYER_CATEGORY	Player category of applicant	enum('Public','Private','Mixed')	Text	Predefined values	**Yes**		
	TRAINING_PREFERENCE	Preferred training arrangement	enum('Group','Private','Flexible')	Text	Predefined values	**Yes**		
	AFFILIATED_SCHOOL	School or club affiliation of applicant	varchar(150)	Text		**No**		
	GUARDIAN_NAME	Name of guardian	varchar(100)	Text		**No**		
	GUARDIAN_CONTACT	Contact number of guardian	varchar(20)	Text		**No**		
	PAYMENT_NOTES	Notes about payment or program arrangement	varchar(255)	Text		**No**		
	REQUESTED_RANK	Requested initial rank of applicant	enum('None', 'I. White Belt','II. Yellow Belt',etc.	Text	Predefined values	**Yes**		
	STATUS	Status of request	enum('Pending','Approved','Rejected')	Text	Predefined values	Yes		
	SUBMITTED_AT	Date and time the request was submitted	datetime	Timestamp		No		

	REVIEWED_AT	Date and time the request was reviewed	datetime	Timestamp		No		
	REVIEWED_BY	User who reviewed the request	int(11)	Numeric		No	**FK**	USERS
	REVIEW_NOTES	Notes or remarks of reviewer	varchar(255)	Text		No		
	MEMBER_ID	Linked to member	int(11)	Numeric		No	**FK**	MEMBERS
TRAININGS	TRAINING_ID	Training ID	int(11)	Numeric	Auto Increment	Yes	**PK**	
	TRAINING_DATE	Training date	date	YYYY-MM-DD		Yes		
	TRAINING_TYPE	Type of training	enum	Text	Public/ Private	Yes		
	START_TIME	Start time	time	HH-MM-SS		No		
	END_TIME	End time	time	HH-MM-SS		No		
	LOCATION	Location of training	varchar(255)	Text		No		
	DESCRIPTION	Training description	varchar(255)	Text		No		
	COACH_ID	Assigned coach	int(11)	Numeric		Yes	**FK**	MEMBERS
	CREATED_BY	User who created the record	int(11)	Numeric		No	**FK**	USERS
	CREATED_AT	Date created	datetime	Timestamp		Yes		

	UPDATED_AT	Date updated	datetime	Timestamp		Yes		
	UPDATED_BY	User who updated the record	int(11)	Numeric		No	**FK**	USERS
	IS_VOIDED	Void flag	tinyint(1)	0/1		Yes		
	VOIDED_AT	Date voided	datetime	Timestamp		No		
	VOIDED_BY	User who voided	int(11)	Numeric		No	**FK**	USERS
	VOID_REASON	Reason	varchar(255)	Text		No		
ATTENDANCE	ATTENDANCE_ID	Attendance ID	int(11)	Numeric	Auto Increment	Yes	**PK**	
	TRAINING_ID	Training reference	int(11)	Numeric		Yes	**FK**	TRAININGS
	MEMBER_ID	Member reference	int(11)	Numeric		Yes	**FK**	MEMBERS
	STATUS	Attendance status	enum	Text	Present/ Late/ Absent	Yes		
	REMARKS	Remarks	varchar(255)	Text		No		
	RECORDED_BY	User who recorded the attendance	int(11)	Numeric		Yes	**FK**	USERS
	RECORDED_AT	Date recorded	datetime	Timestamp		Yes		
	UPDATED_BY	User who updated record	int(11)	Numeric		No	**FK**	USERS
	UPDATED_AT	Date record last updated	datetime	Timestamp		Yes		
	IS_VOIDED	Void flag	tinyint(1)	0/1		Yes		
	VOIDED_AT	Date voided	datetime	Timestamp		No		

	VOIDED_BY	User who voided	int(11)	Numeric		No	**FK**	USERS
	VOID_REASON	Reason	varchar(255)	Text		No		
EVENTS	EVENT_ID	Event ID	int(11)	Numeric	Auto Increment	Yes	**PK**	
	EVENT_NAME	Event name	varchar(100)	Text		Yes		
	EVENT_TYPE	Event type	enum	Text	Tournament / etc.	Yes		
	EVENT_DATE	Event date	date	YYYY-MM-DD		Yes		
	LOCATION	Location of event	varchar(255)	Text		No		
	DESCRIPTION	Description of event	text	Text		No		
	CREATED_AT	Date event was created	datetime	Timestamp		Yes		
	UPDATED_AT	Date event was last updated	datetime	Timestamp		Yes		
	CREATED_BY	User who created the event	int(11)	Numeric		No	**FK**	USERS
	UPDATED_BY	User who last updated event	int(11)	Numeric		No	**FK**	USERS
	IS_VOIDED	Void flag	tinyint(1)	0/1		Yes		
	VOIDED_AT	Date voided	datetime	Timestamp		No		
	VOIDED_BY	User who voided	int(11)	Numeric		No	**FK**	USERS
	VOID_REASON	Reason	varchar(255)	Text		No		
EVENT_ PARTICIPANTS	PARTICIPANT_ID	Participant ID	int(11)	Numeric	Auto Increment	Yes	**PK**	

	EVENT_ID	Event reference	int(11)	Numeric		Yes	**FK**	EVENTS
	MEMBER_ID	Member reference	int(11)	Numeric		Yes	**FK**	MEMBERS
	ROLE	Role of member in event	varchar(255)	Texr		No		
	CREATED_AT	Date record was created	datetime	Timestamp		Yes		
	UPDATED_AT	Date record was last updated	datetime	Timestamp		Yes		
	CREATED_BY	User who created the record	int(11)	Numeric		No	**FK**	USERS
	UPDATED_BY	User who last updated record	int(11)	Numeric		No	**FK**	USERS
	IS_VOIDED	Void flag	tinyint(1)	0/1		Yes		
	VOIDED_AT	Date voided	datetime	Timestamp		No		
	VOIDED_BY	User who voided	int(11)	Numeric		No	**FK**	USERS
	VOID_REASON	Reason	varchar(255)	Text		No		
FUNDS	FUND_ID	Transaction ID	int(11)	Numeric	Auto Increment	Yes	**PK**	
	TRANSACTION_TYPE	Income/ Expense	enum	Text		Yes		
	SOURCE_OR_PURPOSE	Source of income or purpose of expense	varchar(255)	Text		Yes		

	AMOUNT	Transaction amount	decimal(10,2)	Numeric		Yes		
	RELATED_EVENT_ID	Event related to transaction	int(11)	Numeric		No	**FK**	EVENTS
	RECORDED_BY	Recorded by	int(11)	Numeric		Yes	**FK**	USERS
	CREATED_AT	Date transaction was recorded	datettime	Timestamp		Yes		
	UPDATED_BY	User who updated the record	int(11)	Numeric		No	**FK**	USERS
	UPDATED_AT	Date record was last updated	datetime	Timestamp		No		
	IS_VOIDED	Void flag	tinyint(1)	0/1		Yes		

System Architecture Design

The Arnis Connect system uses a centralized browser-based architecture in which all major modules are integrated into a single web application connected to a shared database. This structure supports the core operational needs of the Lapu-Lapu Arnis de Abanico Organization by organizing the system into functional areas such as user and access management, member and application management, training and automated attendance, event management, player suggestion, belt progression, funds management, reports generation, and audit logging. Because these modules use a common database, data entered or updated in one module becomes immediately available to other authorized modules. For example, once an applicant is approved in the application management module, the record is stored as an official member and can then be accessed by the attendance, belt progression, event, and reports modules.

The architecture also incorporates the emerging technologies required by the proposed system. Automated attendance is supported through facial recognition and geolocation verification. The member device accesses the system through a browser with camera and location permissions enabled. Verification codes for initial registration may be delivered through an email service or similar messaging service in production deployment. In addition, the player suggestion function can use stored attendance, rank, event participation, training preference, and member category data to provide better recommendations for event preparation. The overall architecture is therefore designed not only for routine record management but also for secure verification, data sharing, and decision support across the organization.

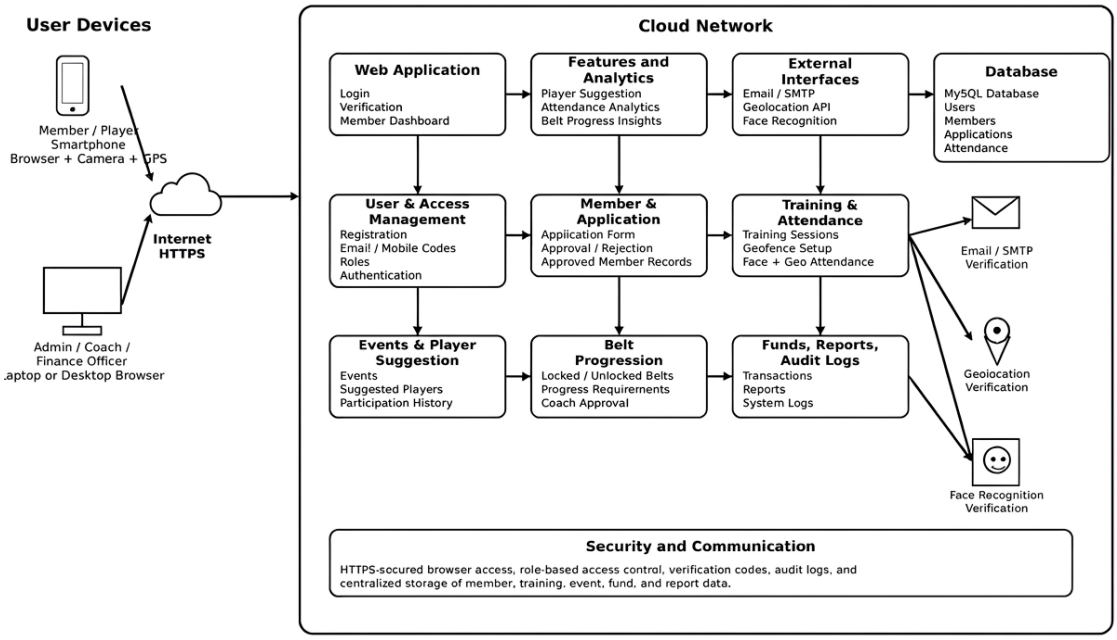


Figure 59
Arnis Connect Cloud-Based System Architecture Diagram

Network Design

The network design of Arnis Connect follows a web-based client-server arrangement that is suitable for both local deployment and online hosting. The client access layer consists of public or private players, coaches, administrators, finance officers, and other authorized users who open the system through a web browser using a smartphone, tablet, laptop, or desktop computer. Since the system includes automated attendance using facial recognition and geolocation, member devices are expected to support camera access and location access. Communication between clients and the application is performed through secure web requests, while the application layer handles user authentication, verification, member application processing, training and attendance operations, events and player suggestions, funds, belt progression, and reports.

At the lower layers of the design, the web server runs the application logic while a centralized MySQL database stores operational records such as users, applications, members, trainings, attendance logs, events, fund transactions, and reports. External or integrated services may be used for verification code delivery, browser-based geolocation, and client-side face recognition support. This layered arrangement helps maintain organization in the system, simplifies maintenance, and ensures that the different modules of Arnis Connect remain connected through a unified and secure data environment.

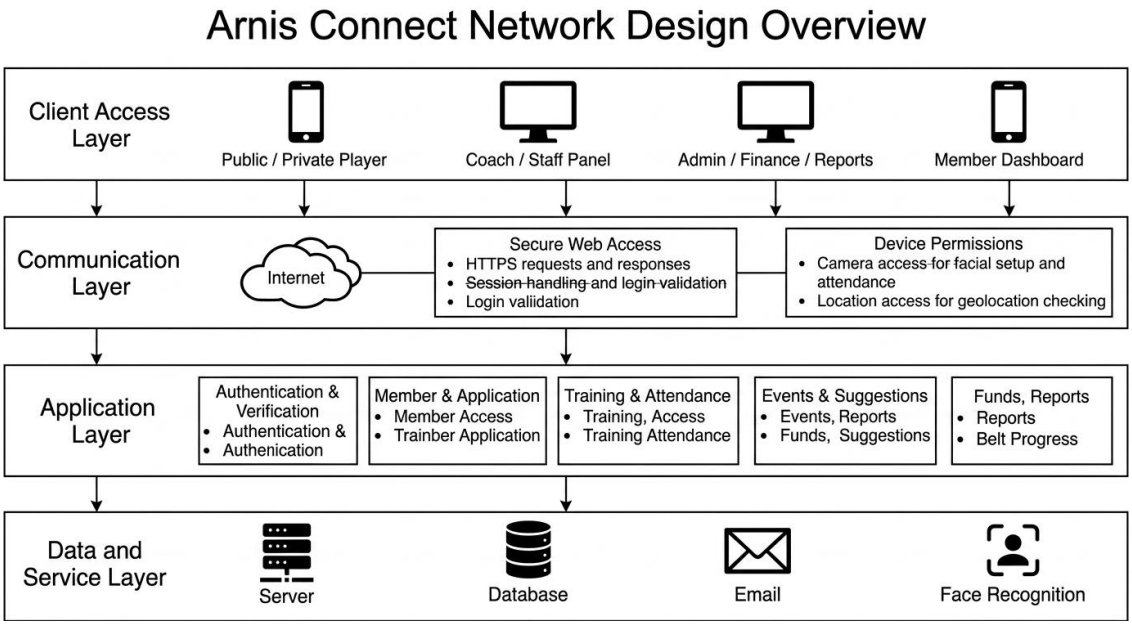


Figure 60
Arnis Connect Network Design Overview

Network Topology

The topology of Arnis Connect can be described as an internet-based star topology with a centralized application server as the main service node. All client devices communicate with the Arnis Connect web server through HTTPS connections, and the web server serves as the central point for processing requests, enforcing user authentication, and coordinating access to the database and integrated services. Member devices connect to the system to register accounts, complete the application form, access the member dashboard, enroll a face profile, and submit automated attendance. Coach and staff devices connect to manage trainings, review attendance, manage members, generate event suggestions, and update belt progression. Administrative and finance users access the same central server to manage funds, users, and reports.

The centralized topology is appropriate for Arnis Connect because it makes system monitoring easier and allows all major functions to remain synchronized in real time. The MySQL database is directly connected to the web server for storing the official records of the organization. The server also interacts with supporting services such as geolocation interfaces, facial recognition support, verification email services, and security controls. Through this topology, Arnis Connect can provide consistent access to all authorized users while preserving data integrity, confidentiality, and availability during daily operation.

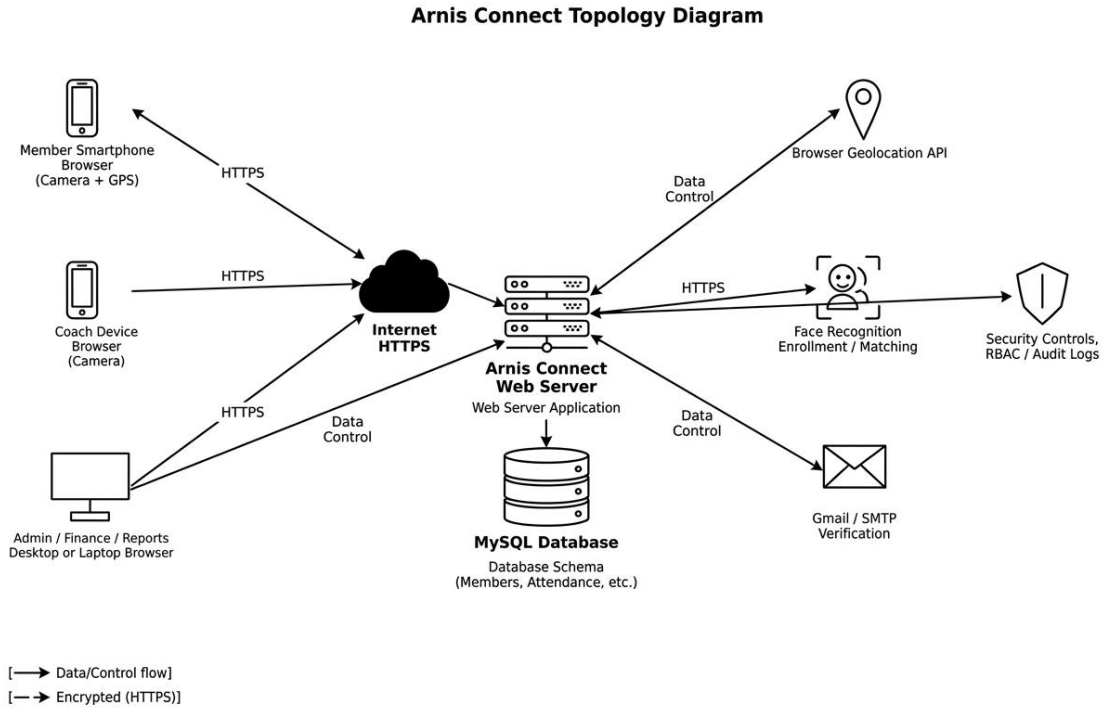


Figure 61
Arnis Connect Topology Diagram

Network Security

The network security design of Arnis Connect was updated to match the revised system features, especially member access, email and mobile verification, automated attendance using facial recognition and geolocation, player suggestion, belt progression, finance monitoring, and role-based access control. Since the system stores personal information, attendance records, location-related attendance data, face profile data, event records, and financial transactions, security controls must be applied across the client device, web application, database, and hosting environment.

Arnis Connect uses secure login authentication, encrypted password storage, verification code checking, role-based access control, session management, input validation, HTTPS communication, audit logging, and backup procedures. For attendance, the system requires the member device to allow camera and location permission so that facial recognition and geofence checking can be performed before attendance is accepted. If the face check or location check fails, or if the attendance appears suspicious, the record should be flagged for coach review instead of

being automatically accepted. This approach supports security while recognizing that no browser-based geolocation system can fully prevent all spoofing attempts.

Table 15
Network Security Table for Arnis Connect

Security Control	Updated Mechanisms	Requirement Reference
Authentication	Username or email and password login with encrypted password storage. Internal users and members must authenticate before accessing protected modules.	SRS - Login and Session Management
Email and Mobile Verification	New member accounts must complete email and mobile verification before the applicant can proceed to the application form.	SRS - Registration and Verification
Authorization (RBAC)	Role-based access control for Admin or Lead Coach, Coach, Finance Officer, Member or Player, and Applicant. Each role can only access allowed modules and actions.	SRS - User Role Management
Password Security	Password hashing, minimum password length, confirmation checking, and no plain-text password display.	SRS - Security Requirements
Session Management	Login sessions, logout function, session timeout, and session regeneration after successful login.	SRS - System Security
Application Approval Security	The approve and reject buttons must update the application status and database records. Duplicate approval of the same applicant must be prevented.	SRS - Registration Management
Attendance Verification	Facial recognition, geolocation distance checking, timestamp logging, and coach review for flagged records.	SRS - Automated Attendance Management
Geolocation Control	The system records latitude, longitude, distance from training location, and geofence status based on the training venue and radius.	SRS - Training and Attendance
Face Profile Protection	Face enrollment and matching data should be limited to authorized attendance verification	SRS - Biometric Data Protection

	functions and protected from unauthorized access.	
Input Validation	All forms, filters, date fields, IDs, and financial inputs must be validated and sanitized before processing.	SRS - Data Validation
SQL Injection Protection	Database operations should use prepared statements or parameterized queries to reduce injection risk.	SRS - Database Security
Transport Security	The deployed system should use HTTPS/TLS to protect credentials, verification codes, attendance data, and finance data during transmission.	SRS - Network Security
Audit Logging	Important actions such as login, approval, member update, attendance verification, finance recording, event update, and report generation are recorded in audit logs.	SRS - Monitoring and Control
Backup and Recovery	The database should be backed up regularly to prevent loss of user, member, attendance, event, and financial records.	SRS - Data Protection
Error Handling	The system should display controlled error messages without exposing database credentials, SQL details, server paths, or internal code information.	SRS - System Reliability

Arnis Connect Network Security Architecture

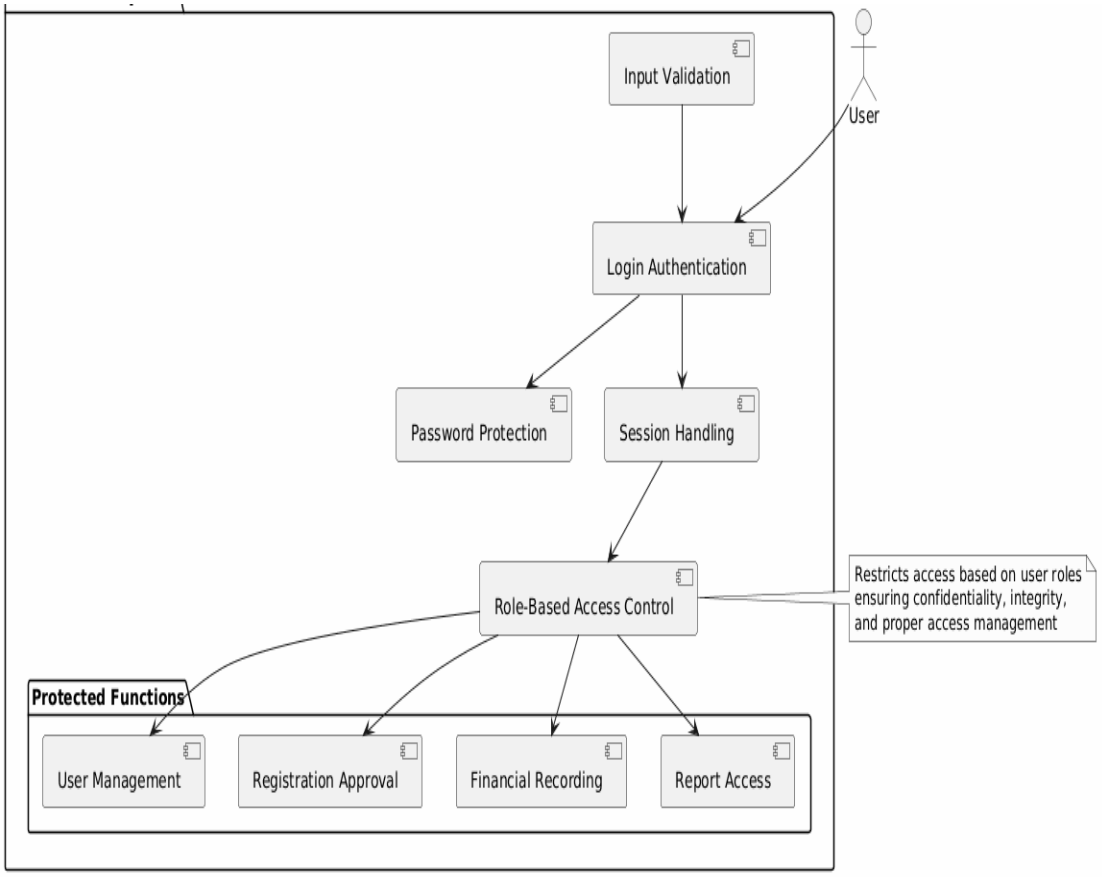


Figure 62
Network Security Figure for Arnis Connect

Network Devices

Arnis Connect is designed to operate through a hosted web environment with minimal physical infrastructure required from the organization. The network devices include the client devices used by applicants, members, coaches, the Finance Officer, and the Admin or Lead Coach, as well as the hosting server, database server, internet connection, browser permissions, and supporting services for verification and backups.

Table 16
Network Devices and Infrastructure Components for Arnis Connect

Component	Specification	Role in Arnis Connect
Member / Player Device	Smartphone, tablet, laptop, or desktop with browser, camera access, and location services.	Used for login, member dashboard, face setup, attendance, belt progress, and event participation viewing.
Applicant Device	Any internet-enabled device with a modern browser and email/mobile access.	Used for account registration, verification, and application form submission.
Coach Device	Laptop, desktop, tablet, or smartphone with a modern browser.	Used to schedule trainings, set geofence details, monitor attendance, review flagged records, manage events, and update belt progress.
Admin / Lead Coach Device	Laptop or desktop recommended; tablet also supported.	Used for member approval, user control, reports, system monitoring, and administrative functions.
Finance Officer Device	Laptop, desktop, or tablet with modern browser.	Used for income, expense, event fund, and financial report management.
Internet Router / Mobile Data	Wi-Fi router, wired broadband, or mobile data connection.	Provides connectivity between users and the hosted web system.
Web Hosting Server	PHP and Apache/Nginx-capable hosting environment.	Hosts the Arnis Connect application files and processes system requests.
MySQL Database Server	Hosted MySQL database managed through phpMyAdmin or hosting database manager.	Stores users, verification records, applications, members, trainings, attendance logs, events, funds, reports, and audit logs.
Email / SMTP Service	Gmail SMTP or hosting-provided email relay when available.	Delivers verification codes and system-related email notices.
Browser Camera and Geolocation APIs	Built-in browser services requiring user permission.	Supports face enrollment, facial verification, location capture, and geofence attendance checking.
Backup Storage	Hosting backup storage, downloaded SQL backup, or secure external backup storage.	Keeps recoverable copies of the database and important uploaded files.

DEVELOPMENT PHASE

This phase involves translating the system design into a working software application. In Arnis Connect, this includes the coding of the modules, configuration of the development environment, database implementation, and integration of frontend and backend components. The development phase produces the actual web-based system that will later be tested and deployed.

Programming Environment

Arnis Connect is developed in a local development environment using each team member's personal computer, with system files organized and optionally synchronized using version control tools such as GitHub. The system is a web-based application developed and tested through standard web browsers, while the backend and database are managed locally using XAMPP before deployment to an online hosting server.

The programming environment of Arnis Connect provides the necessary tools and technologies for efficient system development, testing, and maintenance. It ensures that all software components including frontend, backend, and database are compatible, properly integrated, and aligned with the system's architectural and functional requirements.

Table 16
Programming Environment Components

Programming Environment Component	Technology / Tool	Purpose
Programming Language	PHP, HTML, CSS, JavaScript, SQL	Used to develop the backend logic, frontend interface, client-side interaction, and database operations of the web-based system.
Development Tools / IDE	Visual Studio Code, XAMPP, phpMyAdmin, Google Chrome, Microsoft Edge, draw.io / Lucidchart	Used for coding, local server setup, database management, browser testing, debugging, and system diagram creation.
Frontend Frameworks and Libraries	Bootstrap, DataTables, SweetAlert, Chart.js	Used for responsive layout, enhanced tables, user-friendly confirmation messages, and dashboard visualization.
Emerging Technology Support	Browser Geolocation API, MediaDevices	Used to support automated attendance through geolocation

	Camera API, face recognition JavaScript library or model support	verification and facial recognition enrollment or matching.
Build and Dependency Management	Modular file structure, local assets, selected external libraries	Used to organize system files and manage lightweight dependencies suitable for a capstone-level PHP project.
Version Control	Git, GitHub, organized backup folders	Used for tracking code changes, collaboration, backup, and version management.
Database Administration	MySQL, phpMyAdmin	Used to create tables, manage records, import SQL files, check relationships, and verify data storage.

Programming Language

Arnis Connect is developed using a combination of web-based programming languages that support both frontend and backend system operations. PHP serves as the primary backend language, handling server-side processing such as user authentication, session management, role-based access control, data validation, and execution of core system functionalities including attendance tracking, event management, and financial recording. It also manages communication between the application and the MySQL database.

On the frontend, HTML is used to structure the content and layout of web pages, while CSS is applied to control the visual design, responsiveness, and overall user interface consistency across devices. JavaScript enhances interactivity by enabling dynamic content updates, real-time form validation, and improved user experience during system interaction. SQL is used for database operations, allowing efficient storage, retrieval, and manipulation of structured data within the MySQL database. Together, these languages provide a complete environment for building a functional and responsive web-based system.

Development Tools

The development of Arnis Connect uses Visual Studio Code as the main code editor because it supports PHP, HTML, CSS, JavaScript, and SQL files. XAMPP is used as the local development environment because it provides Apache and MySQL in one package. phpMyAdmin is used to manage database tables, import SQL files, and verify stored records. Google Chrome and

Microsoft Edge are used for browser testing because the system relies on modern browser support for camera, geolocation, responsive layout, and form behavior.

Diagramming tools such as draw.io or Lucidchart are used to produce system diagrams, including the system context diagram, user flow diagrams, functional decomposition diagram, use case diagrams, sequence diagrams, activity diagram, database design, system architecture, network design, topology, and security diagrams.

Build and Dependency Management

The project follows a lightweight approach to build and dependency management suitable for a capstone-level web application. The system does not rely on complex build automation tools; instead, it uses a modular file structure where components are organized by functionality. External resources such as CSS stylesheets and JavaScript libraries are directly linked to the project and are carefully integrated to ensure compatibility and performance.

Dependencies are minimized to maintain system simplicity, reduce potential errors, and ensure ease of deployment in a shared hosting environment. Any added libraries are tested thoroughly to ensure they do not conflict with existing components and that they enhance the overall system functionality without introducing unnecessary complexity.

Version Control

Version control in Arnis Connect is maintained through organized file management and systematic backup practices. Development files are structured clearly to track changes and updates throughout the project lifecycle. The team may also utilize version control systems such as Git or platforms like GitHub to manage code revisions, track development progress, and support collaboration among team members.

This approach allows developers to revert to previous versions if needed, maintain a history of changes, and ensure that code updates are properly documented and synchronized among contributors.

Frameworks and Libraries

Arnis Connect uses selected frameworks and libraries to improve usability, responsiveness, and monitoring. Bootstrap supports a consistent layout across desktops, laptops, tablets, and smartphones. DataTables improves record viewing through searching, sorting, and pagination. SweetAlert provides clearer confirmation messages for actions such as approval, rejection,

deletion, and saving records. Chart.js may be used for summary charts related to members, attendance, events, and funds.

For the attendance feature, the system may use browser-based camera access and a facial recognition library or model for face enrollment and matching. The browser geolocation function is used to check whether a member is within the allowed training location radius before attendance is accepted. These technologies are integrated carefully because they depend on device permission, browser support, internet connection, and user consent.

List of Modules

The major modules of Arnis Connect include:

1. Login and Session Management
2. Account Registration and Verification
3. Member Application and Approval Workflow
4. User and Role Management
5. Member / Player Management
6. Training Management
7. Automated Attendance Management
8. Face Recognition Enrollment and Verification
9. Geolocation and Geofence Validation
10. Belt Progression Management
11. Event Management
12. Player Suggestion Analytics
13. Finance Management
14. Reports Generation
15. Audit Logs and Security Monitoring

Table 17
List of Modules for Arnis Connect

Module	Description
Login and Session Management	Handles user login, logout, session control, password verification, and redirection to the correct dashboard based on role.
Account Registration and Verification	Allows new applicants to create an account using username, password, email/Gmail, and mobile number, then verify using email and mobile verification codes.
Member Application and Approval Workflow	Allows verified applicants to fill out the application form and allows the Administrator or Lead Coach to approve or reject applications. Approved applicants are stored as official members.
User and Role Management	Allows authorized users to manage system accounts, roles, access levels, and account status.
Member / Player Management	Manages member records, public/private player classification, ranks, contact details, status, and training preference.
Training Management	Creates and manages training sessions, schedules, coach assignments, location details, and geofence settings.
Automated Attendance Management	Records attendance using facial recognition, geolocation checking, timestamps, verification status, and coach review for flagged records.
Face Recognition Enrollment and Verification	Allows members to enroll a face profile and allows the system to compare the captured face during attendance.
Geolocation and Geofence Validation	Checks whether the member is within the allowed training radius before attendance is accepted.
Belt Progression Management	Tracks current belt, completed requirements, locked and unlocked levels, progress percentage, and coach assessment.
Event Management	Creates and manages events, event details, participants, documents, results, and related records.
Player Suggestion Analytics	Uses attendance, belt level, player category, event type, achievements, availability, and participation history to support player selection.
Finance Management	Records income and expenses, links transactions to events when needed, and supports financial monitoring.
Reports Generation	Generates member, attendance, event, finance, and summary reports for decision-making.
Audit Logs and Security Monitoring	Stores important user actions such as login events, approvals, updates, attendance submissions, and financial transactions for accountability.

Database Stack

The database stack of Arnis Connect is built around MySQL as the primary relational database management system, selected for its reliability, performance, and compatibility with PHP-based web applications. MySQL supports structured data storage, relational integrity through foreign key constraints, and efficient query execution for handling system operations such as member management, attendance tracking, event participation, and financial recording.

The system utilizes PHP database connectors (such as MySQLi or PDO) to enable secure communication between the application layer and the database. These connectors support prepared statements and parameterized queries, helping prevent SQL injection attacks while ensuring efficient data transactions. Database operations are optimized through structured queries and indexing of key fields to maintain system performance even as data volume increases.

For development and administration, phpMyAdmin is used as the primary database management tool, allowing developers to create tables, define relationships, execute queries, and monitor database performance. In the production environment, the system relies on a hosted MySQL database provided by the web hosting service, which typically includes built-in features such as automated backups, access control, and storage management. This setup ensures data reliability, accessibility, and maintainability throughout the system lifecycle.

Testing Tools (Developer Side)

The development-side testing of Arnis Connect is primarily conducted through manual and functional testing approaches to verify system correctness and usability. Browser-based testing is performed using modern web browsers such as Google Chrome and Microsoft Edge to evaluate user interface behavior, responsiveness, and navigation across different devices and screen sizes.

Form validation checks are implemented and tested to ensure that all user inputs meet the required format and constraints before being processed by the system. Module walkthroughs are conducted for each major system component, including login, registration approval, member management, attendance recording, event management, and finance tracking, to confirm that each module performs according to the defined requirements.

Developers also use issue logging sheets to document identified bugs, errors, and system inconsistencies during testing. This structured approach allows the team to track issues, assign fixes, and verify resolutions efficiently. Additionally, database verification is performed using

phpMyAdmin to ensure that data is correctly stored, updated, and retrieved. This includes checking table relationships, validating record accuracy, and confirming that database transactions reflect expected system behavior.

Overall, these testing tools and methods ensure that the system functions reliably, maintains data integrity, and meets the intended operational requirements before deployment.

Configuration

The configuration of Arnis Connect includes the setup of essential system parameters required for both development and deployment environments. This includes database connection settings such as host, username, password, and database name, which enable the application to securely connect to the MySQL database. Application URL configuration ensures proper routing and accessibility of the system when deployed on a hosting platform.

Session management settings are configured to control user sessions, including login persistence, session timeouts, and secure handling of authentication data. Environment parameters are defined to distinguish between development and production environments, allowing developers to enable debugging features locally while maintaining security and performance in the deployed system.

Hosting-related configurations include file deployment to the web server, database importation, and adjustment of server settings to ensure compatibility with PHP and MySQL. These configurations ensure that the system runs consistently, securely, and efficiently across different environments.

Table 18
Configuration Specifications for Arnis Connect

Configuration Type	Description
Backend Config	PHP configuration files and environment setup (database connection script using config.php or .env equivalent) to handle server-side logic and database connectivity
Frontend Config	HTML, CSS, and JavaScript configuration for UI behavior, layout responsiveness, and client-side interactions
Database Credentials	MySQL database host, username, password, and database name stored securely in configuration files for system access
Authentication Config	Session-based authentication setup including login validation, session timeout, and role-based access control (RBAC)
Deployment Config	Web hosting environment settings including domain configuration, file upload to server, and database import via hosting control panel (e.g., cPanel or phpMyAdmin)
Application URL Config	Base URL configuration to ensure proper routing and accessibility of the system when deployed online
Session Management Config	Settings for session handling such as session lifetime, secure login persistence, and logout behavior
Environment Config	Separation of development and production settings (e.g., enabling debug mode locally and disabling it in live deployment)

Operating System

The development environment of Arnis Connect is compatible with Windows 10 or Windows 11, which provide support for the required development tools such as XAMPP, Visual

Studio Code, and web browsers. These operating systems offer a stable platform for running the local web server, managing files, and executing development tasks.

On the deployment side, the hosting environment typically operates on a Linux-based server, which supports PHP and MySQL and ensures efficient handling of web requests. Client devices accessing the system may run various operating systems, including Windows, Android, and iOS, as long as they support modern web browsers.

Table 19
Operating System for Arnis Connect

Component	Specification	Notes
Development OS	Windows 10 / Windows 11	Supports XAMPP, Visual Studio Code, and web browsers for development
Server OS (Hosting)	Linux-based OS	Managed by hosting provider; supports PHP and MySQL
Client OS	Windows, Android, iOS	Any OS capable of running modern web browsers

Hardware Requirements

The minimum hardware requirement for developing Arnis Connect includes a desktop computer or laptop with sufficient processing power, memory, and storage to run the local development environment. A minimum of 4GB RAM is recommended to ensure smooth operation of the web server, database system, code editor, and browser-based testing tools. Adequate storage is required to accommodate project files, database backups, and development resources.

For deployment, the system relies on a hosted server environment, eliminating the need for high-end hardware on the client side. Users only require standard devices such as desktops, laptops, tablets, or smartphones with internet access and a supported web browser. A stable internet connection is essential to ensure reliable system access, especially for real-time operations such as attendance recording, event management, and report generation.

Table 20
Hardware Requirements for Arnis Connect

Component	Specification	Notes
Development Device	Desktop or Laptop	Used for coding, testing, and database management
Memory (RAM)	Minimum 4GB	Ensures smooth operation of local server and tools
Storage	Sufficient disk space	For project files, backups, and resources
Server Hardware	Hosted environment	Managed by hosting provider (no physical server needed)
Client Devices	Desktop, Laptop, Tablet, Smartphone	Used by admins, coaches, and users
Internet Requirement	Stable connection	Required for system access and real-time operations

TESTING PHASE

This section represents the testing phase of the Arnis Connect capstone project, conducted after all major system modules have been developed and integrated. The testing phase is a systematic process of evaluating the system to verify that it behaves according to the defined Software Requirements Specification (SRS) and meets the functional and usability expectations of its intended users. Software testing plays a critical role in ensuring system quality, reliability, and performance by identifying errors and validating system behavior under different conditions.

The testing process for Arnis Connect is structured into four levels: unit testing, integration testing, system testing, and acceptance testing. Each level builds upon the previous stage, ensuring that the system is validated from individual components up to the fully integrated operational environment.

Unit Testing

Unit testing is conducted to assess the individual modules of Arnis Connect separately before they are integrated with the whole system. Each module is tested on its own to verify if it performs according to its intended function and produces the correct output. This testing focuses on important functions such as login and session handling, account verification, registration approval, member management, training management, automated attendance using facial recognition and geolocation, event management, player suggestion, belt progression, finance recording, and report generation. Unit testing allows the development team to identify and correct errors at the module level before proceeding to integration testing.

Table 21
Unit Testing for Arnis Connect

Component / Module	Test Focus	Expected Outcome	Tools / Method
Login and Session Management	Validate username/email, password, login, logout, and session handling.	Valid users can log in; invalid users are rejected; sessions are controlled properly.	Manual testing, browser testing
Account Registration and Verification	Check account creation, email/mobile fields, code generation, and verification process.	Verified applicants can proceed to login and application form.	Form testing, workflow testing
Application Approval Workflow	Check pending requests, approve button, reject button, and duplicate approval prevention.	Approved applicants are stored as official members and cannot be approved twice.	Workflow testing, database check
Member Management	Create, read, update, search, and filter member records.	Member data is accurately saved and displayed.	Form testing, phpMyAdmin verification
Face Recognition Setup	Check camera access, face enrollment, and stored face profile status.	Member can enroll or update face profile when permission is allowed.	Browser camera testing
Geolocation Validation	Check location permission, latitude/longitude	System identifies whether the member is	Browser geolocation testing

	capture, radius, and distance calculation.	within the training geofence.	
Automated Attendance	Submit attendance using face and location verification.	Attendance is recorded when valid and flagged when verification fails.	Functional testing, DB check
Training Management	Create and update training sessions with schedule, location, coach, and geofence.	Training sessions are properly saved and available to members.	Manual testing
Belt Progression	Check current belt, locked/unlocked next belt, progress, and coach updates.	Member progress is displayed correctly and updated by authorized users.	Functional testing
Event and Player Suggestion	Create events and generate suggested players.	Suggested players are produced based on defined criteria and final selection remains with coach.	Module testing
Finance Management	Record income, expenses, event-linked transactions, and balances.	Financial records are accurately stored and reflected in summaries.	DB verification
Reports and Audit Logs	Generate reports and check logs for key activities.	Reports display accurate data and logs record important actions.	Report validation

Integration Testing

Integration testing evaluates how different modules of Arnis Connect interact with one another to ensure seamless data flow and functional consistency across the system. This phase is conducted after individual modules have passed unit testing.

Table 22
Integration Testing for Arnis Connect

Integration Area	Test Focus	Expected Outcome	Method
Registration -> Verification -> Login	Verified account should allow the applicant to log in.	Only verified applicants can proceed properly to the next step.	Workflow testing
Verified Account -> Application Form	Verified applicant completes and submits member application.	Application is saved as pending with verified email/mobile details.	End-to-end testing
Application Approval -> Members	Approved application becomes an official member record.	Member record is created or linked without duplication.	Database validation
Members -> Face Setup	Approved member enrolls face profile.	Face enrollment status is linked to the correct member.	Browser testing
Training -> Geofence -> Attendance	Training location and radius are used during attendance.	Attendance accepts or flags record based on location validation.	Functional testing
Face Setup -> Attendance	Face profile is used during attendance verification.	Attendance includes face verification status.	Module integration testing
Attendance -> Reports	Attendance records appear in reports and dashboards.	Reports reflect accurate attendance information.	Report validation
Events -> Player Suggestion	Event details use member records and activity history for suggestions.	Suggested players are generated using correct member data.	Functional testing
Finance -> Reports	Income and expense transactions appear in summaries.	Financial reports match stored transactions.	Data validation
User Roles -> Module Access	Role-based access controls feature visibility and permissions.	Users can access only authorized modules.	Role testing

System Testing

System testing involves evaluating the complete and fully integrated Arnis Connect system to ensure that it meets all functional and non-functional requirements. This phase simulates real-world usage scenarios to assess the system's overall performance, usability, and reliability.

Table 23
System Testing for Arnis Connect

Test Area	Description	Expected Outcome	Method
Role-Based Access	Verify permissions for Administrator/Lead Coach, Coach, Finance Officer, Member/Player, and Applicant.	Users access only allowed features based on their role.	Role testing
Registration and Application Flow	Test account registration, verification, login, application form, and approval workflow.	Applicant can complete the full process and approved records are stored correctly.	End-to-end testing
Automated Attendance	Test attendance using camera, face recognition, geolocation, and geofence validation.	Valid attendance is recorded and suspicious records are flagged.	Browser and functional testing
Navigation and UI	Test system menus, buttons, forms, dashboards, and responsiveness.	Interface is understandable and works across supported devices.	Manual testing
Data Processing	Validate data entry, retrieval, updates, and reports.	System displays accurate and updated records.	Functional testing
Compatibility	Test on Chrome, Edge, Firefox, Safari, desktop, and mobile browsers.	System remains usable across supported browsers and devices.	Cross-browser testing

Performance	Observe loading of dashboards, records, reports, and attendance submission.	System responds within an acceptable time.	Observation
Error Handling	Submit invalid or incomplete inputs.	Proper messages are displayed and invalid records are not saved.	Input testing
Security Behavior	Test unauthorized access attempts and session handling.	Protected pages cannot be accessed without proper login and role.	Security testing

Acceptance Testing

Acceptance testing is conducted with intended users of Arnis Connect to determine whether the system is practical, understandable, and acceptable for actual use. Users may include the Administrator or Lead Coach, coaches, finance officer, and selected members or players. Feedback is gathered to determine whether the system supports the workflow of the organization.

Table 24
Acceptance Testing for Arnis Connect

Test Scenario	User Role	Task	Expected Outcome	Result Basis
Login to System	All roles	Enter credentials and access dashboard.	Successful login and correct dashboard redirection.	User feedback
Register and Verify Account	Applicant	Create account and enter verification codes.	Verified account is created.	Workflow validation
Submit Application	Applicant	Fill out and submit member application form.	Application is stored as pending.	Task completion
Approve Application	Admin / Lead Coach	Approve pending request.	Applicant becomes an official member in database.	Database validation

Manage Members	Admin / Lead Coach	Add, edit, view, or update member information.	Member data is saved correctly.	Task completion
Schedule Training	Coach	Create training with location and geofence.	Training appears to members and supports attendance.	Workflow validation
Take Attendance	Member / Player	Use face and location verification during training.	Attendance is recorded or flagged for review.	Accuracy check
Review Attendance	Coach	Review valid and flagged attendance records.	Coach can confirm records as needed.	User feedback
Manage Events and Suggestions	Coach / Admin	Create event and generate suggested players.	Suggested roster supports event preparation.	Usability check
Update Belt Progress	Coach	Update belt requirements and progression status.	Member progress is updated correctly.	Task completion
Record Finance	Finance Officer	Record income and expenses.	Financial records and balances update correctly.	Data validation
Generate Reports	Admin / Finance Officer	Generate member, attendance, event, and finance reports.	Correct report output is displayed.	Report validation

This stage focuses on validating whether the system meets user expectations and organizational requirements. Users are asked to perform typical tasks such as logging in, managing members, recording attendance, organizing events, and generating reports. Feedback is collected to assess usability, functionality, and overall satisfaction with the system.

Any issues identified during acceptance testing are reviewed and addressed before final deployment. This phase ensures that the system is practical, user-friendly, and ready for implementation within the Arnis organization.

DEPLOYMENT PHASE

This section presents the deployment phase of Arnis Connect. In this phase, the fully developed and tested system is prepared for use in its operational environment. Deployment includes uploading system files to a hosting server, importing the database, configuring the database connection, enabling browser-accessible routes, and verifying that user roles, member registration, attendance verification, finance records, and reports work properly in the live environment.

Operational Environment

The operational environment defines the hardware, software, network, and security conditions needed for Arnis Connect to function reliably. The system is designed as a web-based platform that can be accessed through standard browsers on desktops, laptops, tablets, and smartphones. It is intended for the Lapu-Lapu Arnis de Abanico Organization to manage member records, application approval, trainings, attendance, events, belt progression, funds, reports, and system logs in one centralized platform.

Hardware Requirements

Table 25
Hardware Specifications for Arnis Connect

Component	Specification	Notes
Server (Hosting)	Managed web hosting server	Provided by the hosting provider; no physical server is required by the organization.
Client Devices	Desktop, laptop, tablet, smartphone	Used by administrators, coaches, finance officer, members, and applicants.
Client Memory	Minimum 4GB RAM recommended	Supports smooth browser operation and system interaction.
Camera	Required for member attendance devices	Needed for face enrollment and facial recognition attendance.
Location Capability	GPS or browser-supported location service	Needed for geolocation and geofence validation.
Internet Device	Router, Wi-Fi, or mobile data	Required for system access and real-time updates.

The deployment of Arnis Connect uses a hosted server environment instead of a local physical server, which reduces hardware and maintenance needs for the organization. The hosting server handles the system processes and database transactions, while users only need an internet-enabled desktop, laptop, tablet, or smartphone with a modern web browser. For members who will use the automated attendance feature, the device must also support camera and location access for facial recognition and geolocation verification.

Operating System

Table 26
Operating System Requirements for Arnis Connect

Component	Specification	Notes
Server OS	Linux-based (via hosting provider)	Managed automatically by hosting service
Client OS	Windows, Android, iOS, macOS	Any OS supporting modern browsers

The hosting environment typically runs on a Linux-based operating system, which supports PHP and MySQL and is commonly used in web hosting platforms due to its stability and efficiency. This environment ensures compatibility with the system's backend technologies.

Client devices may operate on various platforms, including Windows, Android, iOS, or macOS, as the system is browser-based and platform-independent. This allows users to access the system regardless of their device's operating system.

Runtime Requirements

Table 27
Runtime Requirements for Arnis Connect

Component	Specification	Notes
Backend Runtime	PHP 7.4 or later recommended	Handles server-side logic and database operations.
Database	MySQL	Stores users, verification records, applications, members, trainings, attendance, events, funds, and reports.
Web Server	Apache or Nginx	Managed by hosting provider or XAMPP during local development.
Browser	Chrome, Edge, Firefox, Safari	Required for system access; modern browsers are needed for camera and geolocation features.
Camera Permission	Allowed by member device	Required for face enrollment and attendance verification.
Location Permission	Allowed by member device	Required for geolocation validation during attendance.
Internet	Stable connection	Required for login, data updates, attendance, and report generation.

The runtime environment requires a hosting platform that supports PHP and MySQL. The server must be capable of executing PHP scripts, handling HTTP/HTTPS requests, and managing database connections. A stable internet connection is required to ensure continuous system availability and proper execution of operations such as attendance recording, event management, and report generation.

The system does not require special runtime installation on client devices. Most processing is handled by the server, while browser-supported features such as camera access and geolocation are used on the client side for automated attendance verification.

Network Requirements

Table 28
Network Requirements for Arnis Connect

Component	Specification	Notes
Connectivity	Internet-based access	No local network dependency
Protocol	HTTP/HTTPS	Secure communication via HTTPS preferred
Bandwidth	Stable broadband or mobile data	Required for smooth performance
Access Type	Centralized web access	All users connect to hosted system

Arnis Connect requires a stable internet connection because the system is accessed through a browser and communicates with a centralized database. HTTPS is strongly recommended for the deployed system because login credentials, personal member information, attendance records, face-related verification data, location data, and financial records are sensitive. Location and camera permissions are requested only when the user uses features that require them, such as face setup and automated attendance.

Database Server Environment

Table 29
Database Server Environment for Arnis Connect

Component	Specification	Notes
Database System	MySQL	Centralized database for all system records.
Access Tool	phpMyAdmin or hosting database manager	Used for importing SQL files, backups, and administration.
Main Data Stored	Users, verification codes, applications, members, trainings, attendance, events, funds, reports, audit logs	Supports operational records and monitoring.
Backup	Manual or hosting-provided backup	Recommended regularly to prevent data loss.
Access Control	Restricted database credentials	Database access should be limited to authorized developers or administrators.

The database server environment of Arnis Connect is based on MySQL, which stores all important system data such as user accounts, member records, registration requests, attendance logs, training schedules, event information, belt progression records, financial transactions, and reports. The system may be deployed using a web hosting platform that supports PHP and MySQL. The deployment process includes uploading the system files, importing the SQL database, setting database credentials, checking system routes, testing login, and verifying role-based dashboards. After deployment, the team must also test whether member registration, account verification, application approval, face setup, geolocation attendance, finance recording, and report generation work properly in the hosted environment.

Application Hosting or Deployment Environment

Table 30
Application Hosting or Deployment Environment for Arnis Connect

Component	Specification	Notes
Hosting Platform	PHP-MySQL-compatible hosting provider	The host must support PHP, MySQL, and HTTPS where possible.
Deployment Method	File Manager, FTP, or similar upload method	Uploads PHP files, assets, and configuration files.
Database Setup	Import SQL file using phpMyAdmin	Creates the database tables and initial data.
Domain	Free or custom domain	Used to access the deployed system.
Configuration	Database connection, base URL, session settings, and file paths	Required after deployment to ensure proper system operation.
HTTPS Support	SSL certificate enabled if available	Recommended for camera/location permissions and secure data transmission.

Arnis Connect is deployed using a PHP-MySQL-compatible web hosting provider such as InfinityFree or similar services. The deployment process involves uploading the application files to the hosting server, configuring the database connection, and importing the system database using tools such as phpMyAdmin.

After deployment, the application is accessed through a domain or hosting URL. System configurations are adjusted to match the production environment, including database credentials, file paths, and session settings. This ensures that the system operates correctly in its live environment.

User Environment Requirement

Table 31
User Environment Requirements for Arnis Connect

Component	Specification	Notes
Device	Any internet-enabled device	Desktop, laptop, tablet, or smartphone may be used.
Browser	Modern web browser	Chrome, Edge, Firefox, or Safari recommended.
Access	Login credentials required for internal users and members	Admins, coaches, finance officer, and members must log in.
Public Access	Registration page available	Applicants can create an account and verify before filling out the application form.
Camera Permission	Required for face setup and automated attendance	Members must allow camera access when enrolling face profile or taking attendance.
Location Permission	Required during attendance	Members must allow location access for geofence checking.
User Training	Basic orientation recommended	Users should be informed how to register, approve, record attendance, and protect account credentials.

End users of Arnis Connect only need an internet-enabled device and a supported web browser to access the system. Internal users such as administrators, coaches, and finance officers must have valid login credentials, while applicants can access the public registration page without logging in. Members who will use automated attendance must allow camera and location permissions for facial recognition and geolocation verification. The system should also inform users why these permissions are needed before attendance is recorded.

Security Requirements

Table 32
Security Requirements for Arnis Connect

Component	Specification	Notes
Authentication	Username/email and password login with password hashing.	Secures access to system accounts.
Account Verification	Email/Gmail and mobile verification codes during registration.	Ensures that applicants verify contact information before proceeding to the application form.
Access Control	Role-Based Access Control (RBAC).	Limits access based on user role such as Admin/Lead Coach, Coach, Finance Officer, Member/Player, and Applicant.
Session Handling	Session timeout, protected pages, and logout.	Prevents unauthorized access and direct page access without login.
Input Validation	Validate and sanitize all form inputs.	Prevents invalid data, errors, and injection attacks.
Database Security	Prepared statements, restricted credentials, and regular backups.	Protects stored data and supports recovery.
Application Approval Security	Approval button must update the database and prevent duplicate approval.	Ensures that approved applicants are properly stored as members.
Face Recognition Privacy	Use consent notice, limited access, and secure storage of face profile or face-related data.	Protects sensitive biometric-related information.
Geolocation Privacy	Store only necessary attendance location details and verification results.	Limits unnecessary exposure of user location data.
Attendance Integrity	Use face verification, geofence checking, timestamp logging, and coach review for flagged attendance.	Reduces attendance misuse and supports accountability.

Finance Security	Finance records limited to authorized Finance Officer or Admin users.	Protects income, expense, and balance information.
Audit Logs	Record important actions and system changes.	Supports monitoring, accountability, and troubleshooting.
HTTPS	Enable HTTPS if supported by the hosting provider.	Secures data transmission and supports camera/geolocation features in modern browsers.
Privacy Compliance	Use privacy notices, consent, access limitation, and responsible data handling.	Supports responsible handling of personal, location, attendance, and face-related data.

The updated security requirements of Arnis Connect protect system access, user information, and organizational records. The system uses login authentication, password protection, session management, role-based access control, input validation, and secure database operations. Sensitive functions such as user management, application approval, attendance review, finance recording, reports, and audit logs are limited to authorized users only. Since the system also uses facial recognition and geolocation for attendance, users must be informed why camera and location permissions are required. These measures help maintain data privacy, confidentiality, integrity, and system reliability.

CURRICULUM VITAE

CURRICULUM VITAE



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EXPERTISE

- Web Development (HTML, CSS, JavaScript, PHP)
- Basic Programming
- Basic Video Editing

LANGUAGE

- English
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PROFILE

An Information Systems student with a strong interest in leveraging technology to solve business problems. Possesses foundational knowledge in systems analysis, database management, and software development, along with strong analytical and organizational skills. Demonstrates the ability to learn quickly, adapt to new technologies, and collaborate effectively in team environments. Eager to apply academic knowledge to real-world projects and contribute to organizational success.

EXPERIENCE

None



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- PHP & MySQL
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PROFILE

I am an Information Systems student with hands-on experience in web development, system documentation, and client-based academic projects, along with professional experience in non-voice customer service. During my contract at Foundever under the Gymshark account, I handled customer concerns through chat and email and consistently performed well in quality assurance. I bring a combination of technical knowledge, communication skills, adaptability, and attention to detail, and I am eager to contribute to real-world projects and workplace success.

EXPERIENCE

Customer Service Representative (Non-Voice)
Foundever | Gymshark Account
4-month contract
Handled customer concerns through chat and email for the Gymshark account. Helped customers with orders, returns, delivery, and product-related questions while making sure responses were clear, polite, and accurate. Performed well in quality assurance and maintained good service quality throughout the contract.



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- Presentation Skills
- Networking
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PROFILE

Likes complex puzzle and solving process based problems.

EXPERIENCE

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- Developed and executed sales strategies, resulting in a 25% increase annual revenue. Managed a portfolio of 50+ clients, achieving a 95% customer retention rate.
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EXPERTISE

- Web Development (HTML, CSS, JavaScript, PHP)
- Basic Programming
- Basic Video Editing

LANGUAGE

- English
- Filipino
- Cebuano

PROFILE

I am an Information Systems student with basic knowledge in systems analysis, database management, and software development. Although I am still developing my skills, I am eager to learn more and continuously improve my understanding of technology and how it can be used to solve real-world problems. As a student, I am looking forward to gaining hands-on experience where I can apply my basic knowledge, develop new skills, and build my confidence in the field of Information Systems. I am motivated to grow, learn, and contribute to an organization while continuing to enhance my abilities.

EXPERIENCE

None

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APPENDICES

APPENDIX A

Oath of Confirmation

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COLLEGE OF COMPUTER, INFORMATION AND COMMUNICATIONS TECHNOLOGY

APPENDIX A

Capstone Project 1

Oath of Confirmation

Date: March 4, 2026

OATH OF CONFIRMATION

This is to confirm that I, Renz Kayne P. Luab, a BSIS-3A student, currently enrolled in the course of Capstone Project I, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

RENZ KAYNE P. LUAB / 3-6-2026

Signature of Student Over Printed Name / Date

Date: March 4, 2026

OATH OF CONFIRMATION

This is to confirm that I, Renz Kayne P. Luab, a BSIS-3A student, currently enrolled in the course of Capstone Project 1, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

RENZ KAYNE P. LUAB / 3-6-2026

Signature of Student Over Printed Name / Date

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APPENDIX A

Capstone Project 1

Oath of Confirmation

Date: March 4, 2026

OATH OF CONFIRMATION

This is to confirm that I, Christ Jhon Denver S. Partosa, a BSIS-3A student, currently enrolled in the course of Capstone Project I, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

Christ Jhon Denver S. Partosa / 3-6-2026

Signature of Student Over Printed Name / Date

Date: March 4, 2026

OATH OF CONFIRMATION

This is to confirm that I, Christ Jhon Denver S. Partosa, a BSIS-3A student, currently enrolled in the course of Capstone Project 1, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

Christ Jhon Denver S. Partosa / 3-6-2026

Signature of Student Over Printed Name / Date

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APPENDIX A

Capstone Project 1

Oath of Confirmation

Date: February 03, 2026

OATH OF CONFIRMATION

This is to confirm that I, Ulric Villagracia, a BSIS-3A student, currently enrolled in the course of Capstone Project I, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

Ulric Villagracia

Signature of Student Over Printed Name / Date

Date: February 03, 2026

OATH OF CONFIRMATION

This is to confirm that I, Ulric Villagracia, a BSIS-3A student, currently enrolled in the course of Capstone Project 1, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

ULRIC VILLAGRACIA

Signature of Student Over Printed Name / Date

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APPENDIX A

Capstone Project 1

Oath of Confirmation

Date: March 4, 2026

OATH OF CONFIRMATION

This is to confirm that I, Jannalyn N. Ycoy, a BSIS-3A student, currently enrolled in the course of Capstone Project I, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

JANNALYN N. YCOY 3-6-2026

Signature of Student Over Printed Name / Date

Date: March 4, 2026

OATH OF CONFIRMATION

This is to confirm that I, Jannalyn N. Ycoy, a BSIS-3A student, currently enrolled in the course of Capstone Project 1, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

JANNALYN N. YCOY 3-6-2026

Signature of Student Over Printed Name / Date

APPENDIX B

Capstone Project Team Composition Form

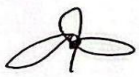
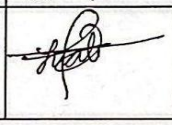
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APPENDIX B

Capstone Project Team Composition Form

Name	Signature	Role Assignment	Email	Contact #
1. **Christ Jhon Denver S. Partosa**		Hustler	christpartosa9@gmail.com	09927854750
2. **Ulric, Villagracia**		Hacker	ULRICV7@gmail-com	09062128264
3. **Jannalyn N. Ycoy**		Hipster	ycoyjannalynnunez@gmail.com	09478598521
4. **Renz Kayne P. Luab**		Hipster	renzkayneluab@gmail.com	09685401841

APPENDIX C

Acceptance of Capstone Adviser

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Appendix C

Acceptance of Capstone Adviser

College of Computer, Information and Communications Technology

Acceptance of Capstone Adviser

This is to certify that I am accepting the ByteWorks to be their Capstone Project adviser in the degree of Bachelor of Science and Information System.

CERRY RITA R. BELLING
(Signature of Adviser over printed name)

Date: 3/6/2026

Noted by:

Jose Maria S. Garcia II, PhD

(Signature of Subject Teacher over printed name)

Date: _____

APPENDIX D

Capstone Project Feasibility Study Model

The feasibility analysis of Arnis Connect was conducted to determine the viability of the proposed web-based sports organization management system for the Lapu-Lapu Arnis de Abanico Organization. Each feasibility dimension evaluates the capability of the system to be successfully developed and implemented within the scope of the academic capstone project and the operational environment of the organization.

Technical Feasibility

Arnis Connect is technically feasible with manageable constraints. The development team possesses sufficient knowledge and experience in PHP, MySQL, HTML, CSS, JavaScript, Bootstrap, and related web development technologies. The proposed system architecture consists of a centralized web-based platform with role-based access control, integrated attendance management, event monitoring, financial recording, analytics processing, geolocation validation, and facial recognition attendance verification. Development tools such as XAMPP, phpMyAdmin, and Visual Studio Code support efficient system development, testing, and deployment.

Frontend libraries including Bootstrap, DataTables, Chart.js, and SweetAlert improve responsiveness, reporting visualization, and user interaction. Advanced technologies such as Python-based analytics and OpenCV facial recognition may also be integrated to support player suggestion analytics, attendance validation, and organizational insights. Although analytics integration, geolocation services, and facial recognition introduce moderate implementation complexity and dependency on internet connectivity and device permissions, these technologies remain achievable within the technical capabilities of the development team. Overall, the technical feasibility of Arnis Connect is assessed as high.

Operational Feasibility

The system is operationally feasible because it aligns with the actual administrative workflow and operational structure of the organization. Arnis Connect supports administrators, coaches, finance officers, members, and applicants through role-based dashboards and centralized organizational management functions. The system simplifies attendance monitoring, member management, registration approval, event coordination, player participation tracking, financial recording, and report generation through an organized and accessible web-based interface.

The centralized nature of the system improves coordination, record accessibility, and organizational monitoring compared to fragmented spreadsheets and manual documentation. Attendance validation through geolocation and facial recognition also strengthens monitoring accuracy and accountability. While some users may initially require orientation and familiarization with advanced system features, the system interface is designed to remain understandable and accessible through standard web browsers. Overall, the operational feasibility of Arnis Connect is assessed as high.

Economic Feasibility

Arnis Connect is economically feasible because it primarily utilizes open-source technologies and existing development resources, minimizing software acquisition and operational costs. The proposed system uses freely available technologies such as PHP, MySQL, Bootstrap, JavaScript libraries, Python-based analytics tools, XAMPP, and Visual Studio Code. Existing computers, webcams, and internet-enabled devices can support development and deployment without requiring expensive infrastructure investments.

The system is expected to reduce administrative inefficiencies caused by manual attendance logs, paper-based records, fragmented spreadsheets, and repetitive reporting tasks. Centralized information management, automated report generation, attendance monitoring, and player analytics improve operational efficiency and reduce workload for administrators, coaches, and finance officers. Although analytics processing and facial recognition integration may introduce additional implementation complexity, the expected long-term benefits outweigh the development costs. Overall, the economic feasibility of Arnis Connect is assessed as high.

Schedule Feasibility

The system is schedule feasible and aligned with the academic timeline of the capstone project. The development of Arnis Connect follows an Agile-based development methodology, allowing iterative development, continuous testing, and phased implementation of system features. Core modules such as authentication, member management, attendance monitoring, event management, and financial recording are prioritized during the early development phases, while advanced features such as predictive analytics, geolocation validation, and facial recognition may be implemented progressively.

Incremental testing and debugging support manageable refinement and issue resolution throughout the development cycle. The web-based deployment process is also relatively simple and achievable within the available academic schedule. Although advanced integrations may require additional testing and optimization, the project timeline remains manageable through phased development and prioritized implementation. Overall, the schedule feasibility of Arnis Connect is assessed as high.

Legal Feasibility

Arnis Connect is legally feasible provided that user information, attendance records, financial data, and facial recognition images are handled responsibly and used strictly for authorized organizational purposes. The system applies authentication mechanisms, role-based access control, and secured record management procedures to reduce unauthorized access and misuse of information.

Collected member and applicant information is intended solely for organizational management, attendance monitoring, event coordination, and financial recording. Facial recognition functionality also requires responsible handling of image data and proper consent procedures from users. Most development tools, libraries, and frameworks used in the project are open-source and legally accessible for academic development purposes. Overall, the legal feasibility of Arnis Connect is assessed as high.

Based on the evaluation across all feasibility dimensions, Arnis Connect is considered viable for development and implementation. The system demonstrates strong technical capability, operational readiness, economic practicality, manageable scheduling, and compliance with legal and ethical considerations. These findings indicate that Arnis Connect presents a sustainable and effective solution for improving administrative coordination, attendance monitoring, event management, player participation tracking, and financial management within the Lapu-Lapu Arnis de Abanico Organization.

APPENDIX E

Capstone Project Comparative Matrix

Features	TeamSnap	TeamLinkt	Teamup	Spond	Arnis Connect
Member profile management	✓	✓	✗	✓	✓
Attendance monitoring	✓	✓	Limited	✓	✓
Event scheduling / planning	✓	✓	✓	✓	✓
Communication / notifications	✓	✓	✓	✓	✓
Financial tools / payments	✓	✓	✗	✓	✓
Role-based access	Limited	✓	Limited	Limited	✓
Registration approval workflow	✗	Partial	✗	✗	✓
Event documentation with participants	Limited	Partial	✗	Limited	✓
Reports generation	Partial	✓	Limited	Limited	✓
Player suggestion feature	✗	✗	✗	✗	✓
Total	62.5%	75%	35%	65%	100%

APPENDIX F

Capstone Project Comparative Matrix

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Appendix F

Research Working Title Form

Name of the Proponents	
1. Christ Jhon Denver S. Partosa	Hustler
2. Ulric Villagracia	Hacker
3. Jannalyn N. Ycoy	Hipster
4. Renz Kayne P. Luab	Hipster
Proposed Research Title: Arnis Connect: An Integrated Web and Mobile-based Management System for Arnis Sports Program	
Date: March 6, 2026	Date:
Submitted by: Christ Jhon Denver S. Partosa	Noted and Approved by:
(Signature of Project Leader over Printed Name)	(Signature of Subject Teacher over Printed Name)

APPENDIX G

Capstone Project 1 Consultation Logs Form



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Appendix G. Capstone 1 Consultation Logs Form

Research/Capstone Project Title

Arnis Connect: An Integrated Web and Mobile-based Management System for Arnis Sports Program

Name of the Proponents

Renz Kayne P. Luab	Jannalyn N. Ycoy
Christ Jhon Denver S. Partosa	
Ulric Villagracia	

Name of the Group

ByteWorks

1st Consultation: CHAPTER 1 must be completely delivered for the adviser's evaluation

- In scope and limitations, list the modules and connect it into the specific objectives
- In the environment, specify why the organization was chosen
- Make the diagrams bigger.

Date	**PM's Signature**	**Adviser's Signature**



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2nd Consultation: Chapters 1 and 2 must be completely delivered for the adviser's evaluation

- Citation must be uniform
- Improve the related systems
- Add percentage on the comparative matrix
- Add more references

Date	PM's Signature	Adviser's Signature

3rd Consultation : CHAPTER 1, 2 and 3 must be completely delivered for adviser's evaluation

- Wireframe, add explanation each diagram/figure
- Make figures/diagrams bigger
- Update works in work breakdown structure
- Make specific objectives more clear

Date	PM's Signature	Adviser's Signature
April 16, 2024		

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4th Consultation : CHAPTER 1, 2, 3 and including initial and finals pages must be completely delivered for advisers' evaluation

Improve citations
Panel names
Table and figure no. / name - Improve
Add respondents
Finalization of paper

April 18, 2026		
Date	**PM's Signature**	**Adviser's Signature**

This is to certify that I have been regularly consulted by my advisees; have reviewed their manuscript of the above-stated study. As their adviser, I therefore submit them ready for Proposal Hearing as the Chapters 1, 2, and 3 and the pertinent parts of their manuscript are complete.

Gerry Ritz R. Densing
(Signature of Adviser over printed name)

APPENDIX H

List of Modules

Module	Description
Login and Session Management	Handles user authentication, login, logout, and session control to ensure secure access to the system
User Management	Allows administrators to create, update, and manage user accounts, roles, and permissions
Member/ Player Management	Manages member records including personal information, member profiles, player categories, belt and rank monitoring, and status updates
Registration Approval Workflow	Processes registration requests submitted by applicants, allowing administrators to review, approve, reject, or convert applicants to full members
Attendance Management	Records and tracks member attendance during training sessions, including attendance monitoring, history, reports, geolocation attendance validation, and facial recognition attendance verification
Training Management	Manages training sessions and schedules, assigns coaches, handles training details, and maintains training records
Event Management	Handles creation, scheduling, documentation, and tracking of events including managing event participants and monitoring participation
Player Suggestion Feature	Provides intelligent participant suggestions for events based on attendance, achievements, participation history, availability, rank eligibility, and predictive player recommendations
Finance Management	Records and manages financial transactions including income, expenses, financial monitoring, summaries, and transaction history
Reports Generation	Generates reports for attendance, members, events, financial records, analytics dashboards, and predictive analytics reports for monitoring and decision-making
Analytics and Emerging Technologies	Provides system capabilities including predictive analytics, geolocation services, and facial recognition technology

APPENDIX I

System Quality Acceptance Questionnaire

Questionnaire for End Users of Arnis Connect

(Lead Coach / Administrator, Coaches / Trainers, Officers / Committee Members, and Active Members / Players)

System Quality Questionnaire

This questionnaire is designed to evaluate Arnis Connect: An Integrated Web and Mobile-Based Management System for Arnis Sports Program. It is based on selected ISO/IEC 25010 software quality characteristics, particularly Functional Suitability, Usability, Reliability, Performance Efficiency, and Security.

Instructions: Please read each statement carefully and honestly. Rest assured that any information you provide will be treated with confidentiality and used only for academic purposes.

Please put a check mark **(✓)** in the column that best represents your level of agreement with each statement.

Numerical Rating	Equivalent
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

Part I: Respondent Profile

Name: _____
Age: _____
Sex: _____

Role / Classification:
[ ] Lead Coach / Administrator
[ ] Coach / Trainer
[ ] Officer / Committee Member
[ ] Active Member / Player

Length of involvement in the organization: _____

Part II: System Quality Evaluation						
No.	**Characteristics**	**Score**				
	Functional Suitability	**1**	**2**	**3**	**4**	**5**
1	The system provides the functions needed for managing member records.					
2	The system correctly supports attendance recording during training sessions.					
3	The system provides the needed functions for event documentation and participant management.					
4	The system provides the necessary functions for recording and monitoring financial transactions.					
5	The system supports the generation of reports for attendance, events, and funds.					
	Usability					
6	The system is easy to learn and understand.					
7	The menus, labels, and page contents of the system are clear and understandable.					
8	It is easy to move from one page or module to another.					
9	The layout and design of the system make it convenient to use.					
10	The system helps users complete tasks without unnecessary confusion.					
	Reliability					
11	The system performs its functions consistently during use.					
12	The system saves records correctly without losing important information.					
13	The system produces accurate records and outputs.					
14	The system remains stable while performing different tasks.					
	Performance Efficiency					
15	The system loads pages within an acceptable amount of time.					

16	The system responds quickly when saving or updating records.					
17	The system performs smoothly during normal use.					
18	The system allows users to complete tasks without unnecessary delay.					
	Security					
19	The system properly restricts access based on user roles.					
20	The login process helps protect the system from unauthorized access.					
21	Sensitive records are accessible only to authorized users.					
22	The system gives confidence that stored information is protected.					

Part III: Open-Ended Questions

23. What part of Arnis Connect do you find most useful?

24. What problems or difficulties did you experience while using the system?

25. What improvements would you recommend for Arnis Connect?

APPENDIX J

Post-Proposal Hearing Compliance Checklist